A Coherent State View of Linear Response

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Abstract

Contents

I. INTRODUCTION

Isolated molecular systems do not interact with their environment and are thus unobservable. In order to have a physical effect and be observed they must respond to external probes, such as electromagnetic fields, and form a composite interacting system whose evolution is described by a joint Hamiltonian. The effect of radiation on the molecular subsystem can be monitored by changes in the molecules density and reduced density matrices. If the radiation is weak the behavior of the interaction is well described by the linear response of the molecular system to the radiation, i.e. is linearly proportional to the radiation strength, while if the radiation is intense the molecular system responds in a nonlinear way. The interaction of the electromagnetic field with molecules can be approximated in a standard semi classical fashion by averaging over the photonic degrees of freedom to produce effective one electron observables. This leads to a reduced Hamiltonian for the molecular subsystem that includes perturbing effective time dependent one body potentials which describe the interaction with the radiation field. In the case of weak intensity radiation, where the linear response regime is sufficient to describe the changes in the molecular system, one needs to only consider responses that are first order in these observables.

The temporal behavior of a state of physical system is completely described by the evolution of its full density matrix, which in term determines the evolution of all of its reduced density matrices. These variations are conveniently expressed in terms of various propagators, which thus provide a comphrehensive formulation of the response of the molecular subsystem to external probes. As is well known exact solutions for quantum evolutions have only been determined for simple model systems, and are not known for multi-electron molecular systems that involve coulombic interactions between electrons. Thus for practical applications the goal is to develop systematic, cost effective, theoretically well founded and non arbitrary approximation schemes, that are accurate enough to explain experimental results. A major focus of the illustrious career of professor Osvaldo Goscinski has been the analysis and construction of approximate propagators, that fulfill these objectives. In particular the one-electron and polarization propagators, where he effectively used the properties of super operators acting on operator manifolds associated with a given reference state to generate various orders of approximation [1][2][3][4][5][6][7]. In this article I will concentrate, for illustrative purposes, on the polarization propagator for a molecular system with a fixed even number of electrons at a fixed geometry. This propagator describes the linear response of the First Order Reduced Density Matrix (FORDM) to electromagnetic radiation.

The polarization propagator is a hermitian matrix of time dependent response functions of the matrix elements of the FORDM. If one takes the Laplace transform of this propagator and expresses the resulting matrix in diagonal form, in terms of normal mode operators, one can clearly observe that the poles of this propagator are excitation energies and the associated residues the transition probabilities for excitations generated by one particle operators between stationary states, $\{|\Psi_k\rangle\}$, of the molecular system. If the initial molecular reference state is stationary (usually the ground state, but not necessarily so) then the poles are the excitation energies and the residues the transition probabilities from that state. If the transition probability is 100% for a given one particle operator then this operator produces an excited state from the initial stationary state and is called a state excitation operator. If such an operator produces the excited state $|\Psi_k\rangle$, then its effect on the reference state $|\Psi_0\rangle$ is the same as the N-particle bra-ket operator $|\Psi_k\rangle\langle\Psi_0|$.

Approximations to the polarization propagator are often based on the three criteria, that

- all one particle operators that diagonalize the propagator matrix are state excitation operators,
- their adjoints annihilate the reference state,
- the reference state is stationary with respect to at least one particle variations.

If an approximation satisfies these conditions one says that it is a consistent approximation, however, in general, usually one or more of these conditions are not satisfied and the approximation is inconsistent. Some approximations can even lead to propagators that do not correspond to a valid N-particle reference state and thus are not even N-representable.

An important point of note is that it is not necessary that the normal mode one-particle operators actually produce excited states from the reference state, in fact, in the exact case they generally do not. Hence there is thus some level of ambiguity about polarization propagator approximations, if one is trying to obtain an approximation to a one-particle linear response function then the annihilation condition is not necessary, though stationarity and N-representability are desirable, while if one is asserting that the polarization propagator is a one-particle approximation to the full N-particle propagator and is used to produce excited states, as well as excitation energies and transition probabilities, then the annihilation condition is necessary for consistency. During the course of this article concrete examples of these points will be presented.

In the late 1970's it was found [8][5][6] [9][10] that an Antisymmetrized Geminal Power (AGP) State was the most general $2M$ electron state that was annihilated by a maximal set of one electron operators, (for odd number of electrons Generalized AGP (GAGP) states [**?**] are needed but their properties can be derived from AGP states). The $2M$-electron AGP $|g^M\rangle$ state is an antisymmetrized product of M two particle geminals

$$|g\rangle = \sum_{1\leq j<k\leq s} g_{jk} |\varphi_j\varphi_k\rangle \tag{1}$$

where $\{g_{ij}\}$ are complex coefficients and $\{|\varphi_j\rangle | 1 \leq j \leq r = 2s\}$ is an orthonormal basis for the one particle Hilbert space $\mathcal{H}^1$. However even when the reference state was an energy optimized AGP state the polarization propagator approximation failed to be consistent, although the numerical results obtained from the application of this approximation scheme agreed well with experimental values of excitation energies and dipole induced transition probabilities i.e. oscillator strengths[11][12][13][14][15][16] [17][18][19][20][21][22][23]

It turns out that the annihilation condition can be naturally expressed in terms of group Coherent States (CS) [24][25][26], where the reference state corresponds to a lowest or highest weight state associated with a representation of the group. The basic ingredients needed for the construction of a family of group CS's in a vector space is the existence of an irreducible representation of a group and a one dimensional representation of a subgroup. The subgroup is called the isotropy or invariance group of the CS's and the family of CS's is generated by the action of the coset formed by factoring the group with respect to (wrt) this subgroup. In the study of electronic systems it is natural to consider the real, compact, [27] Lie group $U(r)$ of unitary transformations of the one electron state space $\mathcal{H}^1$ (of dimension $r = 2s$), that has irreducible representations on all N-electron state spaces $\mathcal{H}^N$. Hence in this article we will primarily consider nonlinear manifolds of the N-electron Hilbert space, $\mathcal{H}^N$, that are produced by families of coherent states generated by cosets formed by one-particle operators. We will construct approximate propagators based on stationary states of the restriction of the Time Dependent Schrödinger Equation (TDSE) to these manifolds. Solutions to such restrictions of the TDSE are obtained by using the Time Dependent Variational Principle (TDVP) [28] or equivalently the Stationary Phase Approximation (SPA) to the path integral expressed in terms of paths on this CS manifold. The resulting Equations of Motion (EOM) are those of a generalized *classical* system described by a curved phase space, a consequence of the fact that a family of group coherent states forms a differentiable manifold [29] whose tangent spaces have an intrinsic symplectic structure [27]. Approximate stationary states of the molecular system then correspond to the special stationary phase "paths" that are critical points of the corresponding classical Hamiltonian. The linear response of these approximate stationary states to external perturbing fields then involves the generators of the Lie algebra of the group, and hence describes the response of the reduced density matrix formed by these operators. Thus the response of the FORDM is determined by generators of one-particle groups.

The path integral construction is powerful method that is often used to evaluate the full quantum propagator between an initial and a final state. In the CS approach path integrals can be formulated in terms of complex valued paths in spaces of group coherent states. These spaces are complex differentiable manifolds, that are also coset spaces of groups. The family of coherent states chosen determines the specific geometric structure of the space of the complex valued paths, $\mathrm{z}(t)$, via a metric function $C\left(\mathrm{z}(t), \mathrm{z}'(t)\right)$. The time development of the paths is described by a classical Hamiltonian function $\mathcal{E}\left(\mathrm{z}, \mathrm{z}', t\right)$. The form and degree of these functions is determined by the family of coherent states, and thus the group-isotropy subgroup pair. If the Lagrangian associated with the Hamiltonian function is quadratic then the path integral can be evaluated exactly, if this is not the case approximations must be invoked.

In detail this article will be composed of the following sections. In section II we discuss the Lie algebra structure of Fermi-Fock space, which is the direct sum of all of the state spaces $\left\{\mathcal{H}^N\right\}$, section III will describe the possible decompositions of the Lie algebra $su\left(2s\right)$ of $SU(2s)$, the one particle group associated with $\mathcal{H}^1$, and thus the factorizations of $SU(2s)$ that lead to manifolds of coherent states in $\mathcal{H}^N$ that can be based on $SU(2s)$. In section IV we study in detail the form of coherent states, in particular the family of AGP states, and the geometric and phase space structure of coherent state manifolds. In section E we briefly describe the coherent state formulation of the path integral and the SPA. In section VII we come to the central topic of this paper; Linear Response Theory in the context of coherent states. We conclude with a discussion in section VIII.

II. LIE ALGEBRAS

The most convenient representation of the generators of the Lie algebra of $SU(2s)$ for fermionic quantum mechanics is by second quantized operators acting in Fermi-Fock space, $\mathcal{H}_F$, which is defined to be the direct sum

$$\mathcal{H}_F = \bigoplus_{0\leq N\leq r} \mathcal{H}^N \tag{2}$$

where

$$\mathcal{H}^N = \bigwedge \mathcal{H}^1 \tag{3}$$

is the N -fold antisymmetric tensor product of the one particle state space $\mathcal{H}^1$, a space generated by an orthonormal basis of one particle functions of position and spin

$$\left\{\varphi_j \middle| 1 \leq j \leq r\right\}. \tag{4}$$

The Banach space [**?**] of bounded operators acting in $\mathcal{H}_F$ is denoted by $\mathcal{B}(\mathcal{H}_F)$, and is spanned by a second quantized operator basis

$$\left\{ a_{j_1}^{\dagger} \cdots a_{j_p}^{\dagger} a_{k_q} \cdots a_{k_1} \middle| 1 \leq j_1 < \cdots j_p \leq r; 1 \leq k_1 < \cdots k_p \leq r; 1 \leq p, q \leq r \right\} \tag{5}$$

where the second quantized operators $\left\{ a_j^{\dagger}, a_j \right\}$ act on the zero particle vacuum $|\phi\rangle$ in the following fashion

$$\begin{aligned} a_j |\phi\rangle &= 0 \\ a_j^{\dagger} |\phi\rangle &= |\varphi_j\rangle \end{aligned} \tag{6}$$

and satisfy the Canonical Anti-commutation Relationships (CAR)

$$\left[a_j^{\dagger}, a_k \right]_+ = \delta_{jk} I \tag{7}$$

The operator space $\mathcal{B}(\mathcal{H}_F)$ contains realizations of many classical Lie groups, algebras and universal enveloping algebras [**?**]. In particular the generators

$$\left\{ a_j^{\dagger}, a_k, a_j^{\dagger} a_k - \frac{1}{2}\delta_{jk} I, a_j^{\dagger} a_k^{\dagger}, a_j a_k; 1 \leq j \leq k \leq r \right\} \tag{8}$$

satisfy the commutation relationships of the lie algebra $so(2r+1, \mathbb{R})$, (see [**?**] for example for the definitions of classical lie groups and algebras), and generate a representation of the universal enveloping algebra $so(2r+1, \mathbb{R})$ in $\mathcal{B}(\mathcal{H}_F)$, which is fact all of $\mathcal{B}(\mathcal{H}_F)$. The subset of these generators

$$\left\{ a_j^{\dagger} a_k - \frac{1}{2}\delta_{jk} I, a_j^{\dagger} a_k^{\dagger}, a_j a_k; 1 \leq j \leq k \leq r \right\} \tag{9}$$

satisfy the commutation relationships of the lie algebra of $so(2r, \mathbb{R})$ and generate a representation of the universal enveloping algebra of $so(2r, \mathbb{R})$ in $\mathcal{B}(\mathcal{H}_F)$. The subset of these generators

$$\left\{ a_j^{\dagger} a_k - \frac{1}{2}\delta_{jk} I; 1 \leq j \leq k \leq r \right\} \tag{10}$$

satisfy the commutation relationships of the lie algebra $su(r)$ and generate a representation of the universal enveloping algebra of $su(r)$ in $\mathcal{B}(\mathcal{H}_F)$. The representation of $so(2r+1, \mathbb{R})$ is irreducible, while the representations of $so(2r, \mathbb{R})$ and $su(r)$ are reducible. The irreducible representations of $so(2r, \mathbb{R})$ are obtained by restricting to the subspace of even or odd number states in $\mathcal{H}_F$, while the irreducible representations of $su(r)$ are obtained by restricting to subspaces of states of fixed integer particle number.

III. ONE PARTICLE GROUPS

In this section we describe the possible direct sum decompositions of the lie algebra, $u(2s)$, of the group $U(2s)$ of unitary transformations of one particle state space, $\mathcal{H}^1$ of dimension $2s$, which lead in a canonical fashion to families of CS's in N-particle state space $\mathcal{H}^N$. These decompositions are based on the subgroup of special orthogonal transformations, $SO\left(2s,\mathbb{R}\right)$, of a real $2s$-dimensional Hilbert space, the subgroup, $U\left(N\right)\times U\left(2s-N\right)$, formed by products of unitary transformations of complex N and $2s-N$ dimensional Hilbert spaces and the subgroup, $USp\left(2s\right)$, of transformations of a complex $2s$-dimensional Hilbert space, that are simultaneously unitary and symplectic. This latter subgroup is isomorphic to the group of unitary transformations, $U\left(s,\mathbb{Q}\right)$ of a quaternionic s-dimensional Hilbert space. We signify the CS's that they induce as G/K-CS's, where $G=SU\left(2s\right)$ and the subgroup $K\in\left\{SO\left(2s,\mathbb{R}\right),SU\left(N\right)\times SU\left(2s-N\right),USp\left(2s\right)\right\}$.

The Lie algebra $u(2s)$ has three direct sum Cartan decompositions [**?**] which are in terms of the Lie algebras, $so\left(2s,\mathbb{R}\right)$, $\left[u\left(N\right)\oplus u\left(2s-N\right)\right]$, and $usp\left(2s\right)$, of its maximal compact (Appendix **??**) subgroups $SO\left(2s,\mathbb{R}\right)$, $\left[U\left(N\right)\times U\left(2s-N\right)\right]$, and $USp\left(2s\right)$, (see [**?**] for the definitions of these groups). These decompositions are

$$u\left(2s\right)=so\left(2s,\mathbb{R}\right)\oplus so\left(2s,\mathbb{R}\right)^{\perp} \tag{11a}$$

$$u\left(2s\right)=\left[u\left(N\right)\oplus u\left(2s-N\right)\right]\oplus\left[u\left(N\right)\oplus u\left(2s-N\right)\right]^{\perp};\quad 0\leq N\leq 2s \tag{11b}$$

$$u\left(2s\right)=usp\left(2s\right)\oplus usp\left(2s\right)^{\perp} \tag{11c}$$

where the orthogonality is with respect to the killing form inner product [**?**] (also see Appendix **??**) of $u(2s)$. The decomposition eqs. $(11a,11b,11c)$ define the following distinct coset decompositions of $U(2s)$

$$U\left(2s\right)\Big/_{U\left(N\right)\otimes U\left(2s-N\right)};\quad 0\leq N\leq 2s \tag{12a}$$

$$U\left(2s\right)\Big/_{USp\left(2s\right)} \tag{12b}$$

$$U\left(2s\right)\Big/_{SO\left(2s,\mathbb{R}\right)} \tag{12c}$$

and thus three classes of families of coherent states (see Section IV).

The motivation of using *maximal* compact subgroups in eqs. $(12a)$, $(12b)$ and $(12c)$ is to produce coset spaces that are irreducible Riemannian Manifolds (a Riemannian Manifold has tangent spaces that are inner product spaces [**?**]) i.e. they cannot be factored into a Riemannian product [**?**] of smaller Riemannian Manifolds. This technical feature, although important in the classification of Riemannian Manifolds, is too restrictive when considering manifolds of CS's, where we also need to examine decompositions generated by subgroups of these maximal compact groups. We thus enlarge the designation G/K-CS to include subgroups of K.

In order for any of the subgroups $\{SO(2s,\mathbb{R}), \{U(N)\otimes U(2s-N)|\, 0\leq N\leq 2s\}, USp(2s)\}$, or their subgroups, to be isotropy groups of a given N-electron state, the state must carry a one dimensional unitary representation of that subgroup and not of any larger subgroup containing that subgroup.

There are *no* states in $\mathcal{H}^N$ that produce a one-dimensional representation of $SO(2s,\mathbb{R})$ and we conjecture (see Appendix **??**) that there are *no* subgroups of $SO(2s,\mathbb{R})$, which *are not* subgroups of $U(N)\otimes U(2s-N)$ or $USp(2s)$, that produce one-dimensional representations on $\mathcal{H}^N$. Thus there are *no* CS manifolds in $\mathcal{H}^N$ based on subgroups of $SO(2s,\mathbb{R})$ that cannot be described in terms of $U(N)\otimes U(2s-N)$ or $USp(2s)$ CS's.

The only subgroup from the set $\{U(N)\otimes U(2s-N)|\, 0\leq N\leq 2s\}$ that produces a one-dimensional representations in $\mathcal{H}^N$ is $U(N)\otimes U(2s-N)$. The

$$U(2s)\Big/U(N)\otimes U(2s-N) \tag{13}$$

-coherent states give the well known Thouless parameterization [**?**] of the manifold of Independent Particle States (IPS) and lead to the standard Random Phase Approximation (RPA) for the linear response function. The subgroup $U(N)\otimes U(2s-N)$ is actually a subgroup of $USp(2s)$ when $N=2M$ and we display in section (IV) that $U(2s)/[U(N)\otimes U(2s-N)]$-CS's can in fact be described in terms of the class of $U(2s)/USp(2s)$-CS's.

We show by construction in section (IV) that the group $USp(2s)$ and its subgroups do have one dimensional representations in $\mathcal{H}^N=\mathcal{H}^{2M}$ and one can form

$$U(2s)/K-\text{coherent states,where } USp(2s)\supseteq K, \tag{14}$$

which are manifolds of general Antisymmetrized Geminal Power (AGP) states (see Appendix **??**). These states produce a generalized RPA for the linear response function (see Section VII). All manifolds of CS's in $\mathcal{H}^{2M}$ based on $U(2s)$ can thus be expressed in terms of the cosets eq. (14) which have the form

$$U(2s) \Big/ \left[USp(2\omega_1) \cdots \times USp(2\omega_p) \times SU(2)^\sigma \times SU(2\xi)\right], \tag{15}$$

where

$$SU(2)^\sigma = \overbrace{SU(2) \times \cdots \times SU(2)}^{\sigma-\text{factors}}, \tag{16}$$

the index set $\{\sigma, \omega_1, \cdots, \omega_\rho, \xi\}$ characterizes the type of AGP reference state and its elements satisfy

$$\begin{aligned} 2\sigma + 2\omega + 2\xi = r = 2s \\ \omega = \sum_{1 \le j \le \rho} \omega_j. \end{aligned} \tag{17}$$

The index set $\{\sigma, \omega_1, \cdots, \omega_\rho, \xi\}$ is determined by the array $\{c_j | 1 \le j \le s\}$ of canonical canonical coefficients [**?**] of the geminal that determines the AGP state according to:

- σ = number of non zero, non equal $\{c_j\}$'s
  - ω_j = number of equal coefficients in the j^{th} set
  - ρ = number of sets of equal $\{c_j\}$'s
  - ξ = number of zero coefficients.

If $\xi = 0$ and $\omega = 0 \Leftrightarrow \rho = 0$, i.e. all the coefficients are different and non zero, then the isotropy subgroup is

$$SU(2)^s = \overbrace{SU(2) \times \cdots \times SU(2)}^{s-\text{factors}} \tag{18}$$

which is the smallest one possible, leading to the biggest coset. In this case the action of coset on the reference AGP generates the manifold of all AGP states. If $\xi \neq 0$ the reference AGP is produced by a degenerate geminal and the action of the associated coset only produces a submanifold of AGP states, all of which correspond to geminals that have the same null space (see Appendix **??**) as the reference AGP. If $\rho \neq 0$ then the geminal has some extreme components and again the coset only generates a sub manifold of AGP's from the reference. If $\sigma = 0$, $\xi = 0$ and $\rho = s$ then the reference state is an extreme AGP (see Appendix **??**) and the action of the coset on the reference produces the manifold of *all* extreme AGP states.

All CS manifolds in $\mathcal{H}^{2M}$ based on the one particle group $U(2s)$ are families of AGP states, some of these manifolds are irreducible Riemannian manifolds and correspond to cosets formed by the maximal compact subgroups $U(2M) \otimes U(2s-2M)$ and $USp(2s)$, while others are reducible and correspond to non maximal compact subgroups $USp(2\omega_1) \cdots \times USp(2\omega_p) \times SU(2)^{\sigma} \times SU(2\xi)$. Each of these manifolds of CS's have certain basic physical properties e.g. $U(2M) \otimes U(2s-2M)$ invariant manifold describes uncorrelated Independent Particle States (IPS's), the $USp(2s)$ invariant manifold describes highly correlated extreme AGP states that are superconducting, while the $USp(2\omega_1) \cdots \times USp(2\omega_\rho) \times SU(2)^{\sigma} \times SU(2\xi)$ invariant manifold for general $\{\omega_1, \cdots, \omega_\rho, \sigma, \xi\}$ describe intermediate types of correlation and linear response properties, see for a particular example [**?**], most of which have not been explored in any depth.

IV. COHERENT STATES

A. Definition and Construction

The coherent state construction is based on the Hilbert space of states $\mathcal{H}$ that carries unitary irreducible representations $\mathcal{R}(G)$ of a compact group G on $\mathcal{H}$ and a one dimensional representation of a subgroup $K \subset G$. The groups G we consider are $U\left(\mathcal{H}^N\right)$ and $U\left(\mathcal{H}^1\right)$ the groups associated with N-electron state space and 1-electron state space respectively. In the finite dimensional case these groups are the classical matrix groups and $U\left(\mathcal{H}^N\right) \equiv U\left(\binom{r}{N}\right)$ and $U\left(\mathcal{H}^1\right) \equiv U(r)$, in general we will use the matric notation even though in an exact treatment $r = \infty$ and one should be aware of possible infinite dimensional pathologies. The irreducible representations we consider are $\mathcal{R}\left(U\left(\binom{r}{N}\right)\right)$ and $\mathcal{R}(U(r))$ that are carried by $\mathcal{H}^N$, unless otherwise stated we will not distinguish between the groups G and K and their representations on $\mathcal{H}^N$.

A state $|\Phi\rangle \in \mathcal{H}^N$ is a reference state for a group G if there is an isotropy subgroup K that is contained in G that has 1-dimensional representation on $\mathcal{H}^N$ defined by

$$K = \left\{ g| \quad \mathcal{R}(g)|\Phi\rangle = e^{i\theta(g)}|\Phi\rangle; g \in G; \theta(g) \in \mathbb{R} \right\}. \tag{19}$$

If such isotropy groups exists G can be factored as

$$G = (G/K)K. \tag{20}$$

B. AGP Coherent States

The group $U(2s)$ has subgroups that leave reference $2M$-electron AGP states, $\left|g^{\frac{N}{2}}\right\rangle$ invariant up to an overall phase factor and thus satisfy eq. (19). AGP states are given [**?**] by

$$\left|g^{\frac{N}{2}}\right\rangle = \sum_{1\le j_1<j_2<\cdots<j_{\frac{N}{2}}\le s} c_{j_1}\cdots c_{j_{\frac{N}{2}}} \left|\eta_{j_1}\eta_{j_1+s}\cdots\eta_{j_N}\eta_{j_N+s}\right\rangle , \tag{21}$$

where $\{|\eta_j\rangle|\, 1\le j\le r\}$ is a complete orthonormal basis for $\mathcal{H}^1$, $\{c_{j_1}|\, 1\le j\le s\}$ are positive real numbers called the canonical coefficients of the two electron geminal $|g\rangle$ which can be expanded in canonical form as

$$|g\rangle = \sum_{1\le j\le s} c_{j_1}\left|\eta_j\eta_{j+s}\right\rangle . \tag{22}$$

The state $\left|g^{\frac{N}{2}}\right\rangle$ can also be expressed in terms of the coset $SO(2r,\mathbb{R})/SU(r)$ of the group $SO(2r,\mathbb{R})$ acting on the zero particle vacuum state $|\phi\rangle$ of $\mathcal{H}_F$ [**?**][**?**][**?**] as

$$\left|g^{\frac{N}{2}}\right\rangle = P_{\frac{N}{2}}\exp\left\{\sum_{1\le j\le s} c_{j_1}b_j^\dagger b_{j+s}^\dagger\right\}|\phi\rangle \tag{23}$$

where $P_{\frac{N}{2}}$ is the orthogonal projector onto $\mathcal{H}^N$ and $\left\{b_j^\dagger\middle|\, 1\le j\le s\right\}$ are the creators for the one particle states $\{|\eta_j\rangle|\, 1\le j\le r\}$ i.e.

$$b_j^\dagger|\phi\rangle = |\eta_j\rangle . \tag{24}$$

1. *Coset Factorization for $U(2s)$*

In general the action of $U(2s)$ on a reference AGP, $\left|g^{\frac{N}{2}}\right\rangle$, state produces a manifold, $\mathcal{M}(\sigma,\omega_1,\cdots,\omega_\rho,\xi)$ of AGP states given by

$$\left\{U(2s)\Big/_{K}\times K\right\}\left|g^{\frac{N}{2}}\right\rangle . \tag{25}$$

where

$$K = USp(2\omega_1)\cdots\times USp(2\omega_p)\times SU(2)^\sigma\times SU(2\xi) . \tag{26}$$

Each element of the manifold can be expressed as

$$\left|g(c)^{\frac{N}{2}}\right\rangle = ck_{\omega_1}\cdots k_{\omega_\rho}k_\sigma k_\xi\left|g^{\frac{N}{2}}\right\rangle = c\left|g^{\frac{N}{2}}\right\rangle , \tag{27}$$

where the isotropy subgroup is given by the products $k_{\omega_1}\cdots k_{\omega_\rho}k_\sigma k_\xi$ where

$$k_{\omega_j}\in USp\left(2\omega_j\right);1\leq j\leq\rho,k_\sigma\in SU\left(2\right)^\sigma,k_\xi\in SU\left(2\xi\right), \tag{28}$$

$c\in U\left(2s\right)/K$ and the reference AGP is generated by a geminal $|g\rangle$ in the class $\{\rho,\sigma,\xi\}$. The most general AGP manifold is produced when $\sigma=s$ ($\Leftrightarrow$ $(\rho=0\ \&\ \xi=0)$) so it is advantages to first consider that case.

a. $SU\left(2\right)^s$ factors The isotropy sub group, K, for an AGP reference state that has non equal and nonzero canonical coefficients $\{c_j|\,1\leq j\leq s\}$, i.e. $\sigma=s$, is the product group $SU\left(2\right)^s$, which produces the coset decomposition

$$U\left(2s\right)=\left[U\left(2s\right)/SU\left(2\right)^s\right]\times SU\left(2\right)^s. \tag{29}$$

One can express coset representatives in a factored form as the product $G_{+1}G_0G_{-1}$, [**?**] where $\{G_{+1},G_0,G_{-1}\}$ are subgroups of the linear group $GL\left(2s,\mathbb{C}\right)$, which is the complexification of $U\left(2s\right)$, and the group G_{-1} leaves the reference AGP state invariant (note that it is different to the real compact isotropy groups). Restricting attention to the connected components of these subgroups, that contain the identity element, one can express this factorization in terms of Lie algebras as

$$U\left(2s\right)=e^{N_+}e^{N_0}e^{N_-}e^{\mathcal{K}} \tag{30}$$

where

$$\mathcal{K}=\overbrace{su\left(2\right)\oplus\cdots\oplus su\left(2\right)}^{s\text{-subspaces}}, \tag{31}$$

$\{N_{+1},N_0,N_{-1}\}$ are Lie subalgebras of $gl\left(2s,\mathbb{C}\right)$, the Lie algebra of $GL\left(2s,\mathbb{C}\right)$. The subalgebras $\{N_{-1},\mathcal{K}\}$ annihilate $\left|g^{\frac{N}{2}}\right\rangle$ i.e.

$$b\left|g^{\frac{N}{2}}\right\rangle=0\quad\forall b\in N_{-1}\oplus\mathcal{K}, \tag{32}$$

while the subalgebras $\{N_{+1},N_0\}$ have the property

$$b\left|g^{\frac{N}{2}}\right\rangle\neq0\quad\forall b\in N_{+1}\oplus N_0. \tag{33}$$

Introducing coordinate systems $\left\{\mathbf{z}_+,\lambda,\mathbf{z}_-,\eta\right\}$ with respect to fixed bases of $\{N_{+1},N_0,N_{-1},\mathcal{K}\}$, that depend on the reference AGP state, one can parameterize group elements $u\in U\left(2s\right)$ as

$$u\left(\tilde{\mathbf{z}},\eta\right)=\tilde{c}\left(\tilde{\mathbf{z}}\right)k\left(\eta\right)=c_{+1}\left(\mathbf{z}_+\right)c_0\left(\lambda\right)c_{-1}\left(\mathbf{z}_-\right)k\left(\eta\right). \tag{34}$$

The action of $U(2s)$ on the AGP state is

$$u(\tilde{\mathbf{z}},\eta)\left|g^{\frac{N}{2}}\right\rangle = \tilde{c}(\tilde{\mathbf{z}})\,k(\eta)\left|g^{\frac{N}{2}}\right\rangle = c_{+1}(\mathbf{z}_+)\,c_0(\lambda)\,c_{-1}(\mathbf{z}_-)\,k(\eta)\left|g^{\frac{N}{2}}\right\rangle = c_{+1}(\mathbf{z}_+)\,c_0(\lambda)\left|g^{\frac{N}{2}}\right\rangle, \tag{35}$$

showing that the non redundant coordinates $\mathbf{z}\equiv(\mathbf{z}_+,\lambda)$ describe a configuration space, that is a sub manifold $\mathcal{M}$ (in general nonlinear) of $\mathcal{H}^N$ with a coordinate system $\{\mathbf{z}\}$, that parameterize the set of $\{|\Phi(\mathbf{z})\rangle\}$ of $U(2s)/SU(2)^s$-coherent states

$$|\Phi(\mathbf{z})\rangle = c_{+1}(\mathbf{z}_+)\,c_0(\lambda)\left|g^{\frac{N}{2}}\right\rangle \equiv c(\mathbf{z})\left|g^{\frac{N}{2}}\right\rangle, \tag{36}$$

and at the same time represents the coset space $U(2s)/SU(2)^s$. The reference AGP is generated by a geminal $|g\rangle$ in the class $\{\rho,\sigma,\xi\}=\{0,s,0\}$

The AGP state dependent bases one chooses are

$$N_{+1} = \mathbb{C}-\text{linspan}\left\{c_j a_k^\dagger a_j - \sigma(j)\,\sigma(k)\,c_k a_{-j}^\dagger a_{-k};\right.$$
$$\left. -s\le j<k\le s, |j|\neq|k|\,;\,\sigma(j)=\text{sign}(j)\right\} \tag{37a}$$

$$N_0 = \mathbb{C}-\text{linspan}\left\{\left(a_j^\dagger a_j + a_{-j}^\dagger a_{-j}\right),\left(a_j^\dagger a_{-j} + a_{-j}^\dagger a_j\right),\left(a_j^\dagger a_j - a_{-j}^\dagger a_{-j}\right),\right.$$
$$\left. i\left(a_j^\dagger a_{-j} - a_{-j}^\dagger a_j\right);1\le j\le s\right\} \tag{37b}$$

$$N_{-1} = \mathbb{C}-\text{linspan}\left\{c_k a_k^\dagger a_j + \sigma(j)\,\sigma(k)\,c_j a_{-j}^\dagger a_{-k}\right.$$
$$\left. -s\le j<k\le s, |j|\neq|k|\,;\,\sigma(j)=\text{sign}(j)\right\} \tag{37c}$$

$$\mathcal{K} = \mathbb{R}-\text{linspan}\left\{i\left(a_j^\dagger a_{-j} + a_{-j}^\dagger a_j\right),i\left(a_j^\dagger a_j - a_{-j}^\dagger a_{-j}\right),\left(a_j^\dagger a_{-j} - a_{-j}^\dagger a_j\right);\quad 1\le j\le s\right\}, \tag{37d}$$

where we have translated the indexing of the CGSO's by

$$|\eta_j\rangle \to |\eta_{j-s}\rangle \quad 1\le j\le 2s. \tag{38}$$

It is useful to note that

$$N_0 = \{\mathcal{K}\oplus\mathcal{N}\}^{\mathbb{C}} \tag{39}$$

where

$$\mathcal{N} = \bigoplus\nolimits_{1\le j\le s}\left\{\mathbb{R}-\text{linspan}\left\{\left.\left(a_j^\dagger a_j + a_{-j}^\dagger a_{-j}\right)\right|\quad 1\le j\le s\right\}\right\}. \tag{40}$$

and $\{\}^{\mathbb{C}}$ denotes complexification. The factors in the group decomposition are

$$c_{+1}(\mathbf{z}_+) = \exp\left\{ \sum_{\substack{-s\le j<k\le s \\ |j|\ne|k|}} z_{+jk}\left\{ c_j a_k^\dagger a_j - \sigma(j)\,\sigma(k)\, c_k a_{-j}^\dagger a_{-k} \right\} \right\} = \exp \sum_{\substack{-s\le j<k\le s \\ |j|\ne|k|}} z_{+jk} q_{jk}^\dagger; \tag{41}$$

$$c_{-1}(\mathbf{z}_-) = \exp\left\{ \sum_{\substack{-s\le j<k\le s \\ |j|\ne|k|}} z_{-jk}\left\{ c_k a_k^\dagger a_j + \sigma(j)\,\sigma(k)\, c_j a_{-j}^\dagger a_{-k} \right\} \right\} = \exp \sum_{\substack{-s\le j<k\le s \\ |j|\ne|k|}} z_{-jk} q_{jk} \tag{42}$$

$$c_0(\lambda) = \exp\left\{ \sum_{\substack{1\le j\le s \\ 1\le k\le 4}} \lambda_{jk} n_{jk} \right\} \tag{43}$$

$$h(\eta) = \exp\left\{ \sum_{\substack{1\le j\le s \\ 1\le k\le 3}} \eta_{jk} u_{jk} \right\} \tag{44}$$

where $\{z_{+jk}, z_{-jk}\} \in \mathbb{C}, \{\lambda_{jk}, \eta_{jk}\} \in \mathbb{R}$, and $\{n_{jk}, u_{jk}\}$ are given by equ's. $(37b)$ and $(37d)$ respectively.

In order that the transformation $c_{+1}(\mathbf{z}_+)\, c_0(\lambda)\, c_{-1}(\mathbf{z}_-)\, k(\eta)$ be unitary the values of $\{\mathbf{z}_+, \lambda, \mathbf{z}_-\}$ must be constrained resulting in dependencies between the variables. The action of such unitary transformations on a normalized reference state produce normalized coherent states that can be expressed in terms of the coset representatives $c_{+1}(\mathbf{z}_+)\, c_0(\lambda)$. However there are dependencies between the variables $\{\mathbf{z}_+\}$ and $\{\lambda\}$ which are often convenient to avoid. One method of achieving this is to consider unnormalized coherent states by modifying the coset representative $c_0(\lambda)$ and explicitly considering the norm of the state in various expectation value expressions.

2. $USp(\omega_j)$ factors

The changes from the $SU(2)^s$ case can be illustrated by an $\sigma = s-2, \rho = 1, \omega_1 = 2$ example, which clearly leads to the general picture for $\rho > 1$. Here we have $s-2$ eigenvalues, $\left\{ c_j^2 \middle| 3 \le j \le s \right\}$ of $gg^\dagger$ that are doubly degenerate and one fourfold degenerate eigenvalue, c_1^2, of $gg^\dagger$, which is associated with two doubly degenerate eigenvalues, $\{c_1, -c_1\}$ of g affiliated with CGSO's that are linear combinations of $\{|\eta_1\rangle, |\eta_2\rangle\}$ and $\{|\eta_{-1}\rangle, |\eta_{-2}\rangle\}$ respectively ([?]). In this case unitary transformations, $u(\tilde{\mathbf{z}}, \eta)$, belonging to $U(2s)$ can be expressed in factored form as

$$u(\tilde{\mathbf{z}}, \eta) = c(\tilde{\mathbf{z}}) k_\sigma(\eta_\sigma) k_\omega(\eta_\omega) = c_{+1}(\mathbf{z}_{\sigma+}) u_{+1}(\mathbf{z}_{\omega+}) u_0(\lambda_\omega) c_0(\lambda_\sigma) u_{-1}(\mathbf{z}_{\omega-}) c_{-1}(\mathbf{z}_-) k_\sigma(\eta_\sigma) k_\omega(\eta_\omega). \tag{45}$$

where $k_\sigma(\eta_\sigma) \in SU(2)^{s-2}$ and $c_{\sigma 0}(\lambda_\sigma) \in \exp N_0$ and are expanded in the basis eqs.(37b) and (37d) respectively with the operators referring to CGSO's $\{|\eta_j\rangle| \, j = \pm 1, \pm 2\}$ deleted. The operators $c_{+1}(\mathbf{z}_{\sigma+}) \in \exp N_{+1}$ and $c_{-1}(\mathbf{z}_{\sigma-}) \in \exp N_{-1}$ are expanded in the basis eqs.(37a) and (37c) with the operators referring to the pairs $\{(|\eta_j\rangle, |\eta_k\rangle)| \, j = \pm 1, k = \pm 1\}$ of CGSO's deleted. The factors $u_{+1}(\mathbf{z}_{\omega+}) \in \exp U_{+11}, u_{-1}(\mathbf{z}_{\omega-}) \in \exp U_{-11}, u_{\omega 0}(\lambda_\omega) \in \exp U_{01}, k_\omega(\eta_\omega) \in USp(4)$, where U_{+1}, U_0, U_{-1} and $usp(4)$ are expanded in the following bases

$$U_{+11} = \mathbb{C}-\text{linspan}\left\{a_k^\dagger a_j + \sigma(j)\sigma(k) a_{-j}^\dagger a_{-k}; \quad j = 1, -1; k = 2, -2\right\} \tag{46a}$$

$$\begin{aligned} U_{01} = \mathbb{C}-\text{linspan}\Big\{&\left(a_j^\dagger a_j + a_{-j}^\dagger a_{-j}\right), \left(a_j^\dagger a_j - a_{-j}^\dagger a_{-j}\right), \left(a_j^\dagger a_{-j} + a_{-j}^\dagger a_j\right), \\ &i\left(a_j^\dagger a_{-j} - a_{-j}^\dagger a_j\right), i\left(a_k^\dagger a_j - \sigma(j)\sigma(k) a_{-j}^\dagger a_{-k} - a_j^\dagger a_k + \sigma(j)\sigma(k) a_{-k}^\dagger a_{-j}\right), \\ &\left(a_k^\dagger a_j - \sigma(j)\sigma(k) a_{-j}^\dagger a_{-k} + a_j^\dagger a_k - \sigma(j)\sigma(k) a_{-k}^\dagger a_{-j}\right); \quad j = 1, -1; k = 2, -2\Big\} \end{aligned} \tag{46b}$$

$$U_{-11} = \mathbb{C}-\text{linspan}\left\{a_k^\dagger a_j - \sigma(j)\sigma(k) a_{-j}^\dagger a_{-k}; \quad j = 1, -1; k = 2, -2\right\} \tag{46c}$$

$$\begin{aligned} usp(2\omega_1) = usp(4) = \mathbb{R}-\text{linspan}\Big\{&i\left(a_j^\dagger a_j - a_{-j}^\dagger a_{-j}\right), i\left(a_j^\dagger a_{-j} + a_{-j}^\dagger a_j\right), \left(a_j^\dagger a_{-j} - a_{-j}^\dagger a_j\right), \\ &\left(a_k^\dagger a_j - \sigma(j)\sigma(k) a_{-j}^\dagger a_{-k} - a_j^\dagger a_k + \sigma(j)\sigma(k) a_{-k}^\dagger a_{-j}\right), \\ &i\left(a_k^\dagger a_j - \sigma(j)\sigma(k) a_{-j}^\dagger a_{-k} + a_j^\dagger a_k - \sigma(j)\sigma(k) a_{-k}^\dagger a_{-j}\right); \quad j = 1, -1; k = 2, -2\Big\}, \end{aligned} \tag{46d}$$

where $\sigma(j) = \text{sign}(j)$. The manifold of $U(2s) / \left[USp(4) \times SU(2)^{s-1}\right]$-coherent states is then given by

$$|\Phi(\mathbf{z})\rangle = c_{+1}(\mathbf{z}_{+\sigma}) u_{+1}(\mathbf{z}_{+\omega}) u_{01}(\lambda_{\omega_1}) c_0(\lambda_\sigma) \left| g^{\frac{N}{2}} \right\rangle, \tag{47}$$

where $\mathbf{z} \equiv (\mathbf{z}_{+\sigma}, \mathbf{z}_{+\omega}, \lambda_\omega, \lambda_\sigma)$ and the reference AGP is generated by a geminal $|g\rangle$ in the class $\{\rho, \sigma, \xi\} = \{1, s-1, 0\}$.

If $\rho > 2$ then one deletes the appropriate operators in the bases $\{N_{+1}, N_0, N_{-1}, \mathcal{K}\}$ defined in eqs. $(37a) - (37d)$ and defines new bases $\{\{U_{+1j}, U_{0j}, U_{-1j}, usp\,(2\omega_j)\}|\, 1 \le j \le \rho\}$ according to eqs. $(46a) - (46d)$. The subalgebras $\{U_{-1j}, usp\,(2\omega_j)|\, 1 \le j \le \rho\}$ annihilate $\left|g^{\frac{N}{2}}\right\rangle$ i.e.

$$b\left|g^{\frac{N}{2}}\right\rangle = 0 \quad \forall b \in \bigoplus_{1\le j\le\rho} U_{-1j} \oplus \bigoplus_{1\le j\le\rho} usp\,(2\omega_j)\,, \tag{48}$$

while the subalgebras $\{U_{+1j}, U_{0j}|\, 1 \le j \le \rho\}$ have the property

$$b\left|g^{\frac{N}{2}}\right\rangle \ne 0 \quad \forall b \in \bigoplus_{1\le j\le\rho} U_{+1j} \oplus \bigoplus_{1\le j\le\rho} U_{0j}. \tag{49}$$

The $U\,(2s) / \left[\prod_{1\le j\le\rho} USp\,(2\omega_j) \times SU\,(2)^{s-\omega}\right]$-coherent state manifold is then given by

$$|\Phi\,(\mathbf{z})\rangle = c_{+1}\,(\mathbf{z}_{\sigma+}) \prod\nolimits_{1\le j\le\rho} \{u_{+1j}\,(\mathbf{z}_{+j})\, u_{0j}\,(\lambda_j)\}\, c_0\,(\lambda_\sigma) \left|g^{\frac{N}{2}}\right\rangle. \tag{50}$$

The manifold of extreme AGP's $\mathcal{M}\,(0, s, 0)$ are $U\,(2s)\,/USp\,(2s)$-coherent state, which have been discussed in super conductivity [**?**] is given by

$$|\Phi\,(\mathbf{z})\rangle = u_{+11}\,(\mathbf{z}_{+1})\, u_{01}\,(\lambda_1) \left|g^{\frac{N}{2}}\right\rangle. \tag{51}$$

3. $SU\,(2\xi)$ factors

The changes from $SU\,(2)^s$ can this time be illustrated by an $\sigma = s - 1, \rho = 0, \xi = 1$ example, which also clearly leads to the general picture for $\xi > 1$. Here we have $s - 1$ eigenvalues, $\left\{c_j^2\right|\, 2 \le j \le s\}$ of $gg^\dagger$ that are doubly degenerate and a doubly degenerate zero eigenvalue of $gg^\dagger$, which is associated with the CGSO's $\{|\eta_{-1}\rangle\,, |\eta_1\rangle\}$. In this case unitary transformations, $u\,(\tilde{\mathbf{z}}, \eta)$, belonging to $U\,(2s)$ can be expressed in factored form as

$$u\,(\tilde{\mathbf{z}}, \eta) = c\,(\tilde{\mathbf{z}})\, k_\sigma\,(\eta_\sigma)\, k_\xi\,(\eta_\xi) = c_{+1}\,(\mathbf{z}_{\sigma+})\, v_0\,(\lambda_\xi)\, c_0\,(\lambda_\sigma)\, c_{-1}\,(\mathbf{z}_{-})\, k_\sigma\,(\eta_\sigma)\, k_\xi\,(\eta_\xi)\,, \tag{52}$$

where $k_\sigma\,(\eta_\sigma) \in SU\,(2)^{s-1}$ and $c_0\,(\lambda_\sigma) \in \exp N_0$ which are expanded in the basis eqs.$(37b)$, $(37d)$ with the operators referring to CGSO's $\{|\eta_j\rangle|\, j = \pm 1\}$ deleted. The definitions of $c_{+1}\,(\mathbf{z}_{\sigma+}) \in \exp N_{+1}$ and $c_{-1}\,(\mathbf{z}_{\sigma-}) \in \exp N_{-1}$ are unchanged and are still expanded in the basis eqs.$(37a)$ and $(37c)$. The factors $v_0\,(\lambda_\xi) \in \exp V_0, k_\xi\,(\eta_\xi) \in SU\,(2)$, where $su\,(2)$ is expanded in the following bases

$$su\,(2) = \mathbb{R}\text{-- linspan} \left\{ i \left(a_1^\dagger a_{-1} + a_{-1}^\dagger a_1\right), i \left(a_1^\dagger a_1 - a_{-1}^\dagger a_{-1}\right), \left(a_1^\dagger a_{-1} - a_{-1}^\dagger a_1\right) \right\} \tag{53}$$

and V_0 is given by

$$\{su(2) \oplus \mathcal{N}\}^{\mathbb{C}} \tag{54}$$

where

$$\mathcal{N} = \mathbb{R}-\text{linspan}\left\{\left(a_1^{\dagger}a_1 + a_{-1}^{\dagger}a_{-1}\right)\right\}. \tag{55}$$

The subalgebra $\{su(2\xi)\}$ annihilates $\left|g^{\frac{N}{2}}\right\rangle$ i.e.

$$b\left|g^{\frac{N}{2}}\right\rangle = 0 \quad \forall b \in su(2\xi). \tag{56}$$

As the class of the reference AGP changes from $(s-1,0,1)$ to$(s-p,0,p)$ it is clear how the definitions of the factors eq. (52) evolve. It is easy to confirm that the $(M,0,s-M)$ cannot in fact exist, while the class $(0,M,s-M)$ does and such reference AGP's are IPS's and the $U(2s)/[USp(2M)\times SU(2s-2M)]$-coherent state manifold is in fact the Thouless $U(2s)/[U(2M)\times U(2s-2M)]$-coherent state manifold.

As an illustration of the preceding forms of the cosets of $U(\mathcal{H}^1)$ we can consider their action on an AGP state $\left|g^{\frac{N}{2}}\right\rangle$ determined by a geminal

$$|g\rangle = \sum_{1\le j\le s} c_{j_1}\,|\eta_j\eta_{j+s}\rangle \tag{57}$$

given by

$$c_{+1}(\mathbf{z}_+)\,c_0(\lambda)\,c_{+1}(\mathbf{z}_{\sigma+})\,v_0(\lambda_\xi)\prod\nolimits_{1\le j\le\rho}\{u_{+1j}(\mathbf{z}_{+j})\,u_{0j}(\lambda_j)\}\,c_0(\lambda_\sigma)\left|g^{\frac{N}{2}}\right\rangle \tag{58}$$

$\sigma =$ number of non zero, non equal $\{c_j\}$'s

- $\omega_j =$ number of equal coefficients in the j^{th} set
  - $\rho =$ number of sets of equal $\{c_j\}$'s
  - $\xi =$ number of zero coefficients.

C. $U\left(\mathcal{H}^N\right)/U(1)$-Coherent States

In order to examine the properties of general state excitation operators in terms of CS's it is necessary and illuminating to consider the *exact* quantum evolution equations expressed in terms of a CS manifold. The resulting classical equations of motion (see Section E 2) defined on such a manifold are equivalent to the full TDSE. The configuration space formed by these coherent sates represents all of the states $|\Psi\rangle \in \mathcal{H}^N$ via a parameterization $\{|\Phi(\mathbf{z})\rangle\}$. The group G in this case is the full unitary group $U\left(\binom{r}{N}\right)$ of N-electron state space and the isotropy subgroup is

$$K = \left\{ g |\, g \in G, g\,|\Phi_1\rangle = e^{i\theta(g)}\,|\Phi_1\rangle \right\} = \left\{ \exp\left(i\lambda\left(g\right)|\Phi_1\rangle\langle\Phi_1|\right) \exp\left(i \sum_{i,j\neq 1} \lambda_{ij}\left(g\right)|\Phi_i\rangle\langle\Phi_j| \right) \right\} \tag{59}$$

The coset representatives are

$$G/K = \left\{ \exp\left(\sum_{i>1} \eta_i\,|\Phi_i\rangle\langle\Phi_1| \right) \exp\left(\eta_1\,|\Phi_1\rangle\langle\Phi_1|\right) \exp\left(-\sum_{i>1} \bar{\eta}_i\,|\Phi_i\rangle\langle\Phi_1| \right) \right\} \tag{60}$$

where $\left\{ |\Phi_i\rangle | \quad 1 \leq i \leq \binom{r}{N} \right\}$ is a complete orthonormal basis for $\mathcal{H}^N$. The family of CS's associated with this coset is defined as

$$|\Phi(\eta)\rangle = \left\{ \exp\left(\sum_{i>1} \eta_i\,|\Phi_i\rangle\langle\Phi_1| \right) \exp\left(\eta_1\,|\Phi_i\rangle\langle\Phi_1|\right) \right\} |\Phi_1\rangle\,. \tag{61}$$

It is convenient to introduce a new equivalent coordinate system $\{\mathbf{z}\}$ by setting

$$\eta_1 = \ln(z_1) \Leftrightarrow z_1 = e^{\eta_1} \tag{62}$$

$$\eta_j = \frac{z_j}{z_1}; \quad 2 \leq j \leq \binom{r}{N} \Leftrightarrow z_j = \eta_j e^{\eta_1} \tag{63}$$

which leads to the following form for the coherent states

$$|\Phi(\mathbf{z})\rangle = \sum_{1\leq j\leq\binom{r}{N}} z_j\,|\Phi_j\rangle \tag{64}$$

Clearly eq. (64) is the familiar CI expansion of a general N-fermion state and the parameters $\{z_j\}$ the configurational interaction coefficients. The normalization of these CS's is given by

$$\langle\,\Phi(\mathbf{z})|\,\Phi(\mathbf{z})\rangle = \sum_{1\leq j\leq\binom{r}{N}} |z_j|^2 = |z_1|^2 + \sum_{2\leq j\leq\binom{r}{N}} |z_j|^2 \tag{65}$$

$$\langle\,\Phi(\eta)|\,\Phi(\eta)\rangle = e^{2\,\mathrm{Re}\,\eta_1} \left\{ 1 + \sum_{2\leq j\leq\binom{r}{N}} |\eta_j|^2 \right\} \tag{66}$$

Hence if one sets η_1 as the function

$$\eta_1 = \eta_1\left(\eta_2, \cdots, \eta_{\binom{r}{N}}\right) = \frac{1}{2}\ln\left(1 - \sum_{2\leq j\leq\binom{r}{N}} e^{2\,\mathrm{Re}\,\eta_j} \right) \tag{67}$$

and z_1 as the function

$$z_1 = z_1\left(z_2, \cdots, z_{\binom{r}{N}}\right) = \left(1 - \sum_{2\leq j\leq \binom{r}{N}} |z_j|^2\right)^{\frac{1}{2}} \tag{68}$$

the family of CS's $\{|\Phi(\mathbf{z})\rangle\} \equiv \{|\Phi(\eta)\rangle\}$ are normalized if the reference state $|\Phi_1\rangle$ is normalized i.e. $\langle \Phi_1 | \Phi_1\rangle = 1$. If one uses the CS parameters $\left\{\eta_2, \cdots, \eta_{\binom{r}{N}}\right\}$ or $\left\{z_2, \cdots, z_{\binom{r}{N}}\right\}$ the CS manifold is the unit sphere in $\mathcal{H}^N$, while if one uses $\{\eta\}$ or $\{\mathbf{z}\}$ then it is all of $\mathcal{H}^N$ and one explicitly considers the normalization in all expressions.

V. DIFFERENTIAL STRUCTURE OF THE COHERENT STATE MANIFOLD

A. General Structure

In the following we gloss over technical, (but mathematical necessary), qualifications and discuss some properties of the differentiable manifold, $\mathcal{M}$ (also see Appendix G) formed by the CS's. The manifold of $G/K-$CS's has a hermitian structure [?] given by the tensor

$$h(\mathbf{z}) = \sum_{1\leq j,k\leq \nu} C_{jk}(\bar{\mathbf{z}}, \mathbf{z})\, d\bar{z}_j \otimes dz_k, \tag{69}$$

where the coefficients $\{C_{jk}(\bar{\mathbf{z}}, \mathbf{z}) | 1 \leq j, k \leq \nu\}$, form a positive, (therefore hermitian), metric matrix defined by

$$C_{jk}(\bar{\mathbf{z}}, \mathbf{z}) = \frac{\partial^2 \Lambda(\bar{\mathbf{z}}, \mathbf{z})}{\partial \bar{z}_j \partial z_k}, \tag{70}$$

called a Kähler metric [? ?], with a Kähler potential $\Lambda(\bar{\mathbf{z}}, \mathbf{z})$ is given by

$$\Lambda(\bar{\mathbf{z}}, \mathbf{z}) = \ln S(\bar{\mathbf{z}}, \mathbf{z}) = \ln \langle \Phi(\mathbf{z}) | \Phi(\mathbf{z})\rangle. \tag{71}$$

The map

$$\Phi : \mathbb{C}^\nu \to \mathcal{M} \tag{72}$$

given by the preceding parameterization of the cosets is constructed to be a diffeomorphism (a 1-1, continuous function that is differentiable) and is thus the inverse of a coordinate chart [?]

$$\Phi^{-1} : \mathcal{M} \to \mathbb{C}^\nu. \tag{73}$$

The coordinate functions

$$z_j : \mathcal{M} \to \mathbb{C} \tag{74}$$

associated with this chart are

$$z_j(m) = \left(\Phi^{-1}\right)_j(m), \qquad m \in \mathcal{M}. \tag{75}$$

Points on the manifold $\mathcal{M}$ are $\Phi(\mathbf{z})$, which defines an association between $\mathcal{M}$ and $\mathbb{C}^\nu$ given by Φ, however as we are in general only considering one representative coordinate chart we will abuse the notation when the meaning is unambiguous and the point $\Phi(\mathbf{z})$ will be denoted as $\mathbf{z}$.

The differential one forms $\{dz_j|\, 1 \le j \le \nu\}$ are the best linear approximations to the coordinate functions $\{z_j|\, 1 \le j \le \nu\}$ at the point $\mathbf{z} \equiv (z_1, \cdots, z_\nu)$, (as is standard the $\mathbf{z}$ dependence of these forms is not explicitly shown) and thus are complex valued linear functionals on the the tangent spaces $\{\mathcal{T}_\mathbf{z}|\, \mathbf{z} \in \mathcal{M}\}$, (which rigorously should be denoted as $\left\{\mathcal{T}_{\Phi(\mathbf{z})}\middle|\, \Phi(\mathbf{z}) \in \mathcal{M}\right\}$), i.e.

$$dz_j : \mathcal{T}_\mathbf{z} \to \mathbb{C} \tag{76}$$

and hence together with $\{d\bar{z}_j\}$ generate the dual spaces, $\{\mathcal{T}_\mathbf{z}^*\}$ of $\{\mathcal{T}_\mathbf{z}\}$. The tangent spaces, $\{\mathcal{T}_\mathbf{z}\}$, are the direct sum of the complex linear space, $\mathcal{T}_\mathbf{z}^{(1,0)}$, spanned by the vectors $\left\{\frac{\partial}{\partial z_j}\middle|\, 1 \le j \le \nu\right\}$,*with* the the complex linear space, $\mathcal{T}_\mathbf{z}^{(0,1)}$, spanned by the vectors $\left\{\frac{\partial}{\partial \bar{z}_j}\middle|\, 1 \le j \le \nu\right\}$, (again the $\mathbf{z}$ dependence is not explicitly shown). Thus the complex dimension of $\mathcal{T}_\mathbf{z}$ is 2ν which should be contrasted to real manifolds where the corresponding dimension of the tangent space is ν (see [**?**]). The bases vectors $\left\{\frac{\partial}{\partial \bar{z}_j}, \frac{\partial}{\partial z_j}\right\}$ are partial differential operators defined by their action

$$\begin{aligned} \frac{\partial}{\partial z_j} &: f \to \frac{\partial f(\mathbf{z})}{\partial z_j} \\ \frac{\partial}{\partial \bar{z}_j} &: f \to \frac{\partial f(\mathbf{z})}{\partial \bar{z}_j} \end{aligned} \tag{77}$$

on functions

$$f : \mathcal{M} \to \mathbb{C}, \tag{78}$$

evaluated at the point $\mathbf{z}$. The action of a general element

$$X = \sum_{1 \le j \le \nu} \left\{ x_j \frac{\partial}{\partial \bar{z}_j} + x_{j+\nu} \frac{\partial}{\partial z_j} \right\} \tag{79}$$

on a function f is to produce the directional derivative

$$X(f) = \sum_{1 \le j \le \nu} \left\{ x_j \frac{\partial f(\mathbf{z})}{\partial \bar{z}_j} + x_{j+\nu} \frac{\partial f(\mathbf{z})}{\partial z_j} \right\} = \mathbf{x} \cdot Df(\mathbf{z}) \tag{80}$$

of f in the direction $\mathbf{x}$, where

$$Df(\mathbf{z}) = \begin{pmatrix} \nabla_{\bar{\mathbf{z}}} f(\mathbf{z}) \\ \nabla_{\mathbf{z}} f(\mathbf{z}) \end{pmatrix}. \tag{81}$$

The basis $\{d\bar{z}_j, dz_j\}$ of the cotangent spaces $\{\mathcal{T}^*_{\mathbf{z}}\}$ and $\left\{\frac{\partial}{\partial \bar{z}_j}, \frac{\partial}{\partial z_j}\right\}$ of the tangent spaces $\{\mathcal{T}_{\mathbf{z}}\}$ are biorthogonal i.e.

$$\begin{aligned} dz_j\left(\frac{\partial}{\partial z_k}\right) &= d\bar{z}_j\left(\frac{\partial}{\partial \bar{z}_k}\right) = \delta_{jk} \\ d\bar{z}_j\left(\frac{\partial}{\partial z_k}\right) &= dz_j\left(\frac{\partial}{\partial \bar{z}_k}\right) = 0. \end{aligned} \tag{82}$$

The differential, df, (a linear functional on $\mathcal{T}_{\mathbf{z}}$ and the best linear approximation to the function f in the neighborhood of the point $\mathbf{z}$), can be expanded as

$$df = \sum_j \left\{ \frac{\partial f(\mathbf{z})}{\partial \bar{z}_j} d\bar{z}_j + \frac{\partial f(\mathbf{z})}{\partial z_j} dz_j \right\}. \tag{83}$$

One can associate with the functions f with vectors, $X_f(\mathbf{z}) \in \mathcal{T}_{\mathbf{z}}$, that produce derivatives in the direction of the gradient of f by the definition

$$X_f(\mathbf{z}) = \sum_j \left\{ \frac{\partial f(\mathbf{z})}{\partial \bar{z}_j} \frac{\partial}{\partial \bar{z}_j} + \frac{\partial f(\mathbf{z})}{\partial z_j} \frac{\partial}{\partial z_j} \right\} \tag{84}$$

and thus extend the action of, h, to functions defined on $\mathcal{M}$ by

$$h(X_f, X_g) = 2 \sum_{1 \le j,k \le \nu} C_{jk}(\bar{\mathbf{z}}, \mathbf{z})\, d\bar{z}_j(X_f) \otimes dz_k(X_g), \tag{85}$$

where again the $\mathbf{z}$ dependence of h is implicit.

If the functions f are holomorphic (i.e. can be expanded in a convergent power series at each point) then

$$\frac{\partial f(\mathbf{z})}{\partial \bar{z}_j} = 0 \quad 1 \le j \le \nu. \tag{86}$$

In order to distinguish holomorphic from non holomorphic functions one introduces the notation $f(\bar{\mathbf{z}}, \mathbf{z})$, (even though $f : \mathcal{M} \to \mathbb{C}$) as

$$\frac{\partial f(\mathbf{z})}{\partial \bar{z}_j} \neq 0 \textit{ and } \frac{\partial f(\mathbf{z})}{\partial z_k} \neq 0 \text{ for some } j \text{ and } k. \tag{87}$$

Rigorously the function f is now considered as the values of $f : \mathcal{M} \times \mathcal{M} \to \mathbb{C}$ restricted to the points $(\bar{\mathbf{z}}, \mathbf{z})$ i.e. $f(\mathbf{z}', \mathbf{z})|_{\mathbf{z}'=\mathbf{z}}$).

The hermitian form h can be decomposed into the sum of a real part, $\operatorname{Re} h$, which is a real symmetric bilinear form, ρ, and i times an imaginary part, $\operatorname{Im} h$, which is an antisymmetric bilinear form ω, as

$$h = \rho + i\omega, \tag{88}$$

which corresponds to decomposing the tensor products $d\bar{z}_j \otimes dz_k$ into symmetric, "$\vee$", and antisymmetric, "$\wedge$" tensor products [30]

$$d\bar{z}_j \otimes dz_k = d\bar{z}_j \vee dz_k + d\bar{z}_j \wedge dz_k. \tag{89}$$

(Notational note: In the following we will use [] delimiters to signify a secondary dependence that is not the primary focus of the properties being described). The symmetric form $\rho[\mathbf{z}]$ can be expanded as

$$\rho[\mathbf{z}] = 2 \sum_{1 \le j,k \le \nu} C_{jk}(\bar{\mathbf{z}}, \mathbf{z})\, d\bar{z}_j \vee dz_k = 2 \sum_{1 \le j,k \le \nu} \{\operatorname{Re} C_{jk}(\bar{\mathbf{z}}, \mathbf{z})\}\, d\bar{z}_j \otimes dz_k \tag{90}$$

and the antisymmetric form $\omega[\bar{\mathbf{z}}, \mathbf{z}]$ as

$$\omega[\mathbf{z}] = -2i \sum_{1 \le j,k \le \nu} C_{jk}(\bar{\mathbf{z}}, \mathbf{z})\, d\bar{z}_j \wedge dz_k = -2i \sum_{1 \le j,k \le \nu} \{\operatorname{Im} C_{jk}(\bar{\mathbf{z}}, \mathbf{z})\}\, d\bar{z}_j \otimes dz_k. \tag{91}$$

The real inner product $\rho[\mathbf{z}]$ produces a Riemannian structure on $\mathcal{M}$, while $\omega[\mathbf{z}]$ produces a symplectic structure.

The symplectic form $\omega[\mathbf{z}]$ acts on tangent vectors $\{X_f, X_g\}$ of $\mathcal{M}$ according to

$$\begin{aligned}
\omega[\mathbf{z}](X_f, X_g) &= -\omega[\mathbf{z}](X_g, X_f) = -2i \sum_{1 \le j,k \le \nu} C_{jk}(\bar{\mathbf{z}}, \mathbf{z})\, d\bar{z}_j \wedge dz_k (X_f \otimes X_g) \\
&= -i \sum_{1 \le j,k \le \nu} C_{jk}(\bar{\mathbf{z}}, \mathbf{z}) \left\{ \frac{\partial f(\mathbf{z})}{\partial \bar{z}_j} \frac{\partial g(\mathbf{z})}{\partial z_k} - \frac{\partial f(\mathbf{z})}{\partial z_k} \frac{\partial g(\mathbf{z})}{\partial \bar{z}_j} \right\} \\
&= -i \sum_{1 \le j,k \le \nu} \left\{ \frac{\partial f(\mathbf{z})}{\partial \bar{z}_j} C_{jk}(\bar{\mathbf{z}}, \mathbf{z}) \frac{\partial g(\mathbf{z})}{\partial z_k} - \frac{\partial f(\mathbf{z})}{\partial z_j} \overline{C_{jk}(\bar{\mathbf{z}}, \mathbf{z})} \frac{\partial g(\mathbf{z})}{\partial \bar{z}_k} \right\},
\end{aligned} \tag{92}$$

which can be expressed in matrix notation as

$$\omega[\mathbf{z}](X_f, X_g) = \begin{pmatrix} \frac{\partial f}{\partial \bar{\mathbf{z}}} & \frac{\partial f}{\partial \mathbf{z}} \end{pmatrix} \begin{pmatrix} \mathbf{0} & i\mathbf{C}(\bar{\mathbf{z}}, \mathbf{z}) \\ -i\overline{\mathbf{C}(\bar{\mathbf{z}}, \mathbf{z})} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \frac{\partial g}{\partial \bar{\mathbf{z}}} \\ \frac{\partial g}{\partial \mathbf{z}} \end{pmatrix}, \tag{93}$$

where

$$\left(\frac{\partial f}{\partial \mathbf{z}} \right)_j = \frac{\partial f}{\partial z_j}. \tag{94}$$

The symplectic form $\omega[\mathbf{z}]$ defines a generalized Poisson Bracket $\{,\}_{\mathbf{z}}$, (which is $\mathbf{z}$ dependent), by

$$\{f,g\}_{\mathbf{z}} = \omega[\mathbf{z}]^{-1}(X_f, X_g) = \begin{pmatrix} \frac{\partial f}{\partial \bar{\mathbf{z}}} & \frac{\partial f}{\partial \mathbf{z}} \end{pmatrix} \begin{pmatrix} \mathbf{0} & i\overline{\mathbf{C}(\bar{\mathbf{z}},\mathbf{z})}^{-1} \\ -i\mathbf{C}(\bar{\mathbf{z}},\mathbf{z})^{-1} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \frac{\partial g}{\partial \bar{\mathbf{z}}} \\ \frac{\partial g}{\partial \mathbf{z}} \end{pmatrix} \tag{95}$$

(Notational Note: Unless it is essential the $\mathbf{z}$ dependence of the Poisson Bracket will be implicit).

The symplectic form and the Poisson Bracket create a generalized phase space structure on $\mathcal{M}$, where real valued functions, $f_{\mathcal{O}}$, defined on $\mathcal{M}$ by the expectation values of self adjoint quantum observables $\mathcal{O}$ as

$$f_{\mathcal{O}}(\bar{\mathbf{z}},\mathbf{z}) = \frac{\langle \Phi(\mathbf{z}) | O\Phi(\mathbf{z})\rangle}{\langle \Phi(\mathbf{z}) | \Phi(\mathbf{z})\rangle}, \tag{96}$$

can be considered as "classical" observables.

B. Coherent State Tangent Spaces

Hilbert spaces are self dual i.e. they are isomorphic to the space of bounded linear functions defined on them, which is depicted in Dirac notation as

$$J : |v\rangle \rightarrow \langle v| \,. \tag{97}$$

The map Φ, from the complex space $\mathbb{C}^{\nu}$ to the CS manifold $\mathcal{M}$, is the inverse of a coordinate chart, Φ^{-1}, of $\mathcal{M}$ (see Appendix G)

$$\Phi^{-1} : \mathcal{M} \rightarrow \mathbb{C}^{\nu}, \tag{98}$$

which is given by

$$\mathbf{z} = \Phi^{-1}\left(\exp\left\{\sum_{1\leq j\leq \nu} z_j Y_j\right\} |\Phi_{ref}\rangle\right) \Leftrightarrow \Phi(\mathbf{z}) = \exp\left\{\sum_{1\leq j\leq \nu} z_j Y_j\right\} |\Phi_{ref}\rangle \,. \tag{99a}$$

where $\{Y_j\}$ are the generators of the coset G/K in $\mathcal{B}\left(\mathcal{H}^N\right)$, as described in section IV. We can extend this chart to the complex conjugate space $\overline{\mathbb{C}^{\nu}}$ by using the isomorphism J, after noting that

$$\bar{\mathbf{z}} = \overline{\Phi^{-1}\left(\exp\left\{\sum_{1\leq j\leq \nu} z_j Y_j\right\} |\Phi_{ref}\rangle\right)} \equiv \overline{\Phi^{-1}}\left(\exp\left\{\sum_{1\leq j\leq \nu} z_j Y_j\right\} |\Phi_{ref}\rangle\right), \tag{100}$$

which gives

$$\bar{\Phi}(\bar{\mathbf{z}}) = \exp\left\{\sum_{1\le j\le\nu} z_j Y_j\right\}|\Phi_{ref}\rangle \tag{101}$$

and finally

$$\Phi(\bar{\mathbf{z}}) = J\left(\exp\left\{\sum_{1\le j\le\nu} z_j Y_j\right\}|\Phi_{ref}\rangle\right) = \langle\Phi_{ref}|\exp\left\{\sum_{1\le j\le\nu} \bar{z}_j Y_j^\dagger\right\}. \tag{102}$$

The derivatives $\left\{\frac{\partial\Phi(\mathbf{z}_0)}{\partial z_j}\middle| 1\le j\le\nu\right\}, \left\{\frac{\partial\Phi(\bar{\mathbf{z}}_0)}{\partial \bar{z}_j}\middle| 1\le j\le\nu\right\}$ are bases for the tangent subspaces $\mathcal{T}_{\mathbf{z}_0}^{(1,0)}$ and $\mathcal{T}_{\mathbf{z}_0}^{(0,1)}$ respectively, (see Appendix G). These basis vectors are easily obtained from eq. (99a) as

$$\frac{\partial\Phi(\mathbf{z}_0)}{\partial z_k} = Y_k \exp\left\{\sum_{1\le j\le\nu} z_{0j} Y_j\right\}|\Phi_{ref}\rangle = Y_k|\Phi(\mathbf{z}_0)\rangle \tag{103}$$

and eq. (**??**) as

$$\frac{\partial\Phi(\bar{\mathbf{z}}_0)}{\partial \bar{z}_k} = \langle\Phi(\mathbf{z}_0)|\,Y_k^\dagger. \tag{104}$$

The full tangent space $\mathcal{T}_{\mathbf{z}_0}$ is given by the direct sum of $\mathcal{T}_{\mathbf{z}_0}^{(1,0)}$ and $\mathcal{T}_{\mathbf{z}_0}^{(0,1)}$ i.e.

$$\mathcal{T}_{\mathbf{z}_0} = \mathcal{T}_{\mathbf{z}_0}^{(1,0)} \oplus \mathcal{T}_{\mathbf{z}_0}^{(0,1)} \tag{105}$$

and the basis $\left\{\left\{\frac{\partial\Phi(\bar{\mathbf{z}}_0)}{\partial\bar{z}_j}\right\},\left\{\frac{\partial\Phi(\mathbf{z}_0)}{\partial z_j}\right\}\middle| 1\le j\le\nu\right\}$ is adapted to this decomposition. The spaces $\mathcal{T}_{\mathbf{z}_0}^{(1,0)}$ and $\mathcal{T}_{\mathbf{z}_0}^{(0,1)}$ can be considered as the images of the space $\mathbb{C}^\nu$ under the linear maps $d\Phi[\mathbf{z}_0]$ and $d\Phi[\bar{\mathbf{z}}_0]$, which are the differentials of the coordinate chart inverse Φ at the points $\mathbf{z}_0$ and $\bar{\mathbf{z}}_0$, i.e.

$$d\Phi[\mathbf{z}_0] : \mathbb{C}^\nu \to \mathcal{T}_{\mathbf{z}_0}^{(1,0)} \tag{106a}$$

$$d\Phi[\mathbf{z}_0](e_k) = \frac{\partial\Phi(\mathbf{z}_0)}{\partial z_j} \tag{106b}$$

and

$$d\Phi[\bar{\mathbf{z}}_0] : \mathbb{C}^\nu \to \mathcal{T}_{\mathbf{z}_0}^{(0,1)} \tag{107a}$$

$$d\Phi[\bar{\mathbf{z}}_0](e_k) = \frac{\partial\Phi(\bar{\mathbf{z}}_0)}{\partial \bar{z}_j}, \tag{107b}$$

where $\{e_k|\,1\le k\le\nu\}$ is a basis for $\mathbb{C}^\nu$. For convenience we will denote the inverse linear coordinate map into $\mathcal{T}_{\mathbf{z}_0}$ as $d\Phi[\bar{\mathbf{z}}_0,\mathbf{z}_0]$, where

$$d\Phi[\bar{\mathbf{z}}_0,\mathbf{z}_0] : \mathbb{C}^\nu\oplus\mathbb{C}^\nu \to d\Phi[\bar{\mathbf{z}}_0](\mathbb{C}^\nu)\oplus d\Phi[\mathbf{z}_0](\mathbb{C}^\nu) \equiv \mathcal{T}_{\mathbf{z}_0}. \tag{108}$$

VI. QUANTUM DYNAMICS ON THE COHERENT STATE MANIFOLD

One is especially interested in the evolution of quantum states induced by a quantum Hamiltonian $H(t)$ restricted to the manifold $\mathcal{M}$ producing "classical" paths z:$\mathbb{R} \rightarrow \mathcal{M}$ that correspond to quantum paths $|\Phi(\mathbf{z}(t))\rangle$. These paths are partially generated, via the principle of stationary action (as discussed in Appendix E), by "classical" Hamiltonians $\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)$ defined as

$$\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t) = \frac{\langle \Phi(\mathrm{z}(t))|\, H(t)\, \Phi(\mathrm{z}(t))\rangle}{\langle \Phi(\mathrm{z}(t))|\, \Phi(\mathrm{z}(t))\rangle}, \tag{109}$$

and are solutions to generalized classical equations of motion

$$\begin{aligned} -i\frac{d\mathrm{z}_j}{dt} &= \{\mathrm{z}_j, \mathcal{E}\} \\ i\frac{d\bar{\mathrm{z}}_j}{dt} &= \{\bar{\mathrm{z}}_j, \mathcal{E}\} \end{aligned} \tag{110}$$

for the Hamiltonian function $\mathcal{E}$. The quantum paths are only partially determined by eq. (110) as one must also solve the independent equation

$$\frac{d\sigma}{dt} = -\operatorname{Im}\left\{\sum_{1\leq j\leq \nu} \mathrm{z}_j \frac{\partial \Lambda}{\partial \mathrm{z}_j}\right\} - \mathcal{E} \tag{111}$$

to determine a time dependent phase, σ, to obtain the quantum path [**?**]

$$\eta_j(t) = e^{\sigma(t)}\mathrm{z}_j(t). \tag{112}$$

If $\mathcal{M}$ is the manifold of $U\left(\mathcal{H}^N\right)/U(1)$-CS's then these equations determine the same paths as the TDSE otherwise they provide approximations to these paths.

To facilitate numerical calculations it is convenient to consider the equations eq. (110) in the matrix form

$$\begin{pmatrix} \frac{d\bar{\mathrm{z}}(t)}{dt} \\ \frac{d\mathrm{z}(t)}{dt} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & i\overline{\mathbf{C}(\bar{\mathrm{z}}(t), \mathrm{z}(t))}^{-1} \\ -i\mathbf{C}(\bar{\mathrm{z}}(t), \mathrm{z}(t))^{-1} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \frac{\partial \mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial \bar{\mathbf{z}}} \\ \frac{\partial \mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial \mathbf{z}} \end{pmatrix}. \tag{113}$$

The metric matrix, $\mathbf{C}(\bar{\mathrm{z}}(t), \mathrm{z}(t))$, and the gradient, $D^1\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)$, of the energy function $\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)$ are given in Appendix A. One should note that the phase space, $\mathcal{M}$, is not in general flat and hence the Poisson Brackets are position dependent, a situation akin to problems in General Relativity. However the equations eq. (113) can in principle be integrated and solutions found by standard differential equation methods [31]. In appendix H we describe a method based on local quadratic approximations that is related to the linear response of the system.

Stationary paths $z_0(t)$ have constant values $\mathbf{z}_c$ i.e.

$$z_0(t) = \mathbf{z}_c \quad \forall t. \tag{114}$$

where $\mathbf{z}_c$ is a critical point of $\mathcal{E}(\bar{z}_0, z_0, t)$. Eq. (113) implies that $D^1\mathcal{E}(\bar{z}_0, z_0, t) = 0\ \forall t$ i.e.

$$\frac{1}{\mathcal{S}(\bar{z}_0(t), z_0(t))}\left\{\frac{\partial\mathcal{H}(\bar{z}_0, z_0, t)}{\partial\mathbf{x}} - \mathcal{E}(\bar{z}_0, z_0, t)\frac{\partial\mathcal{S}(\bar{z}_0(t), z_0(t))}{\partial\mathbf{x}}\right\} = 0 \tag{115}$$

which thus gives the equation

$$\frac{\partial\mathcal{H}(\bar{\mathbf{z}}_c, \mathbf{z}_c, t)}{\partial\mathbf{x}} \equiv \frac{\partial\mathcal{H}(\bar{\mathbf{z}}_c, \mathbf{z}_c)}{\partial\mathbf{x}} = \mathcal{E}(\bar{\mathbf{z}}_c, \mathbf{z}_c)\frac{\partial\mathcal{S}(\bar{\mathbf{z}}_c, \mathbf{z}_c)}{\partial\mathbf{x}} \tag{116}$$

for $\mathbf{z}_c$.If the manifold $\mathcal{M}$ is the manifold of $U(\mathcal{H}^N)/U(1)$-CS's then this equation *is* the Time Independent Schrödinger Equation (TISE), while if the manifold is contained in the manifold of $U(\mathcal{H}^N)/U(1)$-CS's eq. (116) produces non linear equations that determine approximate solutions to the TISE.

The Hessian $D^2\mathcal{E}(\bar{\mathbf{z}}_c, \mathbf{z}_c)$ of the energy at critical points is made up of blocks of the form

$$\frac{\partial^2\mathcal{E}(\bar{\mathbf{z}}_c, \mathbf{z}_c)}{\partial\mathbf{x}\partial\mathbf{y}} = \frac{1}{\mathcal{S}(\bar{\mathbf{z}}_c, \mathbf{z}_c)}\left\{\frac{\partial^2\mathcal{H}(\bar{\mathbf{z}}_c, \mathbf{z}_c)}{\partial\mathbf{x}\partial\mathbf{y}} - \mathcal{E}(\bar{\mathbf{z}}_c, \mathbf{z}_c)\frac{\partial^2\mathcal{S}(\bar{\mathbf{z}}_c, \mathbf{z}_c)}{\partial\mathbf{x}\partial\mathbf{y}}\right\} \tag{117}$$

for $\mathbf{x}, \mathbf{y} = \mathbf{z}, \bar{\mathbf{z}}$ and gives information of the about the nature of critical point $\mathbf{z}_c$ i.e. whether it is a maximum, minimum or saddle point of the energy function $\mathcal{E}$. It also gives information about the linear response of the system to external time dependent potentials, as we discuss section VII.

A. Linearization of the EOM

In the case of a curved phase space there are two stages to linearization. The first stage is choose a specific point in the phase space and work locally in the flat phase space formed by the tangent space about that point, the second is to make a quadratic approximation to the effect of classical Hamiltonian in this flat space. Most often the point chosen on the manifold $\mathcal{M}$ is an equilibrium or critical point of the Hamiltonian [**?**]. The classical Hamiltonian, $\mathcal{E}(\bar{\mathbf{z}}, \mathbf{z})$, is a *real* analytic function and can be expanded as a power series about an arbitrary point $(\bar{\varsigma}_0, \varsigma_0)$ in the manifold $\mathcal{M}$ as

$$\mathcal{E}(\bar{\mathbf{z}}, \mathbf{z}) = \mathcal{E}(\bar{\varsigma}_0, \varsigma_0) + D^1\mathcal{E}[\bar{\varsigma}_0, \varsigma_0](\bar{\mathbf{z}} - \bar{\varsigma}_0, \mathbf{z} - \varsigma_0) + \frac{1}{2}D^2\mathcal{E}[\bar{\varsigma}_0, \varsigma_0](\bar{\mathbf{z}} - \bar{\varsigma}_0, \mathbf{z} - \varsigma_0) + \cdots, \tag{118}$$

where $\bar{\mathbf{z}} - \bar{\varsigma}_0$ and $\mathbf{z} - \varsigma_0$ are vectors in the tangent space $\mathcal{T}_{\varsigma_0}$. If the Hamiltonian is globally analytic this power series converges for all complex coordinates $\mathbf{z}$, otherwise it will converge in some neighborhood of ς_0. This form of $\mathcal{E}$ can be used in the EOM eq. (113), to determine solution paths $\mathrm{z}(t)$ that belong to the manifold $\mathcal{M}$ in terms of coordinate vectors that belong to $\mathcal{T}_{\varsigma_0}$, through the inverse coordinate chart Φ (see Appendix B). The same expansion determines, in general different, paths $\mathrm{z}_F(t)$ in the flat phase space $\mathcal{T}_{\varsigma_0}$ when a fixed Poisson Bracket $\{\}_{\varsigma_0}$, corresponding to a fixed metric of $\mathcal{T}_{\varsigma_0}$, is used in eq. (113).

The first order term eq. (118) is given by

$$D^1\mathcal{E}\left[\bar{\varsigma}_0, \varsigma_0\right]\left(\bar{\mathbf{z}} - \bar{\varsigma}_0, \mathbf{z} - \varsigma_0\right) = \begin{pmatrix} \frac{\partial \mathcal{E}(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}}} & \frac{\partial \mathcal{E}(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z}} \end{pmatrix} \begin{pmatrix} \bar{\mathbf{z}} - \bar{\varsigma}_0 \\ \mathbf{z} - \varsigma_0 \end{pmatrix} \tag{119}$$

and the second order term

$$D^2\mathcal{E}\left[\bar{\varsigma}_0, \varsigma_0\right]\left(\bar{\mathbf{z}} - \bar{\mathbf{z}}_0, \mathbf{z} - \mathbf{z}_0\right) = \begin{pmatrix} \bar{\mathbf{z}} - \bar{\varsigma}_0 & \mathbf{z} - \varsigma_0 \end{pmatrix} \begin{pmatrix} \frac{\partial^2 \mathcal{E}(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}} \partial \mathbf{z}} \\ \frac{\partial^2 \mathcal{E}(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z} \partial \mathbf{z}} \end{pmatrix} \begin{pmatrix} \bar{\mathbf{z}} - \bar{\varsigma}_0 \\ \mathbf{z} - \varsigma_0 \end{pmatrix}. \tag{120}$$

The quadratic approximation, $\mathcal{E}_2\left[\varsigma_0\right]\left(\bar{\mathbf{z}}, \mathbf{z}\right)$, to $\mathcal{E}\left(\bar{\mathbf{z}}, \mathbf{z}\right)$ around the point $\left(\bar{\varsigma}_0, \varsigma_0\right)$ is

$$\mathcal{E}_2\left[\varsigma_0\right]\left(\bar{\mathbf{z}}, \mathbf{z}\right) = \mathcal{E}\left(\bar{\varsigma}_0, \varsigma_0\right) + D^1\mathcal{E}\left[\bar{\varsigma}_0, \varsigma_0\right]\left(\bar{\mathbf{z}} - \bar{\varsigma}_0, \mathbf{z} - \varsigma_0\right) + \frac{1}{2} D^2\mathcal{E}\left[\bar{\varsigma}_0, \varsigma_0\right]\left(\bar{\mathbf{z}} - \bar{\varsigma}_0, \mathbf{z} - \varsigma_0\right) \tag{121}$$

and the EOM eq. (163) can be linearized to give to give tangent space paths that satisfy

$$\dot{\mathrm{z}}_L\left(t\right) = -i\mathbf{C}\left(\bar{\varsigma}_0, \varsigma_0\right)^{-1} \frac{\partial \mathcal{E}_2\left[\varsigma_0\right]\left(\bar{\mathrm{z}}_L\left(t\right), \mathrm{z}_L\left(t\right)\right)}{\partial \bar{\mathbf{z}}}, \tag{122}$$

where

$$\frac{\partial \mathcal{E}_2\left[\varsigma_0\right]\left(\bar{\mathrm{z}}_L\left(t\right), \mathrm{z}_L\left(t\right)\right)}{\partial \bar{\mathbf{z}}} = \frac{\partial \mathcal{E}\left(\bar{\varsigma}_0, \varsigma_0\right)}{\partial \mathbf{z}} + \frac{\partial^2 \mathcal{E}\left(\bar{\varsigma}_0, \varsigma_0\right)}{\partial \bar{\mathbf{z}} \partial \bar{\mathbf{z}}} \cdot \left(\bar{\mathrm{z}}_L\left(t\right) - \bar{\varsigma}_0\right) + \frac{\partial^2 \mathcal{E}\left(\bar{\varsigma}_0, \varsigma_0\right)}{\partial \bar{\mathbf{z}} \partial \mathbf{z}} \cdot \left(\mathrm{z}_L\left(t\right) - \varsigma_0\right). \tag{123}$$

The solution to eq. (122) is given in Appendix and is non holomorphic unless $\frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}} \partial \bar{\mathbf{z}}} = \mathbf{0}$.

If $\left(\bar{\varsigma}_0, \varsigma_0\right)$ is a critical point of $\mathcal{E}_0$ the linearized classical Hamiltonian $\mathcal{E}_2\left[\varsigma_0\right]$ becomes

$$\begin{aligned} \mathcal{E}_2\left[\varsigma_0\right]\left(\bar{\mathbf{z}}, \mathbf{z}\right) &= \mathcal{E}_0\left(\bar{\varsigma}_0, \varsigma_0\right) \\ &+ \frac{1}{2} \begin{pmatrix} \bar{\mathbf{z}} - \bar{\varsigma}_0 & \mathbf{z} - \varsigma_0 \end{pmatrix} \begin{pmatrix} D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0\left(\bar{\varsigma}_0, \varsigma_0\right) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0\left(\bar{\varsigma}_0, \varsigma_0\right) \\ D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0\left(\bar{\varsigma}_0, \varsigma_0\right) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0\left(\bar{\varsigma}_0, \varsigma_0\right) \end{pmatrix} \begin{pmatrix} \bar{\mathbf{z}} - \bar{\varsigma}_0 \\ \mathbf{z} - \varsigma_0 \end{pmatrix} \end{aligned} \tag{124}$$

where

$$D^2_{\mathbf{xy}}\mathcal{E}_0\left(\bar{\varsigma}_0, \varsigma_0\right) = \frac{1}{\mathcal{S}\left(\bar{\varsigma}_0, \varsigma_0\right)} \left\{ D^2_{\mathbf{xy}}\mathcal{H}_0\left(\bar{\varsigma}_0, \varsigma_0\right) - \mathcal{E}_0\left(\bar{\varsigma}_0, \varsigma_0\right) D^2_{\mathbf{xy}}\mathcal{S}\left(\bar{\varsigma}_0, \varsigma_0\right) \right\}. \tag{125}$$

This leads to the linearized EOM for paths $\eta_L(t)$ in the tangent space $\mathcal{T}_{\varsigma_0}$

$$\begin{pmatrix} \frac{d\eta_L(t)}{dt} \\ \frac{d\bar{\eta}_L(t)}{dt} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & i\bar{\mathbf{C}}^{-1}(\bar{\varsigma}_0, \varsigma_0) \\ -i\mathbf{C}^{-1}(\bar{\varsigma}_0, \varsigma_0) & \mathbf{0} \end{pmatrix} \begin{pmatrix} \frac{\partial \mathcal{E}_2[\varsigma_0](\bar{\eta}_L(t),\eta_L(t))}{\partial \bar{\mathbf{z}}} \\ \frac{\partial \mathcal{E}_2[\varsigma_0](\bar{\eta}_L(t),\eta_L(t))}{\partial \mathbf{z}} \end{pmatrix}$$
$$= \begin{pmatrix} \mathbf{0} & i\bar{\mathbf{C}}^{-1}(\bar{\varsigma}_0, \varsigma_0) \\ -i\mathbf{C}^{-1}(\bar{\varsigma}_0, \varsigma_0) & \mathbf{0} \end{pmatrix} \begin{pmatrix} D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) \\ D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\mathbf{z}}_c, \mathbf{z}_c) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) \end{pmatrix} \begin{pmatrix} \eta_L(t) \\ \bar{\eta}_L(t) \end{pmatrix} \tag{126}$$

where

$$\eta_L(t) = \mathrm{z}_L(t) - \varsigma_0 \tag{127}$$

and the approximate solution paths $(\bar{\mathrm{z}}_L(t), \mathrm{z}_L(t))$ with initial value $(\bar{\varsigma}_0, \varsigma_0)$ given by

$$\begin{pmatrix} \bar{\mathrm{z}}_L(t) \\ \mathrm{z}_L(t) \end{pmatrix} = \exp\left\{ \begin{pmatrix} \mathbf{0} & i\bar{\mathbf{C}}^{-1}(\bar{\varsigma}_0, \varsigma_0) \\ -i\mathbf{C}^{-1}(\bar{\varsigma}_0, \varsigma_0) & \mathbf{0} \end{pmatrix} \begin{pmatrix} D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) \\ D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) \end{pmatrix} t \right\} \begin{pmatrix} \varsigma_0 \\ \bar{\varsigma}_0 \end{pmatrix} \tag{128}$$

The solution path $\mathrm{z}_L(t) \in \mathcal{T}_{\varsigma_0}$ to this equation is described in Appendix H and corresponds to the following path of states in the CS tangent space

$$|d\Phi[\varsigma_0](t)\rangle = \sum_{1\le k\le \nu} \left\{ Y_k |\Phi(\varsigma_0)\rangle \mathrm{z}_{Lk}(t) + \langle\Phi(\varsigma_0)| Y_k^\dagger \bar{\mathrm{z}}_{Lk}(t) \right\} = \begin{pmatrix} \mathbf{Y}|\Phi(\varsigma_0)\rangle & \langle\Phi(\varsigma_0)|\mathbf{Y}^\dagger \end{pmatrix} \begin{pmatrix} \mathrm{z}_L(t) \\ \bar{\mathrm{z}}_L(t) \end{pmatrix} \tag{129}$$

VII. LINEAR RESPONSE

If the external radiation fields are of low intensity and the molecular system is initially in a stationary state, it is sufficient to consider the linear response of the molecular state to the applied fields, which can be expressed in terms of a set of linear response functions. In this section we first describe the structure of these functions in the standard quantum mechanical fashion, then in terms of the CS formalism.

A. Hilbert Space Setting

1. *Exact Linear Response*

Time dependent perturbation theory gives that the first order change, $|\delta^1\Psi(t)\rangle$, of the stationary state path $|\Psi_0(t)\rangle$ of the unperturbed Hamiltonian H_0 due to an external potential $F(t)$ is

$$\left|\delta^1\Psi(t)\right\rangle = e^{-iH_0t}\left\{-i\int_0^t F_H(t')\,dt'\right\}|\Psi_0\rangle\,, \tag{130}$$

where

$$F_H(t') = e^{iH_ot'}F(t')\,e^{-iH_ot'}, \tag{131}$$

the stationary path

$$|\Psi_0(t)\rangle = e^{-iH_0t}\,|\Psi_0(t)\rangle = e^{-iE_0t}\,|\Psi_0\rangle \tag{132}$$

and $|\Psi_0\rangle$ is a normalized solution to the TISE for the unperturbed Hamiltonian H_0 corresponding to the energy E_0. It is important to note that during such changes the norm of $|\Psi_0(t)+\delta^1\Psi(t)\rangle$ remains constant to first order as

$$\left\langle\Psi_0(t)+\delta^1\Psi(t)\right|\Psi_0(t)+\delta^1\Psi(t)\Big\rangle \simeq \langle\Psi_0|\,\Psi_0\rangle + \left\langle\delta^1\Psi(t)\right|\Psi_0(t)\Big\rangle + \Big\langle\Psi_0(t)\Big|\,\delta^1\Psi(t)\Big\rangle \tag{133}$$

and

$$\begin{aligned}
&\left\langle\delta^1\Psi(t)\right|\Psi_0(t)\Big\rangle + \Big\langle\Psi_0(t)\Big|\,\delta^1\Psi(t)\Big\rangle = \\
&\qquad\left\langle -i\int_0^t F_H(t')\,dt'\Psi_0\right|\Psi_0\Big\rangle + \left\langle\Psi_0\right|-i\int_0^t F_H(t')\,dt'\;\Psi_0\Big\rangle = \\
&\qquad\qquad i\left\langle\Psi_0\right|\int_0^t F_H(t')\,dt'\;\Psi_0\Big\rangle - i\left\langle\Psi_0\right|\int_0^t F_H(t')\,dt'\;\Psi_0\Big\rangle = 0.
\end{aligned} \tag{134}$$

Using the expression eq. (130) for $|\delta^1\Psi(t)\rangle$ it is straightforward to show that the first order change, $\delta^1\langle A\rangle_0(t)$, of the expectation value of an observable A, from its stationary state value $\langle A\rangle_0$ after a time t is

$$\begin{aligned}
\delta^1\langle A\rangle_0(t) &= \left\{\Big\langle\Psi_0(t)\Big|\,A\delta^1\Psi(t)\Big\rangle + \left\langle\delta^1\Psi(t)\right|A\Psi_0(t)\Big\rangle\right\} \\
&= -\,i\left\{\left\langle e^{-iH_0t}\Psi_0\right|Ae^{-iH_0t}\int_0^t F_H(t')\,dt'\Psi_0\Big\rangle - \left\langle e^{-iH_0t}\int_0^t F_H(t')\,dt'\Psi_0\right|Ae^{-iH_0t}\Psi_0\Big\rangle\right\} \\
&\qquad\qquad = -\,i\int_0^t\langle\Psi_0\,|[A_H(t)\,,F_H(t')]\;\Psi_0\rangle\,dt',
\end{aligned} \tag{135}$$

where

$$A_H(t) = e^{iH_o t} A e^{-iH_o t}. \tag{136}$$

Equation eq. (135) can be rearranged to give

$$\delta^1 \langle A \rangle_0 (t) = -i \int_0^t \langle \Psi_0 | [A_H (t - t'), F(t')] \Psi_0 \rangle \, dt' \tag{137}$$

and finally

$$\delta^1 \langle A \rangle_0 (t) = -i \int_{-\infty}^{\infty} \theta_+ (t - t') \langle \Psi_0 | [A_H (t - t'), F(t')] \Psi_0 \rangle \, dt', \tag{138}$$

where $F(t') = 0$ for $t' < 0$. The response of expectation values of a system initially in a stationary state $|\Psi_0\rangle$ to an external time dependent perturbing potential, $F(t)$,

$$F(t) = \sum_{k,l} f_{kl}(t) \, a_k^\dagger a_l \tag{139}$$

is then given by

$$\delta^1 \langle A \rangle_0 (t) = \int_{-\infty}^{\infty} \sum_{k,l} \left\{ -i\theta_+ (t - t') \left\langle \Psi_0 \middle| \left[A_H (t - t'), a_k^\dagger a_l \right] \Psi_0 \right\rangle \right\} f_{kl}(t') \, dt'. \tag{140}$$

Eq. (140) can be expressed succinctly by defining Response Functions $\{R_{Akl}(\tau)\}$ by

$$R_{Akl}(\tau) = -i\theta_+(\tau) \left\langle \Psi_0 \middle| \left[A_H(\tau), a_k^\dagger a_l \right] \Psi_0 \right\rangle, \tag{141}$$

to give

$$\delta^1 \langle A \rangle_0 (t) = \sum_{k,l} \int_{-\infty}^{\infty} R_{Akl}(t - t') \, f_{kl}(t') \, dt'. \tag{142}$$

The response of any one particle property of a system can be formulated in terms of the response, $\delta\rho(t)$, of the First Order Reduced Density Matrix (FORDM) ρ from its equilibrium value $\rho(0)$, which has matrix elements

$$\rho(0)_{ji} = \left\langle \Psi_0 \right| a_i^\dagger a_j \Psi_0 \left. \right\rangle. \tag{143}$$

In the case of a weak external field one can neglect terms higher than first order in F and use eq. (142) to obtain the linear response of $\rho(0)$ as

$$\delta^1 \rho(t) = \sum_{k,l} \int_{-\infty}^{\infty} R_{\rho kl}(t - t') \, f_{kl}(t') \, dt'. \tag{144}$$

The linear responses $\{\delta\rho_{ij}(t)\}$ of the matrix elements of $\rho(0)$ can be cast in the form

$$\delta\rho_{ij}(t) = \sum_{k,l}\int_{-\infty}^{+\infty} R_{ijkl}(t-t') f_{kl}(t')\,dt', \tag{145}$$

where the one particle linear response functions $\{R_{ijkl}(t)\}$ are defined as

$$R_{ijkl}(t) = -i\theta_{+}(t)\left\langle \Psi_0 \left| \left[a_{Hj}^{\dagger}(t)\,a_{Hi}(t)\,, a_k^{\dagger}a_l\right] \Psi_0 \right.\right\rangle, \tag{146}$$

which can be expressed as

$$R_{ijkl}(t) = -i\theta_{+}(t)\left\{\left\langle \Psi_0 \left| a_j^{\dagger}a_i e^{-i(H_0-E_0)t} a_k^{\dagger}a_l\ \Psi_0 \right.\right\rangle - \left\langle \Psi_0 \left| a_k^{\dagger}a_l e^{i(H_0-E_0)t} a_j^{\dagger}a_i\ \Psi_0 \right.\right\rangle\right\}. \tag{147}$$

Taking the Laplace transform [**?**] of $R_{lkij}(t)$ we obtain

$$R_{ijkl}(\zeta) = \sum_{n\neq 0}\left\{\frac{\left\langle \Psi_0 \left| a_j^{\dagger}a_i\Psi_n \right.\right\rangle\left\langle \Psi_n \left| a_k^{\dagger}a_l\Psi_0 \right.\right\rangle}{\zeta-(E_n-E_0)} - \frac{\left\langle \Psi_0 \left| a_k^{\dagger}a_l\Psi_n \right.\right\rangle\left\langle \Psi_n \left| a_j^{\dagger}a_i\Psi_0 \right.\right\rangle}{\zeta+(E_n-E_0)}\right\}. \tag{148}$$

The Laplace transformed response functions $\{R_{ijkl}(\zeta)\}$ form a hermitian matrix for real ζ, that can be diagonalized to give

$$\tilde{R}_{KL}(\zeta) = g_K(\zeta)\,\delta_{KL} = \sum_{n\neq 0}\left\{\frac{\langle \Psi_0 \left| T_K\Psi_n \right.\rangle\left\langle \Psi_n \left| T_K^{\dagger}\Psi_0 \right.\right\rangle}{\zeta-(E_n-E_0)} - \frac{\langle \Psi_0 \left| T_K\Psi_n \right.\rangle\left\langle \Psi_n \left| T_K^{\dagger}\Psi_0 \right.\right\rangle}{\zeta+(E_n-E_0)}\right\}, \tag{149}$$

where the normal mode excitation operators are

$$T_K^{\dagger} = \sum_{i,j} T_{Kij}a_i^{\dagger}a_j. \tag{150}$$

The poles of the Laplace transform occur at $\pm(E_n-E_0)$ and the residues $\left|\left\langle \Psi_n \left| T_K^{\dagger}\Psi_0 \right.\right\rangle\right|^2$ are transition probabilities that the operator $T_K^{\dagger}$ will cause a transition from the stationary state $|\Psi_0\rangle$ to the stationary state $|\Psi_n\rangle$. In general the excited states can be expressed in terms of the operators $\left\{T_K^{\dagger}\right\}$ as

$$|\Psi_n\rangle = \sum_{0\leq m\leq N}\ \sum_{K_1\cdots K_m} c^n_{K_1\cdots K_m} T_{K_1}^{\dagger}\cdots T_{K_m}^{\dagger}\,|\Psi_0\rangle\,. \tag{151}$$

If the radiation field is weak, which justifies the use of linear response, one is focusing on single photon processes. If such radiation *overwhelmingly* produces excited states $\{|\Psi_K\rangle\}$ then one might assume that the transition probabilities $\left\{\left|\left\langle \Psi_n \left| T_K^{\dagger}\Psi_0 \right.\right\rangle\right|^2\right\}$ are close to 100% and hence

$$|\Psi_K\rangle \approx T_K^{\dagger}\,|\Psi_0\rangle \tag{152}$$

and $\left\{T_K^\dagger\right\}$ are *state* excitation operators. However, for observable processes to be initiated by single photons this transition probability does not have to be 100% , it just needs to be reasonably large, so that interactions between photons and molecules are likely to occur. Thus it is not necessary for exact $T_K^\dagger$, nor approximations to it, to be state excitation operators.

2. Approximate Linear Response

Approximations to linear response can be formulated in terms of approximations to

- $|\Psi_0\rangle$ i.e. restricting the nature of the reference state by any standard approximation ansatz, e.g. Multi Configurational Self Consistent Field (MCSCF), CI, Coupled Cluster, etc...,
- H_0 e.g. express $H_0 = h + V$ where h is an independent particle Hamiltonian and V is a correlation potential and expand in terms of powers of V,
- $|\Psi_n\rangle$ by truncating the expansion eq. (151) at various orders, which is the approach pioneered by Osvaldo Goscinski via the introduction of super operators and operator manifolds [**?**].

B. Linear Response in a Coherent State Setting

In this section we show that *classical* linear response theory applied to the EOM eq. (113) in the CS manifold $\mathcal{M}$ is *equivalent* to quantum linear response theory. This allows one to design, extend and see the implications of approximations more clearly.

1. Full Equations of Motion for any Manifold $\mathcal{M}$

For a manifold of CS's $\{|\Phi(\mathbf{z})\rangle\}$ the classical Hamiltonian, $\mathcal{E}(\bar{\mathbf{z}},\mathbf{z},t)$, corresponding to the Hamiltonian containing the time dependent perturbation is

$$\mathcal{E}(\bar{\mathbf{z}},\mathbf{z},t) = \frac{\langle\Phi(\mathbf{z})|\left(H_0 + F(t)\right)\Phi(\mathbf{z})\rangle}{\langle\Phi(\mathbf{z})|\Phi(\mathbf{z})\rangle} = \mathcal{E}_0(\bar{\mathbf{z}},\mathbf{z}) + \mathcal{F}(\bar{\mathbf{z}},\mathbf{z},t) \tag{153}$$

where the classical observables $\mathcal{E}_0(\bar{\mathbf{z}},\mathbf{z})$ and $\mathcal{F}(\bar{\mathbf{z}},\mathbf{z},t)$ are given by

$$\begin{aligned}\mathcal{E}_0(\bar{\mathbf{z}},\mathbf{z}) &= \frac{\langle\Phi(\mathbf{z})|\,H_0\Phi(\mathbf{z})\rangle}{\langle\Phi(\mathbf{z})|\,\Phi(\mathbf{z})\rangle}\\ \mathcal{F}(\bar{\mathbf{z}},\mathbf{z},t) &= \frac{\langle\Phi(\mathbf{z})|\,F(t)\,\Phi(\mathbf{z})\rangle}{\langle\Phi(\mathbf{z})|\,\Phi(\mathbf{z})\rangle}.\end{aligned} \tag{154}$$

It should be observed that the external potential function, $\mathcal{F}(\bar{\mathbf{z}},\mathbf{z},t)$, is linear in the FORDM

$$\mathcal{F}(\bar{\mathbf{z}},\mathbf{z},t) = \sum_{1\le k,l\le\nu} f_{kl}(t)\,\frac{\left\langle\Phi(\mathbf{z})|\,a_k^{\dagger}a_l\Phi(\mathbf{z})\right\rangle}{\langle\Phi(\mathbf{z})|\,\Phi(\mathbf{z})\rangle} = \sum_{1\le k,l\le\nu} f_{kl}(t)\,\rho_{lk}(\bar{\mathbf{z}},\mathbf{z}), \tag{155}$$

where

$$\rho_{lk}(\bar{\mathbf{z}},\mathbf{z}) = \frac{\left\langle\Phi(\mathbf{z})|\,a_k^{\dagger}a_l\Phi(\mathbf{z})\right\rangle}{\langle\Phi(\mathbf{z})|\,\Phi(\mathbf{z})\rangle}. \tag{156}$$

The restriction of the quantum evolution to the CS manifold is then given by the solution paths $\{\bar{\mathrm{z}},\mathrm{z}\}$ to the equations eq. (110) , that can expressed in matrix form as

$$\begin{pmatrix}\frac{d\bar{\mathrm{z}}(t)}{dt}\\ \frac{d\mathrm{z}(t)}{dt}\end{pmatrix} = \begin{pmatrix}\mathbf{0} & i\bar{\mathbf{C}}^{-1}(\bar{\mathrm{z}}(t),\mathrm{z}(t))\\ -i\mathbf{C}^{-1}(\bar{\mathrm{z}}(t),\mathrm{z}(t)) & \mathbf{0}\end{pmatrix}\begin{pmatrix}\frac{\partial\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial\bar{\mathrm{z}}}\\ \frac{\partial\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial\mathrm{z}}\end{pmatrix}, \tag{157}$$

where the metric matrix along the paths, $\mathbf{C}(\bar{\mathrm{z}}(t),\mathrm{z}(t))$, is given by

$$\begin{aligned}\mathbf{C}(\bar{\mathrm{z}}(t),\mathrm{z}(t)) = \frac{1}{\mathcal{S}((\bar{\mathrm{z}}(t),\mathrm{z}(t)))}&\left\{\frac{\langle\Phi(\mathrm{z}(t))|\,\mathbf{Y}^{\dagger}\mathbf{Y}\Phi(\mathrm{z}(t))\rangle}{\langle\Phi(\mathrm{z}(t))|\,\Phi(\mathrm{z}(t))\rangle}\right.\\ &\left.-\mathcal{S}((\bar{\mathrm{z}}(t),\mathrm{z}(t)))\,\frac{\langle\Phi(\mathrm{z}(t))|\,\mathbf{Y}^{\dagger}\Phi(\mathrm{z}(t))\rangle}{\langle\Phi(\mathrm{z}(t))|\,\Phi(\mathrm{z}(t))\rangle}\frac{\langle\Phi(\mathrm{z}(t))|\,\mathbf{Y}\Phi(\mathrm{z}(t))\rangle}{\langle\Phi(\mathrm{z}(t))|\,\Phi(\mathrm{z}(t))\rangle}\right\}\end{aligned} \tag{158}$$

and the gradient of the classical Hamiltonian by

$$\begin{aligned}&\begin{pmatrix}\frac{\partial\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial\bar{\mathrm{z}}}\\ \frac{\partial\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial\mathrm{z}}\end{pmatrix}\\ &= \frac{1}{\mathcal{S}((\bar{\mathrm{z}}(t),\mathrm{z}(t)))}\begin{pmatrix}\left\{\left\langle\Phi(\mathrm{z}(t))|\,\mathbf{Y}^{\dagger}H(t)\,\Phi(\mathrm{z}(t))\right\rangle - \mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)\left\langle\Phi(\mathrm{z}(t))|\,\mathbf{Y}^{\dagger}\Phi(\mathrm{z}(t))\right\rangle\right\}\\ \left\{\langle\Phi(\mathrm{z}(t))|\,H(t)\,\mathbf{Y}\Phi(\mathrm{z}(t))\rangle - \mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)\langle\Phi(\mathrm{z}(t))|\,\mathbf{Y}\Phi(\mathrm{z}(t))\rangle\right\}\end{pmatrix}.\end{aligned} \tag{159}$$

The evolution paths, $\rho_{lk}(t)$, of the matrix elements of the FORDM $\rho(t)$ can be expressed in terms of the solutions of eq. (157) as $\rho_{lk}(t) = \rho_{lk}(\bar{\mathrm{z}}(t),\mathrm{z}(t))$ or as solutions to the EOM

$$\frac{d\rho_{lk}(t)}{dt} = \{\rho_{lk}(t),\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)\} \tag{160}$$

for the paths determined by eq. (157) . It should be observed that *no matter* the form of the classical Hamiltonian $\mathcal{E}$, solutions to equation eq. (157) describe paths in the CS manifold $\mathcal{M}$ which correspond to paths of states in $\mathcal{H}^N$, consequently any approximations to $\mathcal{E}\left(\bar{\mathbf{z}},\mathbf{z},t\right)$ or $\mathcal{E}_0\left(\bar{\mathbf{z}},\mathbf{z}\right)$ still lead to states in $\mathcal{M}\subset\mathcal{H}^N$.

2. Full Classical Linear Response

As the CS manifold $\mathcal{M}$ is a classical phase space it is natural to directly utilize classical linear response theory, which is based on classical first order time dependent perturbation theory [?]. However as this phase space is curved one has a choice of applying the perturbation theory about a point $\mathbf{z}_0\in\mathcal{M}$ in the manifold $\mathcal{M}$ including perturbations to the metric, or in the tangent space $\mathcal{T}_{\mathbf{z}_0}$. It is easier and more directly comparable to the Quantum Theory to perform the latter. This produces first order changes of the flat phase space paths given by

$$\delta \mathrm{z}_L\left(t\right)=\int_0^t\left\{\mathrm{z}_{0L}\left(t\right),\mathcal{F}\left(\bar{\mathrm{z}}_{0L}\left(t'\right),\mathrm{z}_{0L}\left(t'\right),t'\right)\right\}_{\mathbf{z}_0}dt', \tag{161}$$

where the unperturbed evolution path with initial value $\mathbf{z}_0$ is given as

$$\begin{aligned}&\mathrm{z}_{0L}\left(t\right)=e^{\mathcal{L}_0t}\mathbf{z}_0=\mathbf{z}_0+\\&\sum_{n\geq1}\int_0^t\int_{t_1}^t\cdots\int_{t_{n-1}}^t\Big\{\left\{\cdots\left\{\mathbf{z}_0,\mathcal{E}_0\left(\bar{\mathbf{z}}_0,\mathbf{z}_0\right)\right\}_{\mathbf{z}_0},\cdots,\mathcal{E}_0\left(\bar{\mathbf{z}}_0,\mathbf{z}_0\right)\right\}_{\mathbf{z}_0},\mathcal{E}_0\left(\bar{\mathbf{z}}_0,\mathbf{z}_0\right)\Big\}_{\mathbf{z}_0}dt_1dt_2\cdots dt_n.\end{aligned} \tag{162}$$

In matrix form the equation for the unperturbed paths, $\mathrm{z}_{0L}\left(t\right)$, leading to the solution eq. (**??**), is

$$\begin{aligned}\dot{\mathrm{z}}_{0L}\left(t\right)=\left\{\mathrm{z}_{0L}\left(t\right),\mathcal{E}_0\left(\bar{\mathrm{z}}_{0L}\left(t\right),\mathrm{z}_{0L}\left(t\right)\right)\right\}_{\mathbf{z}_0}&=\begin{pmatrix}\mathbf{0}&\mathbf{0}\\-i\mathbf{C}\left(\bar{\mathbf{z}}_0,\mathbf{z}_0\right)^{-1}&\mathbf{0}\end{pmatrix}\begin{pmatrix}\frac{\partial\mathcal{E}_0\left(\bar{\mathrm{z}}_{0L}(t),\mathrm{z}_{0L}(t)\right)}{\partial\bar{\mathbf{z}}}\\\frac{\partial\mathcal{E}_0\left(\bar{\mathrm{z}}_{0L}(t),\mathrm{z}_{0L}(t)\right)}{\partial\mathbf{z}}\end{pmatrix}\\&=-i\mathbf{C}\left(\bar{\mathbf{z}}_0,\mathbf{z}_0\right)^{-1}\frac{\partial\mathcal{E}_0\left(\bar{\mathrm{z}}_{0L}\left(t\right),\mathrm{z}_{0L}\left(t\right)\right)}{\partial\bar{\mathbf{z}}}.\end{aligned} \tag{163}$$

The solution of equation eq. (163) is discussed in Appendix H.

The linear response of the classical observables $\{\rho_{lk}\}$ about an initial point $\rho_{lk}\left(\bar{\mathbf{z}}_0,\mathbf{z}_0\right)$ is given by

$$\delta\rho_{lk}\left(t\right)=-\int_0^t\left\{\rho_{Hlk}\left(t\right),\mathcal{F}_H\left(\bar{\mathbf{z}},\mathbf{z},t'\right)\right\}_{\mathbf{z}_0}dt' \tag{164}$$

where

$$\rho_{Hji}(t) = e^{\mathcal{L}_0 t}\rho_{ji}(\bar{\mathbf{z}}_0, \mathbf{z}_0)$$
$$\mathcal{F}_H(\bar{\mathbf{z}}, \mathbf{z}, t') = e^{\mathcal{L}_0 t'}\mathcal{F}(\bar{\mathbf{z}}_0, \mathbf{z}_0, t'), \qquad (165)$$

which can be expanded in the same fashion as eq. (162). Using the expansion eq. (155) the response of the density matrix elements eq. (164) becomes

$$\delta\rho_{ji}(t) = \sum_{r,s}\int_0^t \{\rho_{Hji}(t), \rho_{Hsr}(t')\}_{\mathbf{z}_0} f_{rs}(t')\, dt' = \sum_{r,s}\int_{-\infty}^{\infty} R_{ijrs}(t,t') f_{rs}(t')\, dt', \qquad (166)$$

where we have introduced response functions

$$R_{ijrs}(t,t') = \theta_+(t-t')\{\rho_{jiH}(t), \rho_{sr}(t')\}_{\mathbf{z}_0}. \qquad (167)$$

3. *Truncated Classical Linear Response*

Basing the effects of the perturbation on the time evolution in the tangent space on the full unperturbed Hamiltonian $\mathcal{E}_0(\bar{\mathbf{z}}, \mathbf{z})$, is complicated and does not produce RPA equations that correspond to ones obtained from standard quantum theory. However using the quadratic approximation $\mathcal{E}_{02}[\bar{\varsigma}_0, \varsigma_0]$ to $\mathcal{E}_0(\bar{\mathbf{z}}, \mathbf{z})$ and the linear approximations $\{\rho_{1ji}[\bar{\varsigma}_0, \varsigma_0]\}$ to the observables $\{\rho_{ji}(\bar{\mathbf{z}}, \mathbf{z})\}$ at a critical point $(\bar{\varsigma}_0, \varsigma_0)$ of $\mathcal{E}_0(\bar{\mathbf{z}}, \mathbf{z})$ does reproduce the RPA and generalized RPA. The approximations are based on the expansion (see Appendix B)

$$\mathcal{O}[\bar{\varsigma}_0, \varsigma_0](\bar{\eta}, \eta) = \sum_{1\leq n\leq\infty} D^n\mathcal{O}[\bar{\varsigma}_0, \varsigma_0](\bar{\eta}, \eta). \qquad (168)$$

with $\mathcal{E}_{02}(\bar{\varsigma}_0, \varsigma_0)$ given by

$$\mathcal{E}_{02}[\bar{\varsigma}_0, \varsigma_0](\bar{\eta}, \eta) = \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) + D^2\mathcal{E}_0[\bar{\varsigma}_0, \varsigma_0](\bar{\eta}, \eta), \qquad (169)$$

where

$$\frac{\partial^2\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial\bar{\mathbf{z}}\partial\bar{\mathbf{z}}} =$$
$$\frac{1}{\mathcal{S}(\bar{\varsigma}_0, \varsigma_0)}\left\{\left\langle \Phi(\varsigma_0)\right| \mathbf{Y}^\dagger\mathbf{Y}^\dagger H_0\Phi(\varsigma_0)\right\rangle - \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)\left\langle \Phi(\varsigma_0)\right| \mathbf{Y}^\dagger\Phi(\varsigma_0)\right\rangle\left\langle \Phi(\varsigma_0)\right| \mathbf{Y}^\dagger\Phi(\varsigma_0)\right\rangle\}$$
$$\frac{\partial^2\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial\bar{\mathbf{z}}\partial\mathbf{z}} =$$
$$\frac{1}{\mathcal{S}(\bar{\varsigma}_0, \varsigma_0)}\left\{\left\langle \Phi(\varsigma_0)\right| \mathbf{Y}^\dagger H_0\mathbf{Y}\Phi(\varsigma_0)\right\rangle - \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)\left\langle \Phi(\varsigma_0)\right| \mathbf{Y}^\dagger\Phi(\varsigma_0)\right\rangle\left\langle \Phi(\varsigma_0)\right| \mathbf{Y}\Phi(\varsigma_0)\right\rangle\} \qquad (170)$$

and

$$\frac{\partial^2\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial\mathbf{z}\partial\bar{\mathbf{z}}} = \frac{\partial^2\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial\bar{\mathbf{z}}\partial\mathbf{z}}^{\dagger} = \overline{\frac{\partial^2\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial\bar{\mathbf{z}}\partial\mathbf{z}}}$$

$$\frac{\partial^2\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial\mathbf{z}\partial\mathbf{z}} = \overline{\frac{\partial^2\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial\bar{\mathbf{z}}\partial\bar{\mathbf{z}}}}. \tag{171}$$

The linear approximations $\{\rho_{1ji}[\bar{\varsigma}_0,\varsigma_0]\}$ to the FORDM matrix elements are given by

$$\rho_{1ji}[\bar{\varsigma}_0,\varsigma_0](\bar{\eta},\eta) = \rho_{ji}(\bar{\varsigma}_0,\varsigma_0) + D^1\rho_{ji}[\bar{\varsigma}_0,\varsigma_0](\bar{\eta},\eta), \tag{172}$$

where the first order derivatives of ρ_{ji} are

$$\frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\bar{\mathbf{z}}} = \frac{1}{\mathcal{S}(\bar{\varsigma}_0,\varsigma_0)}\left\{\left\langle\Phi(\varsigma_0)\right|\mathbf{Y}^{\dagger}a_i^{\dagger}a_j\Phi(\varsigma_0)\right\rangle - \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)\left\langle\Phi(\varsigma_0)\right|\mathbf{Y}^{\dagger}\Phi(\varsigma_0)\rangle\right\}$$

$$\frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\mathbf{z}} = \frac{1}{\mathcal{S}(\bar{\varsigma}_0,\varsigma_0)}\left\{\left\langle\Phi(\varsigma_0)\right|a_i^{\dagger}a_j\mathbf{Y}\Phi(\varsigma_0)\right\rangle - \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)\left\langle\Phi(\varsigma_0)\right|\mathbf{Y}\Phi(\varsigma_0)\rangle\right\}. \tag{173}$$

The time evolution produced by $\mathcal{E}_{02}[\bar{\varsigma}_0,\varsigma_0]$ on the flat phase space $\mathcal{T}_{\varsigma_0}$ produces coordinate paths

$$\begin{pmatrix}\bar{\eta}(t)\\ \eta(t)\end{pmatrix} = e^{\mathcal{L}_{02}t}(\bar{\varsigma}_0,\varsigma_0)$$
$$= \exp\left\{i\begin{pmatrix}\mathbf{0} & \bar{\mathbf{C}}^{-1}(\bar{\varsigma}_0,\varsigma_0)\\ -\mathbf{C}^{-1}(\bar{\varsigma}_0,\varsigma_0) & \mathbf{0}\end{pmatrix}\begin{pmatrix}D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)\\ D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)\end{pmatrix}t\right\}\begin{pmatrix}\bar{\varsigma}_0\\ \varsigma_0\end{pmatrix}. \tag{174}$$

This leads to

$$\rho_{1jiH}(\tau) = \rho_{ji}(\bar{\varsigma}_0,\varsigma_0) + \begin{pmatrix}\frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\bar{\mathbf{z}}} & \frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\mathbf{z}}\end{pmatrix}e^{i\mathbb{M}t}\begin{pmatrix}\bar{\varsigma}_0\\ \varsigma_0\end{pmatrix}, \tag{175}$$

where $\mathbb{M}$ is defined by eq. (174). This can be equivalently written as

$$\rho_{1jiH}(\tau) = \rho_{1ji}[\bar{\varsigma}_0,\varsigma_0]\{(\bar{\eta}(\tau),\eta(\tau))\} = \rho_{ji}(\bar{\varsigma}_0,\varsigma_0) + D^1\rho_{ji}[\bar{\varsigma}_0,\varsigma_0]e^{i\mathbb{M}t}(\bar{\eta}(0),\eta(0))$$
$$= \rho_{ji}(\bar{\varsigma}_0,\varsigma_0) + \sum_{1\le k\le\nu}\left\{\left(D^1\rho_{ji}[\bar{\varsigma}_0,\varsigma_0]e^{i\mathbb{M}t}\right)_k\bar{\eta}_k(0) + \left(D^1\rho_{ji}[\bar{\varsigma}_0,\varsigma_0]e^{i\mathbb{M}t}\right)_{k+\nu}\eta_k(0)\right\}, \tag{176}$$

which shows that derivatives wrt $\{\bar{\eta},\eta\}$ are

$$\frac{\partial\rho_{1jiH}(\tau)}{\partial\bar{\eta}_k} = \left(D^1\rho_{ji}[\bar{\varsigma}_0,\varsigma_0]e^{i\mathbb{M}\tau}\right)_k \tag{177a}$$

$$\frac{\partial\rho_{1jiH}(\tau)}{\partial\eta_k} = \left(D^1\rho_{ji}[\bar{\varsigma}_0,\varsigma_0]e^{i\mathbb{M}\tau}\right)_{k+\nu}. \tag{177b}$$

As the time evolution is homogeneous the linear response functions eq. (167) become

$$R_{ijrs}(\tau) = \theta_+(\tau)\left\{\rho_{1jiH}(\tau), \rho_{sr}(0)\right\}_{\mathbf{z}_0}, \tag{178}$$

so that

$$\delta\rho_{1ji}(t) = \sum_{r,s}\int_0^t \left\{\rho_{1Hji}(t-t'), \rho_{1sr}(0)\right\}_{\mathbf{z}_0} f_{rs}(t')\,dt' = \sum_{r,s}\int_{-\infty}^{\infty} R_{ijrs}(t-t') f_{rs}(t')\,dt'. \tag{179}$$

Developing the response function eq.(178) we obtain

$$R_{ijrs}(\tau) = \\ -i\theta_+(\tau)\begin{pmatrix}\frac{\partial\rho_{2ji}(\tau)}{\partial\bar{\eta}} & \frac{\partial\rho_{2ji}(\tau)}{\partial\eta}\end{pmatrix}\begin{pmatrix}\mathbf{0} & -\bar{\mathbf{C}}^{-1}(\bar{\varsigma}_0,\varsigma_0)\\ \mathbf{C}^{-1}(\bar{\varsigma}_0,\varsigma_0) & \mathbf{0}\end{pmatrix}\begin{pmatrix}\frac{\partial\rho_{2sr}(0)}{\partial\bar{\eta}}\\ \frac{\partial\rho_{2sr}(0)}{\partial\eta}\end{pmatrix}, \tag{180}$$

where

$$\begin{pmatrix}\frac{\partial\rho_{2ji}(\tau)}{\partial\bar{\eta}} & \frac{\partial\rho_{2ji}(\tau)}{\partial\eta}\end{pmatrix} = \begin{pmatrix}\frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\bar{\mathbf{z}}} & \frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\mathbf{z}}\end{pmatrix}\times \\ \exp\left\{-i\begin{pmatrix}\mathbf{0} & -\bar{\mathbf{C}}^{-1}(\bar{\varsigma}_0,\varsigma_0)\\ \mathbf{C}^{-1}(\bar{\varsigma}_0,\varsigma_0) & \mathbf{0}\end{pmatrix}\begin{pmatrix}D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)\\ D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)\end{pmatrix}t\right\} \tag{181}$$

and

$$\begin{pmatrix}\frac{\partial\rho_{2sr}(0)}{\partial\bar{\eta}}\\ \frac{\partial\rho_{2sr}(0)}{\partial\eta}\end{pmatrix} = \begin{pmatrix}\frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\bar{\mathbf{z}}}\\ \frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\mathbf{z}}\end{pmatrix}. \tag{182}$$

One can take the Laplace transform of eq. (179) to obtain

$$\delta\rho_{1ji}(\zeta) = \sum_{\leq 1r,s\leq\nu} R_{ijrs}(\zeta) f_{rs}(\zeta), \tag{183}$$

where

$$R_{ijrs}(\zeta) = \begin{pmatrix}\frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\bar{\mathbf{z}}} & \frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\mathbf{z}}\end{pmatrix}\times \\ \left(\zeta\mathbb{I}_{2\nu} - \begin{pmatrix}\mathbf{0} & -\bar{\mathbf{C}}^{-1}(\bar{\varsigma}_0,\varsigma_0)\\ \mathbf{C}^{-1}(\bar{\varsigma}_0,\varsigma_0) & \mathbf{0}\end{pmatrix}\begin{pmatrix}D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)\\ D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)\end{pmatrix}\right)^{-1}\times \\ \begin{pmatrix}\mathbf{0} & -i\bar{\mathbf{C}}^{-1}(\bar{\varsigma}_0,\varsigma_0)\\ i\mathbf{C}^{-1}(\bar{\varsigma}_0,\varsigma_0) & \mathbf{0}\end{pmatrix}\begin{pmatrix}\frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\bar{\mathbf{z}}}\\ \frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\mathbf{z}}\end{pmatrix} \tag{184}$$

and $f_{rs}(\zeta)$ is the Laplace transform of $f_{rs}(t)$. Eq. (184) can be equivalently written as

$$R_{ijrs}(\zeta) = \begin{pmatrix} \frac{\partial \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}}} & \frac{\partial \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z}} \end{pmatrix} \times$$
$$\left(\zeta \begin{pmatrix} \mathbf{0} & \mathbf{C}(\bar{\varsigma}_0,\varsigma_0) \\ -\bar{\mathbf{C}}(\bar{\varsigma}_0,\varsigma_0) & \mathbf{0} \end{pmatrix} - \begin{pmatrix} D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) \\ D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) \end{pmatrix} \right)^{-1}$$
$$\times \begin{pmatrix} \frac{\partial \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}}} \\ \frac{\partial \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z}} \end{pmatrix}. \tag{185}$$

which after inserting

$$\begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{1} \\ \mathbf{1} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{0} & \mathbf{1} \\ \mathbf{1} & \mathbf{0} \end{pmatrix} \tag{186}$$

after the first term on the RHS can be rearranged in terms of a hermitian matrices as

$$R_{ijrs}(\zeta) = \begin{pmatrix} \frac{\partial \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z}} & \frac{\partial \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}}} \end{pmatrix} \times$$
$$\left(\zeta \begin{pmatrix} \mathbf{C}(\bar{\varsigma}_0,\varsigma_0) & \mathbf{0} \\ \mathbf{0} & -\bar{\mathbf{C}}(\bar{\varsigma}_0,\varsigma_0) \end{pmatrix} - \begin{pmatrix} D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) \\ D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) \end{pmatrix} \right)^{-1}$$
$$\times \begin{pmatrix} \frac{\partial \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}}} \\ \frac{\partial \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z}} \end{pmatrix}. \tag{187}$$

Defining

$$\mathfrak{C} = \begin{pmatrix} \mathbf{C}(\bar{\varsigma}_0,\varsigma_0) & \mathbf{0} \\ \mathbf{0} & -\bar{\mathbf{C}}(\bar{\varsigma}_0,\varsigma_0) \end{pmatrix}$$
$$\mathfrak{D} = \begin{pmatrix} D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) \\ D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0) \end{pmatrix} \tag{188}$$

eq. (187) can be written as

$$R_{ijrs}(\zeta) = \begin{pmatrix} \frac{\partial \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z}} & \frac{\partial \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}}} \end{pmatrix} \mathfrak{C}^{-\frac{1}{2}} \left(\zeta \mathbb{I}_{2\nu} - \mathfrak{C}^{-\frac{1}{2}} \mathfrak{D} \mathfrak{C}^{-\frac{1}{2}} \right)^{-1} \mathfrak{C}^{-\frac{1}{2}} \begin{pmatrix} \frac{\partial \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}}} \\ \frac{\partial \rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z}} \end{pmatrix}, \tag{189}$$

where

$$\mathfrak{C}^{\frac{1}{2}} = \begin{pmatrix} \mathbf{C}^{\frac{1}{2}} & \mathbf{0} \\ \mathbf{0} & i\bar{\mathbf{C}}^{\frac{1}{2}} \end{pmatrix} \tag{190}$$

and

$$\mathfrak{C}^{-\frac{1}{2}} = \begin{pmatrix} \mathbf{C}^{-\frac{1}{2}} & \mathbf{0} \\ \mathbf{0} & -i\bar{\mathbf{C}}^{-\frac{1}{2}} \end{pmatrix}. \tag{191}$$

Then resolvent term in eq. (189) can be evaluated by first noting that

$$\mathfrak{C}^{-\frac{1}{2}}\mathfrak{D}\mathfrak{C}^{-\frac{1}{2}} = \begin{pmatrix} \mathbf{C}^{-\frac{1}{2}} & \mathbf{0} \\ \mathbf{0} & -i\bar{\mathbf{C}}^{-\frac{1}{2}} \end{pmatrix} \begin{pmatrix} D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0 & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0 \\ D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0 & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0 \end{pmatrix} \begin{pmatrix} \mathbf{C}^{-\frac{1}{2}} & \mathbf{0} \\ \mathbf{0} & -i\bar{\mathbf{C}}^{-\frac{1}{2}} \end{pmatrix}$$

$$= \begin{pmatrix} \mathbb{I}_\nu & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_\nu \end{pmatrix} \begin{pmatrix} \mathbf{C}^{-\frac{1}{2}} & \mathbf{0} \\ \mathbf{0} & i\bar{\mathbf{C}}^{-\frac{1}{2}} \end{pmatrix} \begin{pmatrix} D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0 & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0 \\ D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0 & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0 \end{pmatrix} \begin{pmatrix} \mathbf{C}^{-\frac{1}{2}} & \mathbf{0} \\ \mathbf{0} & i\bar{\mathbf{C}}^{-\frac{1}{2}} \end{pmatrix}^\dagger$$

$$= \begin{pmatrix} \mathbf{C}^{-\frac{1}{2}} D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0 \mathbf{C}^{-\frac{1}{2}} & -i\mathbf{C}^{-\frac{1}{2}} D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0 \bar{\mathbf{C}}^{-\frac{1}{2}} \\ -i\bar{\mathbf{C}}^{-\frac{1}{2}} D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0 \mathbf{C}^{-\frac{1}{2}} & -\bar{\mathbf{C}}^{-\frac{1}{2}} D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0 \bar{\mathbf{C}}^{-\frac{1}{2}} \end{pmatrix} \equiv \begin{pmatrix} \mathbb{I}_\nu & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_\nu \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \overline{\mathbf{B}} & \overline{\mathbf{A}} \end{pmatrix}, \quad (192)$$

where

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \overline{\mathbf{B}} & \overline{\mathbf{A}} \end{pmatrix} = \begin{pmatrix} \mathbf{C}^{-\frac{1}{2}} D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0 \mathbf{C}^{-\frac{1}{2}} & -i\mathbf{C}^{-\frac{1}{2}} D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0 \bar{\mathbf{C}}^{-\frac{1}{2}} \\ i\bar{\mathbf{C}}^{-\frac{1}{2}} D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0 \mathbf{C}^{-\frac{1}{2}} & \bar{\mathbf{C}}^{-\frac{1}{2}} D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0 \bar{\mathbf{C}}^{-\frac{1}{2}} \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \overline{\mathbf{B}} & \overline{\mathbf{A}} \end{pmatrix}^\dagger. \quad (193)$$

Then using the spectral theorem to obtain (see Appendix C)

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \overline{\mathbf{B}} & \mathbf{A} \end{pmatrix} = \sum_{1\le j\le\nu} \eta_j \left\{ \mathbf{v}_j\mathbf{v}_j^\dagger + \mathbf{v}_{j+\nu}\mathbf{v}_{j+\nu}^\dagger \right\}; \qquad \eta_j \in \mathbb{R}, \quad (194)$$

where

$$\mathbf{v}_j = \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}; \mathbf{v}_{j+\nu} = \begin{pmatrix} \overline{\mathbf{b}} \\ \overline{\mathbf{a}} \end{pmatrix}; \quad \mathbf{v}_j^\dagger \mathbf{v}_k = \delta_{jk}. \quad (195)$$

Thus

$$\mathbb{I}_{2\nu} = \sum_{1\le j\le\nu} \left\{ \mathbf{v}_j\mathbf{v}_j^\dagger + \mathbf{v}_{j+\nu}\mathbf{v}_{j+\nu}^\dagger \right\} \quad (196)$$

and

$$\begin{pmatrix} \mathbb{I}_\nu & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_\nu \end{pmatrix} = \sum_{1\le j\le\nu} \left\{ \mathbf{v}_j\mathbf{v}_j^\dagger - \mathbf{v}_{j+\nu}\mathbf{v}_{j+\nu}^\dagger \right\}. \quad (197)$$

Using eqs. (194) and (197) we obtain

$$\mathfrak{C}^{-\frac{1}{2}}\mathfrak{D}\mathfrak{C}^{-\frac{1}{2}} = \begin{pmatrix} \mathbb{I}_\nu & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_\nu \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \overline{\mathbf{B}} & \overline{\mathbf{A}} \end{pmatrix} = \sum_{1\le j\le\nu} \eta_j \left\{ \mathbf{v}_j\mathbf{v}_j^\dagger - \mathbf{v}_{j+\nu}\mathbf{v}_{j+\nu}^\dagger \right\} = \sum_{1\le j\le 2\nu} \eta_j \mathbf{v}_j\mathbf{v}_j^\dagger, \quad (198)$$

where

$$\eta_{j+\nu} = -\eta_j. \quad (199)$$

Hence

$$(\zeta\mathfrak{C} - \mathfrak{D}) = \mathfrak{C}^{\frac{1}{2}} \left(\zeta\mathbb{I}_{2\nu} - \mathfrak{C}^{-\frac{1}{2}}\mathfrak{D}\mathfrak{C}^{-\frac{1}{2}} \right) \mathfrak{C}^{\frac{1}{2}} = \mathfrak{C}^{\frac{1}{2}} \left(\zeta \sum_{1\le j\le 2\nu} \mathbf{v}_j\mathbf{v}_j^\dagger - \sum_{1\le j\le 2\nu} \eta_j\mathbf{v}_j\mathbf{v}_j^\dagger \right) \mathfrak{C}^{\frac{1}{2}} \quad (200)$$

$$= \mathfrak{C}^{\frac{1}{2}} \left(\sum_{1\le j\le 2\nu} (\zeta - \eta_j)\, \mathbf{v}_j\mathbf{v}_j^\dagger \right) \mathfrak{C}^{\frac{1}{2}}, \quad (201)$$

leading to

$$\left(\zeta\mathfrak{C}-\mathfrak{D}\right)^{-1}=\mathfrak{C}^{-\frac{1}{2}}\left(\sum_{1\leq j\leq 2\nu}\left(\zeta-\eta_j\right)^{-1}\mathbf{v}_j\mathbf{v}_j^{\dagger}\right)\mathfrak{C}^{-\frac{1}{2}} \tag{202}$$

and

$$R_{ijrs}\left(\zeta\right)=\begin{pmatrix}\frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\mathbf{z}} & \frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\bar{\mathbf{z}}}\end{pmatrix}\mathfrak{C}^{-\frac{1}{2}}\left(\sum_{1\leq j\leq 2\nu}\left(\zeta-\eta_j\right)^{-1}\mathbf{v}_j\mathbf{v}_j^{\dagger}\right)\mathfrak{C}^{-\frac{1}{2}}\begin{pmatrix}\frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\bar{\mathbf{z}}}\\ \frac{\partial\rho_{ji}(\bar{\varsigma}_0,\varsigma_0)}{\partial\mathbf{z}}\end{pmatrix}. \tag{203}$$

If the critical point $(\bar{\varsigma}_0,\varsigma_0)$ of $\mathcal{E}_0$ is a local minimum the rearranged Hessian $\mathfrak{D}$ is positive definite and is explicitly given by in terms of the coset generators as

$$\begin{aligned}&\begin{pmatrix}D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0\left(\bar{\varsigma}_0,\varsigma_0\right) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0\left(\bar{\varsigma}_0,\varsigma_0\right)\\ D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0\left(\bar{\varsigma}_0,\varsigma_0\right) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0\left(\bar{\varsigma}_0,\varsigma_0\right)\end{pmatrix}=\\ &\qquad\frac{1}{\mathcal{S}\left(\bar{\varsigma}_0,\varsigma_0\right)}\left\{\begin{pmatrix}\left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}^{\dagger}H_0\mathbf{Y}\Phi\left(\varsigma_0\right)\right\rangle & \left\langle\Phi\left(\varsigma_0\right)\right|H_0\mathbf{Y}\mathbf{Y}\Phi\left(\varsigma_0\right)\rangle\\ \left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}^{\dagger}\mathbf{Y}^{\dagger}H_0\Phi\left(\varsigma_0\right)\right\rangle & \overline{\left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}^{\dagger}H_0\mathbf{Y}\Phi\left(\varsigma_0\right)\rangle}\end{pmatrix}\right.\\ &\qquad\qquad\left.-\mathcal{E}_0\left(\bar{\varsigma}_0,\varsigma_0\right)\begin{pmatrix}\left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}^{\dagger}\mathbf{Y}\Phi\left(\varsigma_0\right)\right\rangle & \left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}\mathbf{Y}\Phi\left(\varsigma_0\right)\rangle\\ \left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}^{\dagger}\mathbf{Y}^{\dagger}\Phi\left(\varsigma_0\right)\right\rangle & \overline{\left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}^{\dagger}\mathbf{Y}\Phi\left(\varsigma_0\right)\rangle}\end{pmatrix}\right\}.\end{aligned} \tag{204}$$

The metric matrix $\mathfrak{C}$ is given by

$$\begin{aligned}&\begin{pmatrix}\mathbf{C}\left(\bar{\varsigma}_0,\varsigma_0\right) & \mathbf{0}\\ \mathbf{0} & -\bar{\mathbf{C}}\left(\bar{\varsigma}_0,\varsigma_0\right)\end{pmatrix}=\\ &\qquad\frac{1}{\mathcal{S}\left(\bar{\varsigma}_0,\varsigma_0\right)}\left\{\begin{pmatrix}\left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}^{\dagger}\mathbf{Y}\Phi\left(\varsigma_0\right)\right\rangle & \mathbf{0}\\ \mathbf{0} & -\overline{\left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}^{\dagger}\mathbf{Y}\Phi\left(\varsigma_0\right)\rangle}\end{pmatrix}\right.\\ &-\frac{1}{\mathcal{S}\left(\bar{\varsigma}_0,\varsigma_0\right)}\left.\begin{pmatrix}\left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}\Phi\left(\varsigma_0\right)\rangle\left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}^{\dagger}\Phi\left(\varsigma_0\right)\right\rangle & \mathbf{0}\\ \mathbf{0} & \overline{\left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}\Phi\left(\varsigma_0\right)\rangle\left\langle\Phi\left(\varsigma_0\right)\right|\mathbf{Y}^{\dagger}\Phi\left(\varsigma_0\right)\rangle}\end{pmatrix}\right\}.\end{aligned} \tag{205}$$

and the transformed Hessian

$$\mathfrak{C}^{-\frac{1}{2}}\mathfrak{D}\mathfrak{C}^{-\frac{1}{2}} = \begin{pmatrix} \mathbb{I}_\nu & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_\nu \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \overline{\mathbf{B}} & \overline{\mathbf{A}} \end{pmatrix} = \sum_{1\le j\le \nu} \eta_j \left\{ \mathbf{v}_j \mathbf{v}_j^\dagger - \mathbf{v}_{j+\nu}\mathbf{v}_{j+\nu}^\dagger \right\} = \sum_{1\le j\le 2\nu} \eta_j \mathbf{v}_j \mathbf{v}_j^\dagger = \mathbf{V}\eta\mathbf{V}^\dagger$$

$$= \begin{pmatrix} \mathbb{I}_\nu & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_\nu \end{pmatrix} \sum_{1\le j\le \nu} \eta_j \left\{ \mathbf{v}_j \mathbf{v}_j^\dagger + \mathbf{v}_{j+\nu}\mathbf{v}_{j+\nu}^\dagger \right\}$$

$$= \frac{1}{\mathcal{S}\left(\bar{\varsigma}_0, \varsigma_0\right)} \begin{pmatrix} \mathbb{I}_\nu & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_\nu \end{pmatrix} \left\{ \sum_{1\le j\le \nu} \left(\sigma_j - \mathcal{E}\left(\bar{\varsigma}_0, \varsigma_0\right)\right) \mathfrak{C}^{-\frac{1}{2}} \mathfrak{d}^{\frac{1}{2}} \left\{ \mathbf{y}_j \mathbf{y}_j^\dagger + \mathbf{y}_{j+\nu}\mathbf{y}_{j+\nu}^\dagger \right\} \mathfrak{d}^{\frac{1}{2}} \mathfrak{C}^{-\frac{1}{2}\dagger} \right\}$$

$$= \frac{1}{\mathcal{S}\left(\bar{\varsigma}_0, \varsigma_0\right)} \begin{pmatrix} \mathbb{I}_\nu & \mathbb{O} \\ \mathbb{O} & -\mathbb{I}_\nu \end{pmatrix} \mathfrak{C}^{-\frac{1}{2}} \mathfrak{d}^{\frac{1}{2}} \mathbf{Y} \left(\sigma - \mathcal{E}\left(\bar{\varsigma}_0, \varsigma_0\right) \mathbb{I}_{2\nu}\right) \mathbf{Y}^\dagger \mathfrak{d}^{\frac{1}{2}} \mathfrak{C}^{-\frac{1}{2}\dagger} \tag{206}$$

TO BE COMPLETED

C. Linear Response Described by AGP Coherent States .

We now use the manifold of AGP coherent states in which to approximate linear response functions. Within the context of the classical equations of motion in the associated phase space the linear response of the first order reduced density matrix produced by the radiation field is then given as

$$\delta^1 \rho_1 \left(\mathbf{z}_0, \bar{\mathbf{z}}_0\right)_{ij} (\xi) = \begin{pmatrix} \frac{\partial \rho_{ji}(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z}} & \frac{\partial \rho_{ji}(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}}} \end{pmatrix} \times$$
$$\left(\zeta \begin{pmatrix} \mathbf{0} & -\bar{\mathbf{C}}\left(\bar{\varsigma}_0, \varsigma_0\right) \\ \mathbf{C}\left(\bar{\varsigma}_0, \varsigma_0\right) & \mathbf{0} \end{pmatrix} - \begin{pmatrix} D^2_{\mathbf{z}\bar{\mathbf{z}}} \mathcal{E}_0 \left(\bar{\varsigma}_0, \varsigma_0\right) & D^2_{\mathbf{z}\mathbf{z}} \mathcal{E}_0 \left(\bar{\varsigma}_0, \varsigma_0\right) \\ D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}} \mathcal{E}_0 \left(\bar{\varsigma}_0, \varsigma_0\right) & D^2_{\bar{\mathbf{z}}\mathbf{z}} \mathcal{E}_0 \left(\bar{\varsigma}_0, \varsigma_0\right) \end{pmatrix} \right)^{-1}$$
$$\times \begin{pmatrix} \frac{\partial \rho_{ji}(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}}} & \mathbb{O} \\ \mathbb{O} & \frac{\partial \rho_{ji}(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z}} \end{pmatrix} \begin{pmatrix} \mathbf{f}(\zeta) \\ \overline{\mathbf{f}(\zeta)} \end{pmatrix}, \tag{207}$$

where

$$\frac{\partial \mathcal{F}\left(\bar{\varsigma}_0, \varsigma_0, \xi\right)}{\partial \mathbf{z}} = \sum_{1\le k,l\le \nu} \left. \frac{\partial \left\langle \Phi\left(\varsigma_0\right) \right| a_k^\dagger a_l \Phi\left(\varsigma_0\right) \right\rangle}{\partial \mathbf{z}} \right|_{\mathbf{z}=\mathbf{z}_0} f_{kl}(\zeta). \tag{208}$$

In general the off diagonal blocks $\left\{D^2_{\overline{\mathbf{z}}\overline{\mathbf{z}}}\mathcal{E}_0\left(\bar{\varsigma}_0,\varsigma_0\right), D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0\left(\bar{\varsigma}_0,\varsigma_0\right)\right\}$ are not zero. This can be seen by considering

$$\begin{aligned}\mathcal{E}_0\left(\bar{\varsigma}_0,\varsigma_0\right) = \Bigg\langle \exp\left\{\sum_{\substack{-s\le j<k\le s\\ |j|\ne|k|}} z_{+jk}q^\dagger_{jk}\right\}\exp\left\{\sum_{\substack{-s\le j<k\le s\\ |j|\ne|k|}}\lambda_{jk}u_{jk}\right\}g_0^{\frac{N}{2}}\Bigg| H_0 \\ \Bigg|\exp\left\{\sum_{\substack{-s\le j<k\le s\\ |j|\ne|k|}} z_{+jk}q^\dagger_{jk}\right\}\exp\left\{\sum_{\substack{-s\le j<k\le s\\ |j|\ne|k|}}\lambda_{jk}u_{jk}\right\}g_0^{\frac{N}{2}}\Bigg\rangle\end{aligned} \tag{209}$$

one can see that

$$\frac{\partial^2\mathcal{E}_0\left(\bar{\varsigma}_0,\varsigma_0\right)}{\partial\mathbf{z}_+\partial\mathbf{z}_+} = 0 \Rightarrow \Big\langle\, \Phi\left(\varsigma_0\right)\big|\, H_0 q^\dagger_{i_1j_1}\cdots q^\dagger_{i_mj_m}\Phi\left(\varsigma_0\right)\Big\rangle = 0 \quad \forall m\ge 2 \tag{210}$$

implies that $H_0\left(\overline{\mathbf{z}},\mathbf{z}\right)$ is first order in $\mathbf{z}$ and that H_0 is a linear function of $\left\{q^\dagger_{i_1j_1}\right\}$. From

$$\frac{\partial^2\mathcal{E}_0\left(\bar{\varsigma}_0,\varsigma_0\right)}{\partial\overline{\mathbf{z}}_+\partial\overline{\mathbf{z}}_+} = 0, \frac{\partial^2\mathcal{E}_0\left(\bar{\varsigma}_0,\varsigma_0\right)}{\partial\lambda\partial\lambda} = 0, \frac{\partial^2\mathcal{E}_0\left(\bar{\varsigma}_0,\varsigma_0\right)}{\partial\overline{\lambda}\partial\overline{\lambda}} = 0 \tag{211}$$

one reaches the same conclusion about the operators $\{q_{ij}\}$ and $\{\lambda_{ij}\}$. If this were true then

$$H_0 = E_0\left|g_0^{\frac{N}{2}}\right\rangle\left\langle g_0^{\frac{N}{2}}\right| + \sum_n\left(E_n - E_0\right)Q^\dagger_n\left|g_0^{\frac{N}{2}}\right\rangle\left\langle g_0^{\frac{N}{2}}\right|Q_n \tag{212}$$

where

$$Q^\dagger_n = \sum_{\substack{-s\le j<k\le s\\ |j|\ne|k|}} Q_{njk}q^\dagger_{jk} + \sum_{\substack{-s\le j<k\le s\\ |j|\ne|k|}} U_{njk}u_{jk} \tag{213}$$

Thus if the excited states were single excitations out of the reference state were exact, the off diagonal blocks would be identically zero and the Hamiltonian could be written as a bilinear form in the above excitation operators. In this case the quadratic expansion would be exact and Taylor series quadratic error would be zero. (Note that diagonalizing the Hessian does not accomplish this unless the off diagonal blocks are actually zero, as the eigen excitation operators would then be linear combinations of the excitation operators and their adjoints.) The slower the convergence of the Taylor series, i.e. the greater the quadratic error term, the poorer the approximation of excited states by single particle excitations out of the reference is.

Once the coherent state manifold is defined one is free to use any well defined coordinate system in which to find the critical points of the energy function i.e. classical Hamiltonian, for that manifold. We have published a procedure based on the coordinates based on the following parameterization of AGP states

$$\left|g^{\frac{N}{2}}\right\rangle = P_N \exp\left\{\sum_{1\leq i<j\leq r} g_{ij} a_i^\dagger a_j^\dagger\right\}|\phi\rangle \tag{214}$$

This procedure will be compared to ones based on coordinate system $\{\mathbf{z}\}$.

VIII. SUMMARY AND DISCUSSION

We have described how there are three possible chains of coset decompositions of the one particle unitary group $U(r)$ starting from maximal compact subgroups. Each of these chains produce families of cosets $\{U(r)/K\}$. A coset produces a manifold of coherent states from a reference state $|\Phi_{ref}\rangle$ if this state carries a one dimensional unitary representation of the subgroup K and the Hilbert space that $|\Phi_{ref}\rangle$ belongs to carries an irreducible representation of $U(r)$. As we are considering $N = 2M$ electron states this condition rules out cosets based on the maximal compact groups $SO(r)$ and $U(p)\times U(r-p)$ for $p \neq N$, leaving only $USp(2s)$ and $U(N)\times U(r-N)$. The subgroups $USp(2\omega_1)\times\cdots\times USp(2\omega_p)\times SU(2)^\sigma\times SU(2\xi)$ of $USp(2s)$ also satisfy this CS condition, but there are no subgroups of $U(N)\times U(r-N)$ that do. The reference state for all of these cosets is an AGP state belonging to the class (see section III) $\{\sigma,\omega_1,\cdots,\omega_\rho,\xi\}$ that is determined by the canonical coefficients $\{c_j\}$ of the geminal $|g\rangle$, this index set also classifies the possible isotropy subgroups and hence the cosets. The Lie algebra of $U(r)$ has different direct sum decompositions determined by the different subgroups generating the cosets. The manifolds of AGP CS's $\mathcal{M}(\sigma,\omega_1,\cdots,\omega_\rho,\xi)$ have complex coordinates systems $\mathbf{z}$ wrt the reference state dependent bases adapted to these Lie algebra decompositions and these coordinates parameterize AGP states. Different manifolds contain AGP states that describe different physical properties e.g. extreme AGP's have properties associated with highly correlated systems, such as superconductors.

The manifolds $\mathcal{M}(\sigma,\omega_1,\cdots,\omega_\rho,\xi)$ equipped with the coordinate system $\mathbf{z}$ are particularly useful as they have a natural generalized phase space structure and can be used to construct, via the TDVP, approximations of the TDSE as a classical dynamical system. If one used the manifold constituted by whole Hilbert space $\mathcal{H}^N$ (not an AGP manifold) then the classical

Hamilton equations plus an equation for a time dependent phase would be equivalent to the TDSE, and their solutions would provide solutions to the Schrödinger equation. The classical equations restricted to the AGP manifolds produce paths of AGP states that approximate the exact paths. One can consider the classical linear response of approximate stationary states found on these manifolds to external time dependent perturbations either within the curved phase or in the tangent space at these points [32]. This allows a variety of linear response approximations schemes to be developed that generalize and clarify propagator and Random Phase approximations.

APPENDIX A: MATRICES

The metric matrix $\mathbf{C}\left(\bar{z}\left(t\right),z\left(t\right)\right)=\left\{\frac{\partial^2\Lambda(\bar{z}(t),z(t))}{\partial\bar{z}_j\partial z_k}\right\}$ is given by

$$\mathbf{C}\left(\bar{z}\left(t\right),z\left(t\right)\right)=\frac{1}{\mathcal{S}\left(\bar{z}\left(t\right),z\left(t\right)\right)}\left\{\frac{\partial^2\mathcal{S}\left(\bar{z}\left(t\right),z\left(t\right)\right)}{\partial\bar{\mathbf{z}}\partial\mathbf{z}}\right.$$
$$\left.-\frac{1}{\mathcal{S}\left(\bar{z}\left(t\right),z\left(t\right)\right)}\frac{\partial\mathcal{S}\left(\bar{z}\left(t\right),z\left(t\right)\right)}{\partial\bar{\mathbf{z}}}\otimes\frac{\partial\mathcal{S}\left(\bar{z}\left(t\right),z\left(t\right)\right)}{\partial\mathbf{z}}\right\},\tag{A1}$$

and the gradient of the energy function $\mathcal{E}\left(\bar{z},z,t\right)$ has the form

$$D^1\mathcal{E}\left(\bar{z},z,t\right)=\begin{pmatrix}\frac{\partial\mathcal{E}(\bar{z},z,t)}{\partial\bar{\mathbf{z}}}\\ \frac{\partial\mathcal{E}(\bar{z},z,t)}{\partial\mathbf{z}}\end{pmatrix},\tag{A2}$$

where

$$\frac{\partial\mathcal{E}\left(\bar{z},z,t\right)}{\partial\mathbf{x}}=\frac{1}{\mathcal{S}\left(\bar{z}\left(t\right),z\left(t\right)\right)}\left\{\frac{\partial\mathcal{H}\left(\bar{z},z,t\right)}{\partial\mathbf{x}}-\mathcal{E}\left(\bar{z},z,t\right)\frac{\partial\mathcal{S}\left(\bar{z}\left(t\right),z\left(t\right)\right)}{\partial\mathbf{x}}\right\}\tag{A3}$$

for $\mathbf{x}=\mathbf{z},\bar{\mathbf{z}}$ and $\frac{\partial\mathcal{Y}(\bar{z},z,t)}{\partial\mathbf{x}}$ is a column vector with coordinates $\left\{\frac{\partial\mathcal{Y}(\bar{z},z,t)}{\partial x_k}\middle|\,\mathcal{Y}=\mathcal{H},\mathcal{S},\mathcal{E}\right\}$.

APPENDIX B: FUNCTIONS ON THE TANGENT SPACE

Functions $f:\mathcal{M}\rightarrow\mathbb{C}$ induce linear functions, the differentials $df\left[\mathbf{z}_0\right]:\mathcal{T}_{\mathbf{z}_0}\rightarrow\mathbb{C}$, that are the best linear approximations to f around the point $\mathbf{z}_0\in\mathcal{M}$. Physically this corresponds to classical observables f on a curved phase space $\mathcal{M}$ being locally approximated by the linear observables $df\left[\mathbf{z}_0\right]$ that act on a flat phase space $\mathcal{T}_{\mathbf{z}_0}$. As examples consider

(a) the holomorphic function

$$f_{\chi}(\mathbf{z}) = \langle \chi | \Phi(\mathbf{z}) \rangle \tag{B1}$$

then

$$df_{\chi}[\mathbf{z}_0](\mathbf{0}, \eta) = \sum_{1 \le j \le \nu} \eta_j \langle \chi | Y_j \Phi(\mathbf{z}_0) \rangle , \tag{B2}$$

where η are local coordinates i.e.

$$\eta = \mathbf{z} - \mathbf{z}_0 \tag{B3}$$

(b) the real valued function

$$f_{\mathrm{Re}\,\chi}(\bar{\mathbf{z}}, \mathbf{z}) = \langle \chi | \Phi(\mathbf{z}) \rangle + \langle \Phi(\mathbf{z}) | \chi \rangle \tag{B4}$$

then

$$df_{\mathrm{Re}\,\chi}[\mathbf{z}_0](\bar{\eta}, \eta) = \sum_{1 \le j \le \nu} \left\{ \eta_j \langle \chi | Y_j \Phi(\mathbf{z}_0) \rangle + \bar{\eta}_j \left\langle \Phi(\mathbf{z}_0) \middle| Y_j^{\dagger} \chi \right\rangle \right\} , \tag{B5}$$

(c) and a real valued classical observable

$$\mathcal{O}(\bar{\mathbf{z}}, \mathbf{z}) = \frac{\langle \Phi(\mathbf{z}) | O \Phi(\mathbf{z}) \rangle}{\langle \Phi(\mathbf{z}) | \Phi(\mathbf{z}) \rangle} \tag{B6}$$

then

$$\begin{aligned} d\mathcal{O}[\mathbf{z}_0](\bar{\eta}, \eta) = \sum_{1 \le j \le \nu} \frac{1}{\mathcal{S}(\bar{\mathbf{z}}_0, \mathbf{z}_0)} &\left\{ \eta_j \left\{ \langle \Phi(\mathbf{z}_0) | O Y_j \Phi(\mathbf{z}_0) \rangle - \mathcal{O}(\bar{\mathbf{z}}_0, \mathbf{z}_0) \langle \Phi(\mathbf{z}_0) | Y_j \Phi(\mathbf{z}_0) \rangle \right\} \right. \\ &\left. + \bar{\eta}_j \left\{ \left\langle \Phi(\mathbf{z}_0) \middle| Y_j^{\dagger} O \Phi(\mathbf{z}_0) \right\rangle - \mathcal{O}(\bar{\mathbf{z}}_0, \mathbf{z}_0) \left\langle \Phi(\mathbf{z}_0) \middle| Y_j^{\dagger} \Phi(\mathbf{z}_0) \right\rangle \right\} \right\} . \end{aligned} \tag{B7}$$

This can be expressed in matrix form as

$$d\mathcal{O}[\mathbf{z}_0](\bar{\eta}, \eta) = \begin{pmatrix} \frac{\partial \mathcal{O}(\bar{\mathbf{z}}_0, \mathbf{z}_0)}{\partial \bar{\mathbf{z}}} & \frac{\partial \mathcal{O}(\bar{\mathbf{z}}_0, \mathbf{z}_0)}{\partial \mathbf{z}} \end{pmatrix} \begin{pmatrix} \eta \\ \bar{\eta} \end{pmatrix} \tag{B8}$$

where

$$\begin{aligned} \frac{\partial \mathcal{O}(\bar{\mathbf{z}}_0, \mathbf{z}_0)}{\partial \mathbf{z}} &= \frac{1}{\mathcal{S}(\bar{\mathbf{z}}_0, \mathbf{z}_0)} \left\{ \langle \Phi(\mathbf{z}_0) | O \mathbf{Y} \Phi(\mathbf{z}_0) \rangle - \mathcal{O}(\bar{\mathbf{z}}_0, \mathbf{z}_0) \langle \Phi(\mathbf{z}_0) | \mathbf{Y} \Phi(\mathbf{z}_0) \rangle \right\} \\ \frac{\partial \mathcal{O}(\bar{\mathbf{z}}_0, \mathbf{z}_0)}{\partial \bar{\mathbf{z}}} &= \frac{1}{\mathcal{S}(\bar{\mathbf{z}}_0, \mathbf{z}_0)} \left\{ \left\langle \Phi(\mathbf{z}_0) \middle| \mathbf{Y}^{\dagger} O \Phi(\mathbf{z}_0) \right\rangle - \mathcal{O}(\bar{\mathbf{z}}_0, \mathbf{z}_0) \left\langle \Phi(\mathbf{z}_0) \middle| \mathbf{Y}^{\dagger} \Phi(\mathbf{z}_0) \right\rangle \right\} . \end{aligned} \tag{B9}$$

In general real analytic functions on $\mathcal{M}$ can be expanded as functions on any tangent space $\mathcal{T}_{\mathbf{z}_0}$ as

$$f\left(\overline{\mathbf{z}}_0+\overline{\eta},\mathbf{z}_0+\eta\right)=\sum_{1\leq n\leq\infty}D^n f\left[\overline{\mathbf{z}}_0,\mathbf{z}_0\right]\left(\overline{\eta},\eta\right) \tag{B10}$$

where

$$D^n f\left[\overline{\mathbf{z}}_0,\mathbf{z}_0\right]\left(\overline{\eta},\eta\right)=\sum_{p+q=n}\left(\frac{\partial^n f\left(\overline{\mathbf{z}}_0,\mathbf{z}_0\right)}{\partial^p\overline{\mathbf{z}}\partial^q\mathbf{z}}\right)\cdot\left(\overline{\eta}^{\otimes p}\otimes\eta^{\otimes q}\right) \tag{B11}$$

and

$$\left(\frac{\partial^n f\left(\overline{\mathbf{z}}_0,\mathbf{z}_0\right)}{\partial^p\overline{\mathbf{z}}\partial^q\mathbf{z}}\right)\cdot\left(\overline{\eta}^{\otimes p}\otimes\eta^{\otimes q}\right)=\frac{1}{p!q!}\sum_{\substack{1\leq j_1,\ldots,j_p\leq\nu\\1\leq k_1,\ldots,k_q\leq\nu}}\frac{\partial^n f\left(\overline{\mathbf{z}}_0,\mathbf{z}_0\right)}{\partial\overline{z}_{j_1}\cdots\partial\overline{z}_{j_p}\partial z_{k_1}\cdots\partial z_{k_q}}\overline{\eta}_{j_1}\cdots\overline{\eta}_{j_p}\eta_{k_1}\cdots\eta_{k_q}. \tag{B12}$$

Substituting $\eta=\mathbf{z}-\mathbf{z}_0$ in eq. $(B10)$ one obtains

$$f\left(\overline{\mathbf{z}},\mathbf{z}\right)=\sum_{1\leq n\leq\infty}D^n f\left[\overline{\mathbf{z}}_0,\mathbf{z}_0\right]\left(\overline{\mathbf{z}}-\overline{\mathbf{z}}_0,\mathbf{z}-\mathbf{z}_0\right) \tag{B13}$$

showing that analytic functions can be viewed as functions on any tangent space $\mathcal{T}_{\mathbf{z}_0}$. This is not unexpected as $\mathbb{C}^\nu$ is linearly isomorphic to $\mathcal{T}_{\mathbf{z}_0}$ for any $\mathbf{z}_0$ through the maps $d\Phi\left[\mathbf{z}_0\right]$

$$d\Phi\left[\mathbf{z}_0\right]:\mathbb{C}^\nu\rightarrow\mathcal{T}_{\mathbf{z}_0}^{(1,0)} \tag{B14}$$

and thus can be used as the flat coordinate space. Equations eq. $(B10)$ and eq. $(B13)$ lead to approximations defined as

$$f_M\left[\overline{\mathbf{z}}_0,\mathbf{z}_0\right]\left(\overline{\mathbf{z}},\mathbf{z}\right)=\sum_{1\leq n\leq M}D^n f\left[\overline{\mathbf{z}}_0,\mathbf{z}_0\right]\left(\overline{\mathbf{z}}-\overline{\mathbf{z}}_0,\mathbf{z}-\mathbf{z}_0\right). \tag{B15}$$

In particular

$$\mathcal{O}_2\left[\overline{\mathbf{z}}_0,\mathbf{z}_0\right]\left(\overline{\mathbf{z}},\mathbf{z}\right)=\sum_{1\leq n\leq 2}D^n\mathcal{O}\left[\overline{\mathbf{z}}_0,\mathbf{z}_0\right]\left(\overline{\mathbf{z}}-\overline{\mathbf{z}}_0,\mathbf{z}-\mathbf{z}_0\right) \tag{B16}$$

which can be equivalently expressed in terms of tangent space coordinates as

$$\mathcal{O}_2\left[\overline{\mathbf{z}}_0,\mathbf{z}_0\right]\left(\overline{\eta},\eta\right)=\sum_{1\leq n\leq 2}D^n\mathcal{O}\left[\overline{\mathbf{z}}_0,\mathbf{z}_0\right]\left(\overline{\eta},\eta\right). \tag{B17}$$

In an analogous fashion complex analytic functions can be expanded as

$$f\left(\mathbf{z}\right)=\sum_{1\leq n\leq\infty}D^n f\left[\mathbf{z}_0\right]\left(\mathbf{z}-\mathbf{z}_0\right)=\sum_{1\leq n\leq\infty}D^n f\left[\mathbf{z}_0\right]\left(\eta\right). \tag{B18}$$

APPENDIX C: MATRIX THEOREM

Theorem 1 *A hermitian matrix* $\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \overline{\mathbf{B}} & \overline{\mathbf{A}} \end{pmatrix}$ *has degenerate eigenvectors* $\begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}$ *and* $\begin{pmatrix} \overline{\mathbf{b}} \\ \overline{\mathbf{a}} \end{pmatrix}$.

Proof. *Let*

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \overline{\mathbf{B}} & \overline{\mathbf{A}} \end{pmatrix} \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix} = \lambda \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}; \qquad \lambda \in \mathbb{R} \tag{C1}$$

then

$$\begin{aligned} \mathbf{A}\mathbf{a} + \mathbf{B}\mathbf{b} &= \lambda \mathbf{a} \\ \overline{\mathbf{B}}\mathbf{a} + \overline{\mathbf{A}}\mathbf{b} &= \lambda \mathbf{b}. \end{aligned} \tag{C2}$$

Taking the complex conjugate of eq. (C2) we obtain

$$\begin{aligned} \overline{\mathbf{A}}\overline{\mathbf{a}} + \overline{\mathbf{B}\mathbf{b}} &= \lambda \overline{\mathbf{a}} \\ \mathbf{B}\overline{\mathbf{a}} + \mathbf{A}\overline{\mathbf{b}} &= \lambda \overline{\mathbf{b}}, \end{aligned} \tag{C3}$$

which corresponds to

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \overline{\mathbf{B}} & \overline{\mathbf{A}} \end{pmatrix} \begin{pmatrix} \overline{\mathbf{b}} \\ \overline{\mathbf{a}} \end{pmatrix} = \lambda \begin{pmatrix} \overline{\mathbf{b}} \\ \overline{\mathbf{a}} \end{pmatrix}. \tag{C4}$$

■

APPENDIX D: HERMITIAN FORM OF THE ENERGY HESSIAN

The Hessian of the classical Hamiltonian $\mathcal{E}_0(\bar{\mathbf{z}}, \bar{\mathbf{z}})$ at the point $(\bar{\varsigma}_0, \varsigma_0)$

$$\begin{pmatrix} D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) \\ D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) \end{pmatrix} \tag{D1}$$

can be transformed into a hermitian matrix, $\mathfrak{D}$, by premultiplying by $\begin{pmatrix} \mathbf{0} & \mathbf{1} \\ \mathbf{1} & \mathbf{0} \end{pmatrix}$ i.e.

$$\mathfrak{D} \equiv \begin{pmatrix} D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0 & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0 \\ D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0 & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0 \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \mathbf{1} \\ \mathbf{1} & \mathbf{0} \end{pmatrix} \begin{pmatrix} D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) \\ D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) \end{pmatrix}. \tag{D2}$$

If $(\bar{\varsigma}_0, \varsigma_0)$ is a critical point of the classical Hamiltonian $\mathcal{E}_0(\bar{\mathbf{z}}, \mathbf{z})$ then

$$\mathfrak{D} = \frac{1}{\mathcal{S}(\bar{\varsigma}_0, \varsigma_0)} \left\{ \begin{pmatrix} \frac{\partial^2 \mathcal{H}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{H}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \mathbf{z}} \\ \frac{\partial^2 \mathcal{H}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{H}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \mathbf{z}} \end{pmatrix} - \mathcal{E}(\bar{\varsigma}_0, \varsigma_0) \begin{pmatrix} \frac{\partial^2 \mathcal{S}(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{S}(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \mathbf{z}} \\ \frac{\partial^2 \mathcal{S}(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{S}(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \mathbf{z}} \end{pmatrix} \right\}$$

$$\equiv \frac{1}{\mathcal{S}(\bar{\varsigma}_0, \varsigma_0)} \left\{ \mathfrak{H} - \mathcal{E}(\bar{\varsigma}_0, \varsigma_0)\, \mathfrak{d} \right\}, \tag{D3}$$

where the positive definite metric matrix $\mathfrak{d}$ is defined as

$$\mathfrak{d} = \begin{pmatrix} \frac{\partial^2 \mathcal{S}(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{S}(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \mathbf{z}} \\ \frac{\partial^2 \mathcal{S}(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{S}(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \mathbf{z}} \end{pmatrix}. \tag{D4}$$

As

$$\left\{ \mathfrak{H} - \mathcal{E}(\bar{\varsigma}_0, \varsigma_0)\, \mathfrak{d} \right\} = \mathfrak{d}^{\frac{1}{2}} \left\{ \mathfrak{d}^{-\frac{1}{2}} \mathfrak{H} \mathfrak{d}^{-\frac{1}{2}} - \mathcal{E}(\bar{\varsigma}_0, \varsigma_0)\, \mathbb{I}_{2\nu} \right\} \mathfrak{d}^{\frac{1}{2}}, \tag{D5}$$

it is natural to consider the spectral expansions

$$\mathfrak{d}^{-\frac{1}{2}} \mathfrak{H} \mathfrak{d}^{-\frac{1}{2}} = \sum_{1 \le j \le \nu} \sigma_j \left\{ \mathbf{y}_j \mathbf{y}_j^\dagger + \mathbf{y}_{j+\nu} \mathbf{y}_{j+\nu}^\dagger \right\}; \qquad \eta_j \in \mathbb{R} \tag{D6}$$

and

$$\mathbb{I}_{2\nu} = \sum_{1 \le j \le \nu} \left\{ \mathbf{y}_j \mathbf{y}_j^\dagger + \mathbf{y}_{j+\nu} \mathbf{y}_{j+\nu}^\dagger \right\} \tag{D7}$$

to obtain

$$\left\{ \mathfrak{H} - \mathcal{E}(\bar{\varsigma}_0, \varsigma_0)\, \mathfrak{d} \right\} = \mathfrak{d}^{\frac{1}{2}} \left\{ \sum_{1 \le j \le \nu} \sigma_j \left\{ \mathbf{y}_j \mathbf{y}_j^\dagger + \mathbf{y}_{j+\nu} \mathbf{y}_{j+\nu}^\dagger \right\} - \mathcal{E}(\bar{\varsigma}_0, \varsigma_0) \sum_{1 \le j \le \nu} \left\{ \mathbf{y}_j \mathbf{y}_j^\dagger + \mathbf{y}_{j+\nu} \mathbf{y}_{j+\nu}^\dagger \right\} \right\} \mathfrak{d}^{\frac{1}{2}}$$

$$= \mathfrak{d}^{\frac{1}{2}} \left\{ \sum_{1 \le j \le \nu} \left(\sigma_j - \mathcal{E}(\bar{\varsigma}_0, \varsigma_0) \right) \left\{ \mathbf{y}_j \mathbf{y}_j^\dagger + \mathbf{y}_{j+\nu} \mathbf{y}_{j+\nu}^\dagger \right\} \right\} \mathfrak{d}^{\frac{1}{2}}. \tag{D8}$$

Thus producing the spectral expansion

$$\mathfrak{D} = \begin{pmatrix} D^2_{\mathbf{z}\bar{\mathbf{z}}} \mathcal{E}_0 & D^2_{\mathbf{z}\mathbf{z}} \mathcal{E}_0 \\ D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}} \mathcal{E}_0 & D^2_{\bar{\mathbf{z}}\mathbf{z}} \mathcal{E}_0 \end{pmatrix} = \frac{1}{\mathcal{S}(\bar{\varsigma}_0, \varsigma_0)} \mathfrak{d}^{\frac{1}{2}} \left\{ \sum_{1 \le j \le \nu} \left(\sigma_j - \mathcal{E}(\bar{\varsigma}_0, \varsigma_0) \right) \left\{ \mathbf{y}_j \mathbf{y}_j^\dagger + \mathbf{y}_{j+\nu} \mathbf{y}_{j+\nu}^\dagger \right\} \right\} \mathfrak{d}^{\frac{1}{2}}. \tag{D9}$$

APPENDIX E: PATH INTEGRALS

The probability, $P_{\Phi_i \to \Phi_f}(\tau)$, that a quantum process characterized by a Hamiltonian H starting from a given initial state $|\Phi_i\rangle$ at time $t = 0$ will be in a given arbitrary state $|\Phi_f\rangle$ at a time $t = \tau$ is

$$P_{\Phi_i \to \Phi_f}(\tau) = \left| \langle \Phi_f | U(\tau)\, \Phi_i \rangle \right|^2, \tag{E1}$$

where the evolution operator $U(t)$ satisfies the equation

$$\frac{d}{dt}U(t) = -iH(t)U(t) \tag{E2}$$

and produces solution paths $\{\psi(t)\}$ of states

$$\psi(t) = |\Psi(t)\rangle = T\left\{e^{-i\int_0^t H(t')dt'}\right\}|\Phi_i\rangle \tag{E3}$$

indexed by the initial values $\{|\Phi_i\rangle\}$ to the TDSE, where T is the time ordering operator. The transition amplitude $\langle\Phi_f|U(\tau)\,\Phi_i\rangle$ is called the propagator between the states $|\Phi_i\rangle$ and $|\Phi_f\rangle$. This transition amplitude can be expressed *formally* as a path integral over the space of paths

$$\mathfrak{P}(\Phi_i,\Phi_f,\tau) = \left\{\psi|, \psi(t)\in\mathcal{H}^N \quad t\in(0,\tau), \psi(0) = |\Phi_i\rangle, \psi(\tau) = |\Phi_f\rangle\right\} \tag{E4}$$

as a path integral

$$\langle\Psi_f|U(\tau)\,\Psi_i\rangle = \int_{\mathfrak{P}\left(\Phi_i,\Phi_f,\tau\right)} e^{i\mathcal{A}\left(\psi,\bar{\psi},\tau\right)}d\mu(\psi), \tag{E5}$$

where $d\mu(\psi)$ is a measure [?] on the space $\mathfrak{P}(\Phi_i,\Phi_f,\tau)$. The action, $\mathcal{A}$, is defined in terms of a Lagrangian, $\mathcal{L}$, as

$$\mathcal{A}\left(\psi,\bar{\psi},\tau\right) = \int_0^\tau \mathcal{L}\left(\psi,\bar{\psi},t\right)dt, \tag{E6}$$

where the Lagrangian is defined as

$$\mathcal{L}\left(\psi,\bar{\psi},t\right) = \frac{\left\langle\psi(t)\left|\left(i\frac{\partial}{\partial t} - H(t)\right)\psi(t)\right.\right\rangle}{\left\langle\psi(t)|\psi(t)\right\rangle} \equiv \mathcal{K}\left(\psi,\bar{\psi},t\right) - \mathcal{E}\left(\psi,\bar{\psi},t\right) \tag{E7}$$

and we have introduced a classical kinetic energy term $\mathcal{K}$ and Hamiltonian $\mathcal{E}$

$$\mathcal{K}\left(\psi,\bar{\psi},t\right) = \frac{\left\langle\psi(t)\left|i\frac{\partial}{\partial t}\psi(t)\right.\right\rangle}{\left\langle\psi(t)|\psi(t)\right\rangle} \tag{E8a}$$

$$\mathcal{E}\left(\psi,\bar{\psi},t\right) = \frac{\left\langle\psi(t)|H(t)\,\psi(t)\right\rangle}{\left\langle\psi(t)|\psi(t)\right\rangle}. \tag{E8b}$$

The notation used in eqs. (**??**) - (**??**) emphasizes the difference between the value of a path at a particular point, which is a vector in $\mathcal{H}^N$, and a path, which in this case, is a map from the closed real interval $[0,\tau]$ to $\mathcal{H}^N$. The path integral method has and is the focus of a huge literature containing discussions of technical details, mathematical rigor, practical implementations and computational strategies, all of which have been avoided in eqs. (**??**) – (**??**).

1. Path Integrals using Coherent States

One of the fundamental properties of a family of group CS's $\{|\Phi(\mathbf{z})\rangle\}$ associated with a coset G/K that form a manifold in the Hilbert space $\mathcal{H}^N$ is the existence of a resolution of the identity that is a consequence of the irreducibility of G on $\mathcal{H}^N$

$$I = \int |\Phi(\mathbf{z})\rangle \langle \Phi(\mathbf{z})| \, d\mu(\mathbf{z}), \tag{E9}$$

where

$$|\Phi(\mathbf{z})\rangle = C(\mathbf{z}) |\Phi_{ref}\rangle \tag{E10}$$

and $C(\mathbf{z})$ is parameterization of the representation of the coset G/K on $\mathcal{H}^N$. The parameters z belong to a complex manifold $\mathcal{M}$, in the simplest case these manifolds are finite dimensional vector spaces and $\mathbf{z} \equiv (z_1, \cdots, z_\nu)$, but in general they are complex domains [**?**] (restricted regions of $\mathbb{C}^\nu$), or even complex valued function spaces. If one pre and post multiplies the evolution operator $U(\tau)$ with the resolution eq. $(E9)$ we obtain

$$\begin{aligned} U(\tau) &= \int |\Phi(\mathbf{z}')\rangle \langle \Phi(\mathbf{z}') | U(\tau)\, \Phi(\mathbf{z})\rangle \langle \Phi(\mathbf{z})| \, d\mu(\mathbf{z}')\, d\mu(\mathbf{z}) \\ &\equiv \int P(\mathbf{z}', \mathbf{z}, \tau) |\Phi(\mathbf{z}')\rangle \langle \Phi(\mathbf{z})| \, d\mu(\mathbf{z}')\, d\mu(\mathbf{z}), \end{aligned} \tag{E11}$$

where

$$P(z', z, \tau) = \langle \Phi(z') | U(\tau)\, \Phi(z)\rangle \tag{E12}$$

form an overcomplete set of evolution operator matrix elements, also called the propagator kernel. Any transition amplitude $\langle \Phi_f | U(\tau)\, \Phi_i\rangle$ can be expressed in terms of this kernel and holomorphic wavefunctions [**?**] $\{\Phi_i(\mathbf{z}), \Phi_f(\mathbf{z})\}$, both of which obviously depend on the family of CS's being used, as

$$\langle \Phi_f | U(\tau)\, \Phi_i\rangle = \int P(\mathbf{z}', \mathbf{z}, \tau)\, \overline{\Phi_f(\mathbf{z}')} \Phi_i(\mathbf{z})\, d\mu(\mathbf{z}')\, d\mu(\mathbf{z}), \tag{E13}$$

where the holomorhpic wavefunction, $\Psi(\mathbf{z})$, of $|\Psi\rangle$ wrt the CS family $\{|\Phi(\mathbf{z})\rangle\}$ is defined as

$$\Psi(\mathbf{z}) = \langle \Phi(\mathbf{z}) | \Psi\rangle. \tag{E14}$$

Using the path integral construction the propagator kernel becomes an integral over parameter paths $\{\mathrm{z}\}$ with fixed end points, $\mathrm{z}(0) = \mathbf{z}_i; \quad \mathrm{z}(\tau) = \mathbf{z}_f$

$$P(\mathbf{z}_f, \mathbf{z}_i, \tau) = \int_{\mathfrak{P}(\mathbf{z}_i, \mathbf{z}_f, \tau)} e^{i\mathcal{A}(\bar{\mathrm{z}}, \mathrm{z}, \tau)} d\mu(\mathrm{z}), \tag{E15}$$

where the space of parameter paths is

$$\mathfrak{P}\left(\mathbf{z}_i, \mathbf{z}_f, \tau\right) = \left\{ \mathrm{z} \right| , \mathrm{z}\left(t\right) \in \mathcal{M}, \quad t \in \left[0, \tau\right], \mathrm{z}\left(0\right) = \mathbf{z}_i, \mathrm{z}\left(\tau\right) = \mathbf{z}_f \right\}, \tag{E16}$$

the action

$$\mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) = \int_0^\tau \mathcal{L}\left(\bar{\mathrm{z}}, \mathrm{z}, t\right) dt \tag{E17}$$

and the Lagrangian is defined as

$$\mathcal{L}\left(\bar{\mathrm{z}}, \mathrm{z}, t\right) = \frac{\left\langle \Phi\left(\mathrm{z}\left(t\right)\right) \middle| \left(i\frac{\partial}{\partial t} - H\left(t\right)\right) \Phi\left(\mathrm{z}\left(t\right)\right) \right\rangle}{\left\langle \Phi\left(\mathrm{z}\left(t\right)\right) \middle| \Phi\left(\mathrm{z}\left(t\right)\right) \right\rangle} \equiv \mathcal{K}\left(\bar{\mathrm{z}}, \mathrm{z}, t\right) - \mathcal{E}\left(\bar{\mathrm{z}}, \mathrm{z}, t\right). \tag{E18}$$

From this definition one can observe that the Lagrangian $\mathcal{L}$ can be considered as a function of $\mathrm{v}\left(t\right) = \dot{\mathrm{z}}\left(t\right)$, so that

$$\mathcal{L} = \mathcal{L}\left(\bar{\mathrm{z}}, \mathrm{z}, \bar{\mathrm{v}}, \mathrm{v}, t\right), \tag{E19}$$

and $\left\{\bar{\mathrm{z}}, \mathrm{z}, \bar{\mathrm{v}}, \mathrm{v}\right\}$ can be considered as independent variables that determine $\mathcal{L}$. This is important in the derivation of equations of motion for $\left\{\bar{\mathrm{z}}, \mathrm{z}\right\}$ resulting from the Principle of Stationary Action $\delta^1 \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) = 0$, (see any text discussing the calculus of variations e.g. [**?**]). For notational convenience we will not explicitly show the dependence of $\mathcal{L}$ on $\mathrm{v}\left(t\right)$.

2. Stationary Phase Approximation

The action functional $\mathcal{A}$ can be expanded around any path z in terms of the incremental paths $\delta \mathrm{z} \in \mathfrak{P}_0\left(\mathbf{z}_i, \mathbf{z}_f, \tau\right) \subset \mathfrak{P}\left(\mathbf{z}_i, \mathbf{z}_f, \tau\right)$, that are zero at the end points i.e. $\delta \mathrm{z}\left(0\right) = \delta \mathrm{z}\left(\tau\right) = 0$, as

$$\mathcal{A}\left(\bar{\mathrm{z}} + \delta\bar{\mathrm{z}}, \mathrm{z} + \delta\mathrm{z}, \tau\right) = \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) + \delta^1 \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) + \delta^2 \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) + \cdots. \tag{E20}$$

The first and second order changes in the action are given by

$$\delta^1 \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) = \begin{pmatrix} D_{\bar{\mathrm{z}}} \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) & D_{\mathrm{z}} \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) \end{pmatrix} \begin{pmatrix} \delta\bar{\mathrm{z}} \\ \delta\mathrm{z} \end{pmatrix} \tag{E21}$$

$$\delta^2 \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) = \frac{1}{2} \begin{pmatrix} \delta\bar{\mathrm{z}} & \delta\mathrm{z} \end{pmatrix} \begin{pmatrix} D^2_{\bar{\mathrm{z}}\bar{\mathrm{z}}} \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) & D^2_{\bar{\mathrm{z}}\mathrm{z}} \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) \\ D^2_{\mathrm{z}\bar{\mathrm{z}}} \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) & D^2_{\mathrm{z}\mathrm{z}} \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, \tau\right) \end{pmatrix} \begin{pmatrix} \delta\bar{\mathrm{z}} \\ \delta\mathrm{z} \end{pmatrix}, \tag{E22}$$

where the gradient $D_{\mathrm{x}} \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, t\right)$ and Hessian $D^2_{\mathrm{xy}} \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, t\right)$ of the action are composed of the functional derivatives

$$\begin{aligned} D_{\mathrm{x}} \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, t\right) &= \left\{ \frac{\delta \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, t\right)}{\delta \mathrm{x}\left(t\right)} \right\}; \mathrm{x}\left(t\right) = \bar{\mathrm{z}}\left(t\right), \mathrm{z}\left(t\right), \\ D^2_{\mathrm{xy}} \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, t\right) &= \left\{ \frac{\delta^2 \mathcal{A}\left(\bar{\mathrm{z}}, \mathrm{z}, t\right)}{\delta \mathrm{x}\left(t\right) \delta \mathrm{y}\left(t\right)} \right\}; \mathrm{x}\left(t\right) = \bar{\mathrm{z}}\left(t\right), \mathrm{z}\left(t\right); \mathrm{y}\left(t\right) = \bar{\mathrm{z}}\left(t\right), \mathrm{z}\left(t\right), \end{aligned} \tag{E23}$$

which are defined by

$$D_{\mathrm{x}}\mathcal{A}\left(\bar{\mathrm{z}},\mathrm{z},\tau\right)=\lim_{\eta\to 0}\frac{1}{\eta}\mathcal{A}\left(\bar{\mathrm{z}}+\eta\delta\mathrm{x},\mathrm{z}+\eta\delta\mathrm{x},\tau\right). \tag{E24}$$

The matrix product in eq. $(E22)$ is defined in terms of integration as

$$D_{\bar{\mathrm{z}}}\mathcal{A}\left(\bar{\mathrm{z}},\mathrm{z},\tau\right)\left(\delta\bar{\mathrm{z}}\right)=\int_0^\tau\frac{\delta\mathcal{A}\left(\bar{\mathrm{z}},\mathrm{z},t\right)}{\delta\bar{\mathrm{z}}\left(t\right)}\delta\bar{\mathrm{z}}\left(t\right)dt. \tag{E25}$$

The paths $\zeta\in\mathfrak{P}\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right)$ that make the action stationary to all incremental variation, $\delta\mathrm{z}\in\mathfrak{P}_0\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right)$, i.e.

$$\delta^1\mathcal{A}\left(\bar{\zeta},\zeta,\tau\right)=\begin{pmatrix} D_{\bar{\mathrm{z}}}\mathcal{A}\left(\bar{\zeta},\zeta,\tau\right) & D_{\mathrm{z}}\mathcal{A}\left(\bar{\zeta},\zeta,\tau\right)\end{pmatrix}\begin{pmatrix}\delta\bar{\mathrm{z}}\\ \delta\mathrm{z}\end{pmatrix}=0\quad\forall\delta\bar{\mathrm{z}},\delta\mathrm{z} \tag{E26}$$

are called classical paths.

The changes in the action can also be given in terms of the first and second order variations of the Lagrangian as

$$\begin{aligned}\delta^1\mathcal{A}\left(\bar{\mathrm{z}},\mathrm{z},\tau\right)&=\int_0^\tau\delta^1\mathcal{L}\left(\bar{\mathrm{z}},\mathrm{z},t\right)dt\\ \delta^2\mathcal{A}\left(\bar{\mathrm{z}},\mathrm{z},\tau\right)&=\int_0^\tau\delta^2\mathcal{L}\left(\bar{\mathrm{z}},\mathrm{z},t\right)dt.\end{aligned} \tag{E27}$$

where at a fixed time t, the Lagrangian $\mathcal{L}$ is considered to be a function of the independent variables $z=\mathrm{z}\left(t\right)$, $\bar{z}=\bar{\mathrm{z}}\left(t\right)$, $v=\mathrm{v}\left(t\right)$ and $\bar{v}=\bar{\mathrm{v}}\left(t\right)$, i.e. the values of the path and path velocities, in contrast to the action that is a function of the complete path. This leads to the following expressions for the variations of the Lagrangian as

$$\delta^1\mathcal{L}\left(\bar{z},z,t\right)=\begin{pmatrix} D_{\bar{\mathbf{z}}}\mathcal{L}\left(\bar{z},z,t\right) & D_{\bar{v}}\mathcal{L}\left(\bar{z},z,t\right) & D_z\mathcal{L}\left(\bar{z},z,t\right) & D_v\mathcal{L}\left(\bar{z},z,t\right)\end{pmatrix}\begin{pmatrix}\delta\bar{z}\\ \delta\bar{v}\\ \delta z\\ \delta v\end{pmatrix} \tag{E28}$$

$$\begin{aligned}\delta^2\mathcal{L}\left(\bar{z},z,t\right)=&\frac{1}{2}\begin{pmatrix}\delta\bar{z} & \delta\bar{v} & \delta z & \delta v\end{pmatrix}\times\\ &\begin{pmatrix} D^2_{zz}\mathcal{L}\left(\bar{z},z,t\right) & D^2_{z\bar{v}}\mathcal{L}\left(\bar{z},z,t\right) & D^2_{zz}\mathcal{L}\left(\bar{z},z,t\right) & D^2_{zv}\mathcal{L}\left(\bar{z},z,t\right)\\ D^2_{vz}\mathcal{L}\left(\bar{z},z,t\right) & D^2_{v\bar{v}}\mathcal{L}\left(\bar{z},z,t\right) & D^2_{vz}\mathcal{L}\left(\bar{z},z,t\right) & D^2_{vv}\mathcal{L}\left(\bar{z},z,t\right)\\ D^2_{zz}\mathcal{L}\left(\bar{z},z,t\right) & D^2_{z\bar{v}}\mathcal{L}\left(\bar{z},z,t\right) & D^2_{zz}\mathcal{L}\left(\bar{z},z,t\right) & D^2_{zv}\mathcal{L}\left(\bar{z},z,t\right)\\ D^2_{vz}\mathcal{L}\left(\bar{z},z,t\right) & D^2_{v\bar{v}}\mathcal{L}\left(\bar{z},z,t\right) & D^2_{vz}\mathcal{L}\left(\bar{z},z,t\right) & D^2_{vv}\mathcal{L}\left(\bar{z},z,t\right)\end{pmatrix}\begin{pmatrix}\delta\bar{z}\\ \delta\bar{v}\\ \delta z\\ \delta v\end{pmatrix},\end{aligned} \tag{E29}$$

where

$$D_{\mathbf{x}}\mathcal{L}\left(\bar{z},z,\bar{v},v,t\right)\left(\delta\mathbf{x}\right)=\frac{\partial\mathcal{L}\left(\bar{z},z,\bar{v},v,t\right)}{\partial\mathbf{x}}\delta\mathbf{x}=\sum_{1\leq j\leq\nu}\frac{\partial\mathcal{L}\left(\bar{z},z,\bar{v},v,t\right)}{\partial x_j}\delta x_j;\quad \mathbf{x}=\mathbf{z},\bar{\mathbf{z}},\mathbf{v},\bar{\mathbf{v}}. \tag{E30}$$

If one assumes that the action, $\mathcal{A}$, is an analytical functional then one can expand it as a convergent power series about any point $\mathrm{z}\in\mathfrak{P}\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right)$ in terms of functionals,$\{\delta^j\mathcal{A}\}$, of $\{\delta\bar{\mathrm{z}},\delta\mathrm{z}\}$ as

$$\mathcal{A}\left(\bar{\mathrm{z}}+\delta\bar{\mathrm{z}},\mathrm{z}+\delta\mathrm{z},\tau\right)=\sum_{0\leq j\leq\infty}\delta^j\mathcal{A}\left(\delta\bar{\mathrm{z}},\delta\mathrm{z}\right), \tag{E31}$$

where we dropped explicit reference to the time interval τ in the variations. The path integral eq. $(E33)$ can then be reformulated as

$$\begin{aligned}\int_{\mathfrak{P}\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right)}e^{i\mathcal{A}(\bar{\mathrm{z}},\mathrm{z},\tau)}d\mu\left(\mathrm{z}\right)&=\int_{\mathfrak{P}_0\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right)}\exp\left\{i\sum_{0\leq j\leq\infty}\delta^j\mathcal{A}\left(\delta\bar{\mathrm{z}},\delta\mathrm{z}\right)\right\}d\mu\left(\delta\mathrm{z}\right)\\&=e^{i\mathcal{A}\left(\bar{\zeta},\zeta,\tau\right)}\int_{\mathfrak{P}_0\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right)}\exp\left\{i\sum_{1\leq j\leq\infty}\delta^j\mathcal{A}\left(\delta\bar{\mathrm{z}},\delta\mathrm{z}\right)\right\}d\mu\left(\delta\mathrm{z}\right).\end{aligned} \tag{E32}$$

Using the theory of oscillatory integrals the path integral eq. $(E15)$ can be asymptotically approximated [**?**] as

$$\int_{\mathfrak{P}\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right)}e^{i\mathcal{A}(\bar{\mathrm{z}},\mathrm{z},\tau)}d\mu\left(\mathrm{z}\right)\approx\sum_{\lambda}K\left(\bar{\zeta}_\lambda,\zeta_\lambda\right)e^{i\mathcal{A}\left(\bar{\zeta}_\lambda,\zeta_\lambda,\tau\right)} \tag{E33}$$

where $\{\zeta_\lambda\}$ are stationary, i.e. classical, paths of $\mathcal{A}\left(\bar{\mathrm{z}},\mathrm{z},\tau\right)$, $\mathrm{z}\in\mathfrak{P}\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right)$, and are thus stationary phases of the exponent in the integral eq. $(E15)$. Comparing eq. $(E32)$ and eq. $(E33)$ one can observe that the prefactor term can be expressed as

$$K\left(\bar{\zeta}_\lambda,\zeta_\lambda\right)=\int_{\mathfrak{P}_0\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right)}\exp\left\{i\sum_{2\leq j\leq\infty}\delta^j\mathcal{A}\left(\delta\bar{\mathrm{z}}_\lambda,\delta\mathrm{z}_\lambda\right)\right\}d\mu\left(\delta\mathrm{z}_\lambda\right), \tag{E34}$$

where we have split the domain of the integral eq. $(E32)$ into a union of neighborhoods, $\mathfrak{P}_{0\lambda}\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right)$, about stationary points $\{\zeta_\lambda\}$ and hence into a sum of integrals in terms of the incremental paths $\{\delta\mathrm{z}_\lambda\}$

$$\delta\mathrm{z}_\lambda=\mathrm{z}-\zeta_\lambda;\quad \mathrm{z}\in\mathfrak{P}\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right), \tag{E35}$$

and expansions about stationary points.

The second order term, $K^{(2)}\left(\bar{\zeta}_\lambda,\zeta_\lambda\right)$, in the expansion eq. $(E34)$ can be exactly evaluated as a Gaussian integral to give [**?**]

$$K^{(2)}\left(\bar{\zeta}_\lambda,\zeta_\lambda\right)=\int_{\mathfrak{P}_{0\lambda}\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right)}\exp\left\{i\delta^2_{\zeta_\lambda}\mathcal{A}\left(\delta\bar{\mathrm{z}}_\lambda,\delta\mathrm{z}_\lambda\right)\right\}d\mu\left(\delta\mathrm{z}_\lambda\right)=\sqrt{\det\left(i\frac{\partial^2\mathcal{A}\left(\bar{\zeta}_\lambda,\zeta_\lambda,\tau\right)}{\partial\mathbf{z}_f\partial\mathbf{z}_i}\right)},\tag{E36}$$

where the Hessian is with respect to *variation* of the end points of the paths, evaluated at the *fixed* endpoints $\{\mathbf{z}_i,\mathbf{z}_f\}$. In general it is not possible to obtain closed expressions for higher order terms in eq. $(E34)$. The Stationary Phase Approximation (SPA) to the path integral is defined to be the quadratic approximation given by

$$\int_{\mathfrak{P}\left(\mathbf{z}_i,\mathbf{z}_f,\tau\right)}e^{i\mathcal{A}(\bar{\mathrm{z}},\mathrm{z},\tau)}d\mu\left(\mathrm{z}\right)\approx\sum_\lambda K^{(2)}\left(\bar{\zeta}_\lambda,\zeta_\lambda\right)e^{i\mathcal{A}\left(\bar{\zeta}_\lambda,\zeta_\lambda,\tau\right)}\tag{E37}$$

where

$$\delta\mathcal{A}\left(\bar{\zeta}_\lambda,\zeta_\lambda,\tau\right)=0.\tag{E38}$$

In order to obtain the exact N-particle propagator one needs to express the path integral over the manifold of $SU\left(\mathcal{H}^N\right)/U\left(1\right)$-CS's, $\{|\Phi\left(\mathbf{z}\right)\rangle\}$, that form the unit sphere

$$\||\Psi\rangle\|=1\tag{E39}$$

in $\mathcal{H}^N$ (see section **??**). Thus the integral eq. $(E15)$ ranges over all possible paths of normalized vectors in $\mathcal{H}^N$ that begin at $|\Phi\left(\mathbf{z}_i\right)\rangle$ and end at $|\Phi\left(\mathbf{z}_f\right)\rangle$ with duration τ. In terms of these paths the functional $\mathcal{A}$ and the function $\mathcal{L}$ are quadratic and the second order expansions are *exact* thus making the SPA exact.

The SPA uses paths, with fixed end points $\mathrm{z}\left(\tau\right)=\mathbf{z}_f$, $\mathrm{z}\left(0\right)=\mathbf{z}_i$, that satisfy the stationary action condition $\delta\mathcal{A}\left(\bar{\zeta}_\lambda,\zeta_\lambda,\tau\right)=0$ given by solutions to the generalized classical equation eq. (110) that can be expressed as

$$-i\sum C_{jk}\frac{d\mathrm{z}_k}{dt}=\frac{\partial\mathcal{E}}{\partial\bar{\mathrm{z}}_j}\tag{E40}$$

$$i\sum\bar{C}_{jk}\frac{d\bar{\mathrm{z}}_k}{dt}=\frac{\partial\mathcal{E}}{\partial z_j}\tag{E41}$$

over the phase space formed by the manifold, $\mathcal{M}$, of states $\{|\Phi\left(z\right)\rangle\}$ and are evolution paths for the classical Hamiltonian $\mathcal{E}$. If the Lagrangian is quadratic the classical paths $\{\zeta_\lambda\}$ produce an exact propagator

$$P\left(\mathbf{z}_f,\mathbf{z}_i;\tau\right)=\sum_\lambda\sqrt{\det\left(i\frac{\partial^2\mathcal{A}\left(\bar{\zeta}_\lambda,\zeta_\lambda,\tau\right)}{\partial\mathbf{z}_f\partial\mathbf{z}_i}\right)}e^{i\mathcal{A}\left(\bar{\zeta}_\lambda,\zeta_\lambda,\tau\right)},\tag{E42}$$

otherwise truncating at second order produces an approximate propagator. One should note that the degree of the Lagrangian is determined by the degree of the classical Hamiltonian $\mathcal{E}$.

The SPA approximation can be improved by using a higher order approximations to the classical Hamiltonian around stationary paths, produced by considering more terms of the Taylor series. Clearly the nature of the manifold $\mathcal{M}$ controls the nature and order properties of the series expansion of $\mathcal{L}$, so determines the rate of convergence. When $\mathcal{M}$ is the unit sphere in $\mathcal{H}^N$ the expansion is quadratic, while if $\mathcal{M}$ is a smaller manifold the expansions are of higher degree and the rate of convergence is determined by the nature of the states constituting the manifold.

APPENDIX F: ORTHOGONAL MOTION

If the norm of the states $|\Phi(\mathrm{z}(t))\rangle$ are constrained to be normalized i.e.

$$\mathcal{S}(\bar{\mathrm{z}}(t), \mathrm{z}(t)) = \langle \Phi(\mathrm{z}(t)) | \Phi(\mathrm{z}(t)) \rangle = 1 \quad \forall t \tag{F1}$$

then

$$\frac{d}{dt} \langle \Phi(\mathrm{z}(t)) | \Phi(\mathrm{z}(t)) \rangle = \left\langle \dot{\Phi}(\mathrm{z}(t)) | \Phi(\mathrm{z}(t)) \right\rangle + \left\langle \Phi(\mathrm{z}(t)) | \dot{\Phi}(\mathrm{z}(t)) \right\rangle = 0, \tag{F2}$$

which gives in the case $U\left(\mathcal{H}^N\right)/U(1)$-CS's

$$\frac{d\mathrm{z}(t)}{dt}^{\dagger} \mathrm{z}(t) + \mathrm{z}(t)^{\dagger} \frac{d\mathrm{z}(t)}{dt} = \begin{pmatrix} \frac{d\bar{\mathrm{z}}(t)}{dt} & \frac{d\mathrm{z}(t)}{dt} \end{pmatrix} \begin{pmatrix} \mathrm{z}(t) \\ \bar{\mathrm{z}}(t) \end{pmatrix} = 0. \tag{F3}$$

A sufficient condition for this to hold is that

$$\begin{pmatrix} \mathbf{0} & \mathbf{P}_{\mathrm{z}(t)} \\ \mathbf{P}_{\bar{\mathrm{z}}(t)} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \frac{d\bar{\mathrm{z}}(t)}{dt} \\ \frac{d\mathrm{z}(t)}{dt} \end{pmatrix} = 0, \tag{F4}$$

which of course is equivalent to

$$\begin{pmatrix} \mathbf{0} & \mathbf{P}^{\perp}_{\mathrm{z}(t)} \\ \mathbf{P}^{\perp}_{\bar{\mathrm{z}}(t)} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \frac{d\bar{\mathrm{z}}(t)}{dt} \\ \frac{d\mathrm{z}(t)}{dt} \end{pmatrix} = \begin{pmatrix} \frac{d\bar{\mathrm{z}}(t)}{dt} \\ \frac{d\mathrm{z}(t)}{dt} \end{pmatrix}. \tag{F5}$$

The classical equations of motion eq. (110) for the evolution of for $SU\left(\mathcal{H}^N\right)/U(1)$-CS's is

$$\frac{1}{\mathcal{S}(\bar{\mathrm{z}}(t), \mathrm{z}(t))} \begin{pmatrix} \mathbf{0} & -i\mathbf{P}^{\perp}_{\mathrm{z}(t)} \\ i\mathbf{P}^{\perp}_{\bar{\mathrm{z}}(t)} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \frac{d\bar{\mathrm{z}}(t)}{dt} \\ \frac{d\mathrm{z}(t)}{dt} \end{pmatrix} = \begin{pmatrix} \frac{\partial \mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial \bar{\mathbf{z}}} \\ \frac{\partial \mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial \mathbf{z}} \end{pmatrix}, \tag{F6}$$

which is equivalent to

$$\begin{pmatrix} i\frac{d\bar{\mathrm{z}}(t)}{dt} \\ -i\frac{d\mathrm{z}(t)}{dt} \end{pmatrix} = \begin{pmatrix} \frac{\partial\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial\mathbf{z}} \\ \frac{\partial\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial\bar{\mathbf{z}}} \end{pmatrix} \tag{F7}$$

and eq. $(F5)$ if

$$\mathrm{z}\,(t)^{\dagger}\,\frac{\partial\mathcal{E}\,(\bar{\mathrm{z}},\mathrm{z},t)}{\partial\mathbf{z}} = 0 \quad \forall t. \tag{F8}$$

The condition expressed in eq. $(F8)$ is satisfied as

$$\begin{aligned}
\begin{pmatrix} \frac{\partial\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial\bar{\mathbf{z}}} \\ \frac{\partial\mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial\mathbf{z}} \end{pmatrix} &= \frac{1}{\mathcal{S}\,(\bar{\mathrm{z}}\,(t)\,,\mathrm{z}\,(t))} \begin{pmatrix} \left\{\frac{\partial\mathcal{H}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial\bar{\mathbf{z}}} - \mathcal{E}\,(\bar{\mathrm{z}},\mathrm{z},t)\,\frac{\partial\mathcal{S}(\bar{\mathrm{z}}(t),\mathrm{z}(t))}{\partial\bar{\mathbf{z}}}\right\} \\ \left\{\frac{\partial\mathcal{H}(\bar{\mathrm{z}},\mathrm{z},t)}{\partial\mathbf{z}} - \mathcal{E}\,(\bar{\mathrm{z}},\mathrm{z},t)\,\frac{\partial\mathcal{S}(\bar{\mathrm{z}}(t),\mathrm{z}(t))}{\partial\mathbf{z}}\right\} \end{pmatrix} \\
&= \frac{1}{\mathcal{S}\,(\bar{\mathrm{z}}\,(t)\,,\mathrm{z}\,(t))} \begin{pmatrix} \mathbb{O} & \mathbb{H}\,(t) - \mathcal{E}\,(\bar{\mathrm{z}},\mathrm{z},t)\,\mathbb{I} \\ \bar{\mathbb{H}}\,(t) - \mathcal{E}\,(\bar{\mathrm{z}},\mathrm{z},t)\,\mathbb{I} & \mathbb{O} \end{pmatrix} \begin{pmatrix} \bar{\mathrm{z}}\,(t) \\ \mathrm{z}\,(t) \end{pmatrix},
\end{aligned} \tag{F9}$$

where the matrix $\mathbb{H}\,(t)$ is the time dependent full CI matrix given by

$$H\,(t)_{KL} = \langle\Phi_K|H\,(t)\,\Phi_L\rangle\,, \tag{F10}$$

and hence

$$\mathrm{z}\,(t)^{\dagger}\,(\mathbb{H}\,(t) - \mathcal{E}\,(\bar{\mathrm{z}},\mathrm{z},t)\,\mathbb{I})\,\mathrm{z}\,(t) = 0. \tag{F11}$$

$$= -i\sum_{r,s}\int_0^t \left\{\frac{\partial\mathcal{A}\,(\mathbf{z}_c,\bar{\mathbf{z}}_c)}{\partial\bar{\mathbf{z}}} e^{-i\left(D^2\mathcal{H}_0(\mathbf{z}_c,\bar{\mathbf{z}}_c)-\mathcal{E}_0(\mathbf{z}_c,\bar{\mathbf{z}}_c)D^2\mathcal{S}(\mathbf{z}_c,\bar{\mathbf{z}}_c)\right)t}\frac{\partial\rho_{sr}\,(\mathbf{z}_c,\bar{\mathbf{z}}_c)}{\partial\mathbf{z}}\right. \tag{F12}$$

$$\left. -\frac{\partial\rho_{sr}\,(\mathbf{z}_c,\bar{\mathbf{z}}_c)}{\partial\bar{\mathbf{z}}} e^{i\left(D^2\mathcal{H}_0(\mathbf{z}_c,\bar{\mathbf{z}}_c)-\mathcal{E}_0(\mathbf{z}_c,\bar{\mathbf{z}}_c)D^2\mathcal{S}(\mathbf{z}_c,\bar{\mathbf{z}}_c)\right)t}\frac{\partial\mathcal{A}\,(\mathbf{z}_c,\bar{\mathbf{z}}_c)}{\partial\mathbf{z}}\right\} f_{rs}\,(t')\,dt'. \tag{F13}$$

APPENDIX G: VELOCITY FORM OF TANGENT VECTORS.

Let Φ^{-1} be a complex coordinate chart for the ν-dimensional manifold $\mathcal{M}$ i.e.

$$\Phi^{-1} : \mathcal{M} \to \mathbb{C}^{\nu} \tag{G1}$$

and

$$\Phi : \mathbb{C}^{\nu} \to \mathcal{M}. \tag{G2}$$

Let $\mathcal{C}^{\infty}\,(\mathcal{M})$ be the space of smooth complex valued function, i.e. infinitely differentiable, defined on $\mathcal{M}$ and $\Phi\,(\mathrm{z}\,(t))$ be a path in $\mathcal{M}$ i.e.

$$\Phi \circ \mathrm{z} : \mathbb{R} \to \mathcal{M}, \tag{G3}$$

where $\mathrm{z}(t) \in \mathbb{C}^{\nu}$. Two paths, $\Phi(\mathrm{z}_1(t)), \Phi(\mathrm{z}_2(t))$ that pass through the point $\Phi(\mathbf{z}_0)$ at $t = 0$ i.e. $\Phi(\mathrm{z}_1(0)), \Phi(\mathrm{z}_2(0)) = \Phi(\mathbf{z}_0)$ are tangent at this point if the velocities of the corresponding $\mathbb{C}^{\nu}$ paths are equal i.e.

$$\frac{d\mathrm{z}_1(0)}{dt} = \frac{d\mathrm{z}_2(0)}{dt}. \tag{G4}$$

The set of all paths tangent at a point form an equivalence class and the set of equivalence classes form a linear space called the tangent space $\mathcal{T}^{(1,0)}_{\mathbf{z}_0}$ at the point $\Phi(\mathbf{z}_0)$. A tangent vector, $X_{\mathrm{z}(0)}$, belonging to the tangent space $\mathcal{T}^{(1,0)}_{\mathbf{z}_0}$ acts on the space, $\mathcal{C}^{\infty}(\mathcal{M})$, of infinitely differentiable complex value functions, f, defined on $\mathcal{M}$ to produce

$$X_{\mathrm{z}(0)}(f) = \frac{df(\Phi(\mathrm{z}(0)))}{dt} \equiv \frac{df(\mathrm{z}(0))}{dt} \quad \forall f \in \mathcal{C}^{\infty}(\mathcal{M}), \tag{G5}$$

where $\mathbf{z}_0 = \mathrm{z}(t_0)$ and $f(\mathrm{z}(t))$ implicitly means $f(\Phi(\mathrm{z}(t)))$. The path in eq. $(G5)$ is a representative of the equivalence class produced by the relationship eq. $(G4)$. Eq. (**??**) can be developed to give

$$X_{\mathrm{z}(0)}(f) = \frac{df(\Phi(\mathrm{z}(0)))}{dt} = \sum_{1\leq j\leq v} \frac{\partial f(\Phi(\mathrm{z}(0)))}{\partial z_j}\frac{d\mathrm{z}_j(0)}{dt} \quad \forall f \in \mathcal{C}^{\infty}(\mathcal{M}), \tag{G6}$$

which gives

$$\begin{aligned} X_{\mathrm{z}(0)}(f) &= \sum_{1\leq j\leq v} \frac{d\mathrm{z}_j(0)}{dt}\frac{\partial f(\Phi(\mathrm{z}(0)))}{\partial z_j} \equiv \left\{\sum_{1\leq j\leq v} \frac{d\mathrm{z}_j(0)}{dt}\frac{\partial}{\partial z_j}\right\} f(\Phi(\mathrm{z}(0))) \\ &\equiv \left\{\sum_{1\leq j\leq v} \frac{d\mathrm{z}_j(0)}{dt}\frac{\partial}{\partial z_j}\right\} f(\mathrm{z}(0)) \quad \forall f \in \mathcal{C}^{\infty}(\mathcal{M}) \end{aligned} \tag{G7}$$

Eq. $(G7)$ shows that tangent vectors can also be defined as

$$X_{\mathbf{z}_0} = \sum_{1\leq j\leq v} \alpha_j \frac{\partial}{\partial z_j} \qquad \alpha_j \in \mathbb{C}, \quad 1 \leq j \leq \nu \tag{G8}$$

and act on $\mathcal{C}^{\infty}(\mathcal{M})$ by

$$X_{\mathbf{z}_0}(f) = \left\{\sum_{1\leq j\leq v} \alpha_j \frac{\partial}{\partial z_j}\right\} f(\mathbf{z}_0) \quad \forall f \in \mathcal{C}^{\infty}(\mathcal{M}). \tag{G9}$$

If we let the basis of $\mathbb{C}^{\nu}$ be $\{\mathbf{e}_j | 1 \leq j \leq \nu\}$, giving the expansion

$$\mathbf{z} = \sum_{1\leq j\leq \nu} z_j \mathbf{e}_j, \tag{G10}$$

we can consider paths $\mathrm{e}_j(t)$, with coordinates $\{\mathrm{e}_{jk}(t)\}$ defined as

$$\mathrm{e}_j(t) = \mathbf{z}_0 + t\mathbf{e}_j = (\mathrm{e}_{j1}(t), \cdots, \mathrm{e}_{jj}(t), \cdots, \mathrm{e}_{j\nu}(t)) = (z_{01}, \cdots, z_{0j-1}, z_{0j} + t, z_{0j+1}, \cdots, z_{0\nu}) \tag{G11}$$

producing tangent vectors

$$X_{\mathrm{e}_j(0)}(f) = \frac{df(\Phi(\mathrm{e}_j(0)))}{dt} = \sum_{1\leq j\leq v} \frac{\partial f(\Phi(\mathbf{z}_0))}{\partial z_k}\frac{d\mathrm{e}_{jk}(0)}{dt} = \frac{\partial f(\mathbf{z}_0)}{\partial z_j}, \tag{G12}$$

showing that tangent basis vectors can be expressed equivalently as

$$X_{\mathrm{e}_j(t_0)} \equiv \frac{\partial}{\partial z_j} \qquad 1 \leq j \leq \nu. \tag{G13}$$

In particular eq. $(G12)$ must hold for linear functions $l \in \mathcal{C}_L^\infty(\mathcal{M}) \subset \mathcal{C}^\infty(\mathcal{M})$ that belong to the cotangent space $\mathcal{T}_{\mathbf{z}_0}^{(1,0)*}$ giving that

$$X_{\mathrm{e}_j(0)}(l) = \left.\frac{dl(\Phi(\mathrm{e}_j(t)))}{dt}\right|_{t=0} = l\left(\left.\frac{d\Phi(\mathrm{e}_j(t))}{dt}\right|_{t=0}\right) \equiv l\left(\frac{d\Phi(\mathrm{e}_j(0))}{dt}\right)$$
$$| \in \mathbb{C} \quad \forall l \in \mathcal{C}_L^\infty(\mathcal{M}); 1 \leq j \leq \nu \tag{G14}$$

and equivalently

$$\frac{\partial l(\Phi(\mathbf{z}_0))}{\partial z_j} = l\left(\frac{\partial \Phi(\mathbf{z}_0)}{\partial z_j}\right) \in \mathbb{C} \quad \forall l \in \mathcal{C}_L^\infty(\mathcal{M}); 1 \leq j \leq \nu. \tag{G15}$$

Hence one can conclude that

$$\left\{ \left. \frac{\partial \Phi(\mathbf{z}_0)}{\partial z_j} \right| 1 \leq j \leq \nu \right\} \tag{G16}$$

forms a basis for $\mathcal{T}_{\mathbf{z}_0}^{(1,0)}$.

The basis given in eq. $(G16)$ allows one to consider the tangent space $\mathcal{T}_{\mathbf{z}_0}^{(1,0)}$ in terms of the differential, $d\Phi[\mathbf{z}_0]$, of the inverse of the coordinate chart

$$\Phi : \mathbb{C}^\nu \to \mathcal{M} \tag{G17}$$

at the point $\mathbf{z}_0 \in \mathbb{C}^\nu$ which maps $\mathbb{C}^\nu \to \mathcal{T}_{\mathbf{z}_0}$. The differential, $d\Phi[\mathbf{z}_0]$, maps any $\delta\mathbf{z} \in \mathbb{C}^\nu$ to the tangent space $\mathcal{T}_{\mathbf{z}_0}^{(1,0)}$ producing

$$d\Phi[\mathbf{z}_0](\delta\mathbf{z}) = \sum_{1\leq j\leq v} \frac{\partial \Phi(\mathbf{z}_0)}{\partial z_j}\delta z_j \quad \in \mathcal{T}_{\mathbf{z}_0}. \tag{G18}$$

APPENDIX H: SOLUTION TO EOM

The EOM in matrix form for paths on the manifold $\mathcal{M}$ is

$$\begin{pmatrix} \dot{\bar{z}}(t) \\ \dot{z}(t) \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \overline{\mathbf{C}(\bar{z}(t), z(t))}^{-1} \\ -i\mathbf{C}(\bar{z}(t), z(t))^{-1} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \frac{\partial \mathcal{E}_0(\bar{z}(t), z(t))}{\partial \bar{\mathbf{z}}} \\ \frac{\partial \mathcal{E}_0(\bar{z}(t), z(t))}{\partial \mathbf{z}} \end{pmatrix}. \tag{H1}$$

In general $\mathcal{E}_0$ can be expanded about a point $(\bar{\varsigma}_0, \varsigma_0)$ as

$$\mathcal{E}_0(\bar{\mathbf{z}}, \mathbf{z}) = \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) + D^1\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)(\bar{\mathbf{z}} - \bar{\varsigma}_0, \mathbf{z} - \varsigma_0) + \frac{1}{2}D^2\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)(\bar{\mathbf{z}} - \bar{\varsigma}_0, \mathbf{z} - \varsigma_0) + \cdots, \tag{H2}$$

where

$$D^1_\varsigma\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)(\bar{\mathbf{z}} - \bar{\varsigma}_0, \mathbf{z} - \varsigma_0) = \begin{pmatrix} \frac{\partial \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}}} & \frac{\partial \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z}} \end{pmatrix} \begin{pmatrix} \bar{\mathbf{z}} - \bar{\varsigma}_0 \\ \mathbf{z} - \varsigma_0 \end{pmatrix} \tag{H3}$$

and

$$D^2_\varsigma\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)(\bar{\mathbf{z}} - \bar{\mathbf{z}}_0, \mathbf{z} - \mathbf{z}_0) = \begin{pmatrix} \bar{\mathbf{z}} - \bar{\varsigma}_0 & \mathbf{z} - \varsigma_0 \end{pmatrix} \begin{pmatrix} \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}}\partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}}\partial \mathbf{z}} \\ \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z}\partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z}\partial \mathbf{z}} \end{pmatrix} \begin{pmatrix} \bar{\mathbf{z}} - \bar{\varsigma}_0 \\ \mathbf{z} - \varsigma_0 \end{pmatrix}. \tag{H4}$$

If we truncate $\mathcal{E}_0(\bar{\mathbf{z}}, \mathbf{z})$ in eq. $(H2)$ at second order to form $\mathcal{E}_L(\bar{\mathbf{z}}, \mathbf{z})$

$$\mathcal{E}_L(\bar{\mathbf{z}}, \mathbf{z}) = \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0) + D^1_\varsigma\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)(\bar{\mathbf{z}} - \bar{\varsigma}_0, \mathbf{z} - \varsigma_0) + \frac{1}{2}D^2_\varsigma\mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)(\bar{\mathbf{z}} - \bar{\varsigma}_0, \mathbf{z} - \varsigma_0) \tag{H5}$$

we obtain a linearized gradient

$$D^1_{\mathbf{z}}\mathcal{E}_L(\bar{\mathbf{z}}, \mathbf{z}) = \begin{pmatrix} \frac{\partial \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}}} \\ \frac{\partial \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z}} \end{pmatrix} + \begin{pmatrix} \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}}\partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}}\partial \mathbf{z}} \\ \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z}\partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z}\partial \mathbf{z}} \end{pmatrix} \begin{pmatrix} \bar{\mathbf{z}} - \bar{\varsigma}_0 \\ \mathbf{z} - \varsigma_0 \end{pmatrix}. \tag{H6}$$

Then the linear approximation to the EOM becomes

$$\begin{pmatrix} \dot{\bar{z}}_L(t) \\ \dot{z}_L(t) \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \overline{\mathbf{C}(\bar{\varsigma}_0, \varsigma_0)}^{-1} \\ -i\mathbf{C}(\bar{\varsigma}_0, \varsigma_0)^{-1} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \frac{\partial \mathcal{E}_L(\bar{z}_L(t), z_L(t))}{\partial \bar{\mathbf{z}}} \\ \frac{\partial \mathcal{E}_L(\bar{z}_L(t), z_L(t))}{\partial \mathbf{z}} \end{pmatrix} \tag{H7}$$

giving the following equation for paths in the tangent space $\mathcal{T}_{\varsigma_0}$

$$\begin{pmatrix} \dot{\bar{z}}_L(t) \\ \dot{z}_L(t) \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \overline{\mathbf{C}(\bar{\varsigma}_0, \varsigma_0)}^{-1} \\ -i\mathbf{C}(\bar{\varsigma}_0, \varsigma_0)^{-1} & \mathbf{0} \end{pmatrix} \times$$
$$\left\{ \begin{pmatrix} \frac{\partial \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}}} \\ \frac{\partial \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z}} \end{pmatrix} + \begin{pmatrix} \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}}\partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \bar{\mathbf{z}}\partial \mathbf{z}} \\ \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z}\partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0, \varsigma_0)}{\partial \mathbf{z}\partial \mathbf{z}} \end{pmatrix} \right\} \begin{pmatrix} \bar{z}_L(t) - \bar{\varsigma}_0 \\ z_L(t) - \varsigma_0 \end{pmatrix} \tag{H8}$$

which we can abbreviate to

$$\dot{\mathrm{X}}(t) \equiv C^{-1} D_{\varsigma}^{1} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) + C^{-1} D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) \mathrm{X}(t), \tag{H9}$$

leading to

$$\dot{\mathrm{X}}(t) \equiv C^{-1} D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) \left\{ D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0)^{-1} D_{\varsigma}^{1} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) + \mathrm{X}(t) \right\}. \tag{H10}$$

We can then make the substitution

$$\mathrm{Y}(t) = D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0)^{-1} D_{\varsigma}^{1} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) + \mathrm{X}(t) \tag{H11}$$

obtaining

$$\dot{\mathrm{Y}}(t) = C^{-1} D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) \mathrm{Y}(t) \tag{H12}$$

which has the solution

$$\mathrm{Y}(t) = \left\{ \exp C^{-1} D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) t \right\} \mathrm{Y}(0). \tag{H13}$$

This gives

$$D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0)^{-1} D_{\varsigma}^{1} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) + \mathrm{X}(t) = \left\{ \exp C^{-1} D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) t \right\} \left\{ D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0)^{-1} D_{\varsigma}^{1} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) + \mathrm{X}(0) \right\} \tag{H14}$$

leading to

$$\begin{aligned} \mathrm{X}(t) &= \left\{ \exp C^{-1} D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) t \right\} \left\{ D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0)^{-1} D_{\varsigma}^{1} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) + \mathrm{X}(0) \right\} - D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0)^{-1} D_{\varsigma}^{1} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) \\ &\equiv \left\{ \exp C^{-1} D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) t \right\} \left\{ \mathrm{X}(0) - X_{\min} \right\} + X_{\min}, \end{aligned} \tag{H15}$$

where

$$X_{\min} = -D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0)^{-1} D_{\varsigma}^{1} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) \tag{H16}$$

If we choose

$$\mathrm{X}(0) = \begin{pmatrix} \bar{\mathrm{z}}_L(0) - \bar{\varsigma}_0 \\ \mathrm{z}_L(0) - \varsigma_0 \end{pmatrix} = \mathbf{0}, \tag{H17}$$

so that the initial point of the path z_L is ς_0 i.e.

$$\begin{pmatrix} \bar{\mathrm{z}}_L(0) \\ \mathrm{z}_L(0) \end{pmatrix} = \begin{pmatrix} \bar{\varsigma}_0 \\ \varsigma_0 \end{pmatrix}, \tag{H18}$$

then this becomes

$$\mathrm{X}(t) = X_{\min} - \exp\left\{ C(\bar{\varsigma}_0, \varsigma_0)^{-1} D_{\varsigma}^{2} \mathcal{E}_0 (\bar{\varsigma}_0, \varsigma_0) t \right\} X_{\min} \tag{H19}$$

and

$$\begin{pmatrix} \overline{\mathrm{z}}_L(t) \\ \mathrm{z}_L(t) \end{pmatrix} =$$

$$\left\{ \exp \begin{pmatrix} \mathbf{0} & \overline{\mathbf{C}(\bar{\varsigma}_0, \varsigma_0)}^{-1} \\ -i\mathbf{C}(\bar{\varsigma}_0, \varsigma_0)^{-1} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \mathbf{z}} \\ \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \mathbf{z}} \end{pmatrix} t \right\}$$

$$\times \begin{pmatrix} \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \mathbf{z}} \\ \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \mathbf{z}} \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}}} \\ \frac{\partial \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z}} \end{pmatrix}$$

$$- \begin{pmatrix} \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \mathbf{z}} \\ \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \mathbf{z}} \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}}} \\ \frac{\partial \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z}} \end{pmatrix} + \begin{pmatrix} \bar{\varsigma}_0 \\ \varsigma_0 \end{pmatrix}. \tag{H20}$$

This shows that

$$\dot{\mathrm{X}}(t) = \begin{pmatrix} \dot{\bar{\mathrm{z}}}_L(t) - \bar{\varsigma}_0 \\ \dot{\mathrm{z}}_L(t) - \varsigma_0 \end{pmatrix} = \mathbf{0} \tag{H21}$$

$$\Rightarrow \begin{pmatrix} \dot{\bar{\mathrm{z}}}_L(t) \\ \dot{\mathrm{z}}_L(t) \end{pmatrix} = \begin{pmatrix} \bar{\varsigma}_0 \\ \varsigma_0 \end{pmatrix} \tag{H22}$$

at a critical point $\begin{pmatrix} \bar{\varsigma}_0 \\ \varsigma_0 \end{pmatrix}$ of $\mathcal{E}_0$.

If $\begin{pmatrix} \bar{\varsigma}_0 \\ \varsigma_0 \end{pmatrix}$ is a critical point of $\mathcal{E}_0$ then

$$D^1_{\mathbf{z}} \mathcal{E}_L(\bar{\mathbf{z}}, \mathbf{z}) = \begin{pmatrix} \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \mathbf{z}} \\ \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \mathbf{z}} \end{pmatrix} \begin{pmatrix} \bar{\mathbf{z}} - \bar{\varsigma}_0 \\ \mathbf{z} - \varsigma_0 \end{pmatrix} \tag{H23}$$

and the linearized EOM for any classical observable becomes

$$\begin{pmatrix} \dot{\overline{\mathrm{O}}}_L(t) \\ \dot{\mathrm{O}}_L(t) \end{pmatrix} =$$

$$\begin{pmatrix} \frac{\partial O}{\partial \bar{\mathbf{z}}} & \frac{\partial O}{\partial \mathbf{z}} \end{pmatrix} \begin{pmatrix} \mathbf{0} & \overline{\mathbf{C}(\bar{\varsigma}_0, \varsigma_0)}^{-1} \\ -i\mathbf{C}(\bar{\varsigma}_0, \varsigma_0)^{-1} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \bar{\mathbf{z}} \partial \mathbf{z}} \\ \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \bar{\mathbf{z}}} & \frac{\partial^2 \mathcal{E}_0(\bar{\varsigma}_0,\varsigma_0)}{\partial \mathbf{z} \partial \mathbf{z}} \end{pmatrix} \begin{pmatrix} \overline{\mathrm{z}}_L(t) - \bar{\varsigma}_0 \\ \mathrm{z}_L(t) - \varsigma_0 \end{pmatrix} \tag{H24}$$

APPENDIX I: THE EXACT TDSE IN CS FORM

In order to facilitate a direct comparison of approximations to exact linear response functions it is useful to derive the exact linear response function in terms of coherent states based on the coset $SU\left(\mathcal{H}^N\right)/U\left(1\right)$. First we describe the TDSE in this setting. In this case the overlap kernel, $\mathcal{S}\left(\bar{\mathbf{z}},\mathbf{z}\right)$, and the classical Hamiltonian, $\mathcal{E}\left(\bar{\mathbf{z}},\mathbf{z},t\right)$, are defined for an arbitrary quantum Hamiltonian $H\left(t\right)$ as

$$\begin{aligned}\mathcal{S}\left(\bar{\mathbf{z}},\mathbf{z}\right) &= \left\langle \Phi\left(\mathbf{z}\right)|\Phi\left(\mathbf{z}\right)\right\rangle = \sum_{1\leq K\leq\binom{r}{N}} \left|z_k\right|^2 \\ \mathcal{E}\left(\bar{\mathbf{z}},\mathbf{z},t\right) &= \frac{\left\langle \Phi\left(\mathbf{z}\right)|H\left(t\right)\Phi\left(\mathbf{z}\right)\right\rangle}{\left\langle \Phi\left(\mathbf{z}\right)|\Phi\left(\mathbf{z}\right)\right\rangle} = \frac{\sum_{1\leq k,l\leq\binom{r}{N}} \bar{z}_k z_l \left\langle \Phi_k|H\left(t\right)\Phi_l\right\rangle}{\sum_{1\leq k\leq\binom{r}{N}}\left|z_k\right|^2} \\ &\equiv \frac{\sum_{1\leq k,l\leq\binom{r}{N}} \bar{z}_k z_l H_{kl}\left(t\right)}{\sum_{1\leq k\leq\binom{r}{N}}\left|z_k\right|^2}\end{aligned} \tag{I1}$$

and the Kähler metric matrix for this coset manifold is given by

$$\begin{aligned}C_{kl}\left(\bar{\mathbf{z}},\mathbf{z}\right) = \frac{\partial^2 \ln\mathcal{S}\left(\bar{\mathbf{z}},\mathbf{z}\right)}{\partial\bar{z}_k\partial z_l} &= \frac{1}{\mathcal{S}\left(\bar{\mathbf{z}},\mathbf{z}\right)}\left\{\frac{\partial^2\mathcal{S}\left(\bar{\mathbf{z}},\mathbf{z}\right)}{\partial\bar{z}_k\partial z_l} - \frac{1}{\mathcal{S}\left(\bar{\mathbf{z}},\mathbf{z}\right)}\frac{\partial\mathcal{S}\left(\bar{\mathbf{z}},\mathbf{z}\right)}{\partial\bar{z}_k}\frac{\partial\mathcal{S}\left(\bar{\mathbf{z}},\mathbf{z}\right)}{\partial z_l}\right\} \\ &= \frac{1}{\mathcal{S}\left(\bar{\mathbf{z}},\mathbf{z}\right)}\left\{\delta_{kl} - \frac{z_k\bar{z}_l}{\mathcal{S}\left(\bar{\mathbf{z}},\mathbf{z}\right)}\right\},\end{aligned} \tag{I2}$$

which can be written as

$$\mathbf{C}\left(\bar{\mathbf{z}},\mathbf{z}\right) = \frac{1}{\mathcal{S}\left(\bar{\mathbf{z}},\mathbf{z}\right)}\left\{\mathbf{I}-\mathbf{P}_z\right\} = \frac{1}{\mathcal{S}\left(\bar{\mathbf{z}},\mathbf{z}\right)}\mathbf{P}_z^{\perp}, \tag{I3}$$

where $\mathbf{P}_z$ is the orthogonal projector onto the vector $\mathbf{z}\in\mathbb{C}^{\binom{r}{N}}$. Clearly this metric matrix is singular, with a one dimensional kernel, and does not define a symplectic form on the coset manifold, however it does define a non degenerate symplectic form on the manifold that is perpendicular to $\mathbf{z}$. Evolutions determined by the EOM eq. (110) constrained to this manifold conserve the norm (see Appendix F) of states. Hence all states on an evolution path are normalized if the initial state is normalized i.e.

$$\mathcal{S}\left(\bar{\mathrm{z}}\left(t\right),\mathrm{z}\left(t\right)\right) = \mathcal{S}\left(\bar{\mathrm{z}}\left(0\right),\mathrm{z}\left(0\right)\right) = 1. \tag{I4}$$

The Poisson Bracket corresponding to this constrained dynamics is defined by the generalized inverse of the metric matrix resulting in the equations of motion

$$\begin{pmatrix}\frac{d\bar{z}(t)}{dt}\\ \frac{dz(t)}{dt}\end{pmatrix} = \begin{pmatrix}\mathbf{0} & i\mathbf{P}_z^{\perp}\\ -i\mathbf{P}_{\bar{z}}^{\perp} & \mathbf{0}\end{pmatrix}\begin{pmatrix}\frac{\partial\mathcal{E}(\bar{z},z,t)}{\partial\bar{\mathbf{z}}}\\ \frac{\partial\mathcal{E}(\bar{z},z,t)}{\partial\mathbf{z}}\end{pmatrix} = \begin{pmatrix}-i\frac{\partial\mathcal{E}(\bar{z},z,t)}{\partial\mathbf{z}}\\ i\frac{\partial\mathcal{E}(\bar{z},z,t)}{\partial\bar{\mathbf{z}}}\end{pmatrix}, \tag{I5}$$

that become

$$\begin{pmatrix} i\frac{d\bar{\mathrm{z}}(t)}{dt} \\ -i\frac{d\mathrm{z}(t)}{dt} \end{pmatrix} = \begin{pmatrix} \mathbb{O} & \bar{\mathbb{H}}(t) - \mathcal{E}(\bar{\mathrm{z}}, \mathrm{z},t)\,\mathbb{I} \\ \mathbb{H}(t) - \mathcal{E}(\bar{\mathrm{z}}, \mathrm{z},t)\,\mathbb{I} & \mathbb{O} \end{pmatrix} \begin{pmatrix} \mathrm{z}(t) \\ \bar{\mathrm{z}}(t) \end{pmatrix}. \tag{I6}$$

The diagonal blocks of the Hessian $D^2\mathcal{E}(\bar{\mathrm{z}}, \mathrm{z},t)$ are zero for case of $U\left(\mathcal{H}^N\right)/U(1)$-CS's and one observes that

$$\begin{pmatrix} D^2_{\bar{\mathbf{z}}\bar{\mathbf{z}}}\mathcal{E}(\bar{\mathrm{z}}, \mathrm{z},t) & D^2_{\bar{\mathbf{z}}\mathbf{z}}\mathcal{E}(\bar{\mathrm{z}}, \mathrm{z},t) \\ D^2_{\mathbf{z}\bar{\mathbf{z}}}\mathcal{E}(\bar{\mathrm{z}}, \mathrm{z},t) & D^2_{\mathbf{z}\mathbf{z}}\mathcal{E}(\bar{\mathrm{z}}, \mathrm{z},t) \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \bar{\mathbb{H}}(t) - \mathcal{E}(\bar{\mathrm{z}}, \mathrm{z},t)\,\mathbb{I} \\ \mathbb{H}(t) - \mathcal{E}(\bar{\mathrm{z}}, \mathrm{z},t)\,\mathbb{I} & \mathbf{0} \end{pmatrix}, \tag{I7}$$

but for other manifolds this is not in general true.

The formal solution of eq. $(I6)$ is given by

$$\mathrm{z}(t) = Te^{-i\int_0^t\{\mathbb{H}(t') - \mathcal{E}(\bar{\mathrm{z}},\mathrm{z},t')\mathbb{I}\}dt'}\mathrm{z}(0), \tag{I8}$$

where T is the time ordering operator. However, the time dependent CI coefficients vector, $\mathrm{z}(t)$, given by eq. $(I8)$ that determines the state

$$|\Phi(\mathrm{z}(t))\rangle = \sum_j \mathrm{z}_j(t)\,|\Psi_j\rangle, \tag{I9}$$

where $\{|\Psi_j\rangle\}$ is a fixed orthonormal basis of $\mathcal{H}^N$, is *not* a solution of the TDSE. In order to produce a solution one must multiply the CI coefficients, and thus the state, by a time dependent phase to obtain coefficients

$$\eta(t) = e^{-i\int_0^t \mathcal{E}(\bar{z},z,t)dt'}\mathrm{z}(t) \tag{I10}$$

and state

$$|\Psi(\eta(t))\rangle = e^{-i\int_0^t \mathcal{E}(\bar{z},z,t)dt'}\,|\Phi(\mathrm{z}(t))\rangle. \tag{I11}$$

From eq. $(I8)$ one can see that

$$|\Psi(\eta(t))\rangle = e^{-i\int_0^t \mathcal{E}(\bar{z},z,t)dt'}\,|\Phi(\mathrm{z}(t))\rangle = Te^{-i\int_0^t H(t')dt'}\,|\Phi(\mathrm{z}(0))\rangle, \tag{I12}$$

showing that

$$\sigma(t) = \int_0^t \mathcal{E}(\bar{\mathrm{z}}, \mathrm{z},t)\,dt' \tag{I13}$$

is the solution to the time dependent phase equation eq. (111).

The initial state, $|\Phi(\mathrm{z}(0))\rangle$, is usually chosen to be a stationary state, $|\Psi_0\rangle$, of the unperturbed time independent Hamiltonian H_0, which correspond to a critical point $\mathbf{z}_c$ of $\mathcal{E}_0(\bar{\mathbf{z}},\mathbf{z})$ as

$$\frac{d\mathrm{z}_0(t)}{dt}=0\Rightarrow \mathrm{z}_0(t)=\mathbf{z}_c\forall t \tag{I14}$$

and hence

$$\frac{d\mathrm{z}_0(t)}{dt}=0\Leftrightarrow D^1\mathcal{E}_0(\bar{\mathbf{z}}_c,\mathbf{z}_c)=0. \tag{I15}$$

This leads to

$$\begin{pmatrix}0\\0\end{pmatrix}=\begin{pmatrix}\mathbb{O} & \bar{\mathbb{H}}_0-\mathcal{E}(\bar{\mathbf{z}}_c,\mathbf{z}_c)\mathbb{I}\\ \mathbb{H}_0-\mathcal{E}(\bar{\mathbf{z}}_c,\mathbf{z}_c)\mathbb{I} & \mathbb{O}\end{pmatrix}\begin{pmatrix}\mathbf{z}_c\\ \bar{\mathbf{z}}_c\end{pmatrix} \tag{I16}$$

which displays that $\mathbf{z}_c$ is an eigenvector of $\mathbb{H}_0$ and $|\Phi(\mathbf{z}_c)\rangle$ an eigenstate of H_0.

It is interesting to observe that the actual stationary state CI coefficients and stationary state that satisfy the TDSE are given from eqs. $(I10)$ and $(I11)$ as

$$\eta_0(t)=e^{-i\mathcal{E}(\bar{\mathbf{z}}_c,\mathbf{z}_c)t}\mathbf{z}_c \tag{I17}$$

and

$$|\Psi(\eta_0(t))\rangle=e^{-i\mathcal{E}(\bar{\mathbf{z}}_c,\mathbf{z}_c)t}|\Phi(\mathbf{z}_c)\rangle \tag{I18}$$

respectively.

APPENDIX J: LINEAR RESPONSE FOR THE EXACT TDSE

In order to obtain the linear response of stationary states of H_0 to a time dependent external field $F(t)$ one considers terms up to first order in $F(t)$

$$|\Phi(\mathrm{z}(t))\rangle=Te^{-i\int_0^t H(t')dt'}|\Phi(\mathrm{z}(0))\rangle\simeq e^{-iH_0t}\left\{1-i\int_0^t F(t')\,dt'\right\}|\Phi(\mathrm{z}(0))\rangle, \tag{J1}$$

where

$$H(t)=H_0+F(t) \tag{J2}$$

and the initial state is a stationary state, $|\Psi_0\rangle$, of H_0

$$|\Psi_0\rangle=|\Phi(\mathrm{z}(0))\rangle=|\Phi(\mathrm{z}_0(0))\rangle=|\Phi(\mathbf{z}_c)\rangle. \tag{J3}$$

In the CS setting the expectation values of quantum observables, A, correspond to classical observables, $\mathcal{A}$ defined on the phase space $\mathcal{M}$ and for $SU\left(\mathcal{H}^N\right)/U(1)$-CS's they have the form

$$\mathcal{A}\left(\bar{\mathbf{z}},\mathbf{z}\right)=\frac{\left\langle \Phi\left(\mathbf{z}\right)\right| A\left(t\right)\Phi\left(\mathbf{z}\right)\rangle}{\left\langle \Phi\left(\mathbf{z}\right)\right| \Phi\left(\mathbf{z}\right)\rangle}=\frac{\sum_{1\leq k,l\leq\binom{r}{N}} z_k\bar{z}_l A_{lk}}{\sum_{1\leq k\leq\binom{r}{N}} \left|z_k\right|^2}=\frac{\mathbf{z}^\dagger \mathbb{A}\mathbf{z}}{\mathbf{z}^\dagger\mathbf{z}}. \tag{J4}$$

The linear response of the expectation value of an operator A from its value in the stationary state $|\Psi_0\rangle$ is given by eq. (142) in the Hilbert space setting and in terms of the approximate CS parameter path given by first order perturbation theory as

$$\xi\left(t\right)\simeq e^{-i\mathbb{H}_0 t}\left\{1-i\int_0^t \mathbb{F}_H\left(t'\right)dt'\right\}\mathbf{z}_c \tag{J5}$$

where

$$\mathbb{F}_H\left(t'\right)=e^{i\mathbb{H}_o t'}\mathbb{F}\left(t'\right)e^{-i\mathbb{H}_o t'}. \tag{J6}$$

After one substitutes eq. (128) into eq. (**??**), noting that the denominator of eq. (**??**) is constant up to first order in F, one can deduce that the evolution of a classical observable, $\mathcal{A}$, is determined up to first order in F around the point $\left(\bar{\mathbf{z}}_c,\mathbf{z}_c\right)$ by

$$\begin{aligned}\mathcal{A}\left(\bar{\xi}\left(t\right),\xi\left(t\right)\right)\approx{}&\frac{\mathbf{z}_c^\dagger e^{i\mathbb{H}_0 t}\mathbb{A}e^{-i\mathbb{H}_0 t}\mathbf{z}_c}{\mathbf{z}_c^\dagger\mathbf{z}_c}-\\ &i\frac{1}{\mathbf{z}_c^\dagger\mathbf{z}_c}\mathbf{z}_c^\dagger\left\{e^{i\mathbb{H}_0 t}\mathbb{A}e^{-i\mathbb{H}_0 t}\int_0^t \mathbb{F}_H\left(t'\right)dt'-\int_0^t \mathbb{F}_H\left(t'\right)dt' e^{i\mathbb{H}_0 t}\mathbb{A}e^{-i\mathbb{H}_0 t}\right\}\mathbf{z}_c.\end{aligned} \tag{J7}$$

Eq. (**??**) directly shows that the linear response of the observable $\mathcal{A}$ is

$$\begin{aligned}\delta^1\mathcal{A}\left(\bar{\mathbf{z}}_c,\mathbf{z}_c,t\right)&=\frac{1}{\mathbf{z}_c^\dagger\mathbf{z}_c}\mathbf{z}_c^\dagger\left\{-i\int_0^t\left[e^{i\mathbb{H}_0 t}\mathbb{A}e^{-i\mathbb{H}_0 t},\mathbb{F}_H\left(t'\right)\right]dt'\right\}\mathbf{z}_c\\ &=\frac{1}{\mathbf{z}_c^\dagger\mathbf{z}_c}\left\{-i\sum_{r,s}\int_0^t\left\{\mathbf{z}_c^\dagger\left[e^{i\mathbb{H}_0 t}\mathbb{A}e^{-i\mathbb{H}_0 t},e^{i\mathbb{H}_0 t'}\mathbf{a}_r^\dagger\mathbf{a}_s e^{-i\mathbb{H}_0 t'}\right]\mathbf{z}_c\right\}f_{rs}\left(t'\right)dt'\right\}.\end{aligned} \tag{J8}$$

Noticing that $e^{-i\mathbb{H}_0 t}\mathbf{z}_c=e^{-iE_0 t}\mathbf{z}_c$ eq. (**??**) leads to

$$\delta^1\mathcal{A}\left(\bar{\mathbf{z}}_c,\mathbf{z}_c,t\right)=$$
$$\frac{1}{\mathbf{z}_c^\dagger\mathbf{z}_c}\left\{-i\sum_{r,s}\int_0^t\left\{\mathbf{z}_c^\dagger e^{i\mathbb{H}_0(t-t')}\mathbb{A}e^{-i\mathbb{H}_0(t-t')}\mathbf{a}_r^\dagger\mathbf{a}_s\mathbf{z}_c-\mathbf{z}_c^\dagger\mathbf{a}_r^\dagger\mathbf{a}_s e^{i\mathbb{H}_0(t-t')}\mathbb{A}e^{-i\mathbb{H}_0(t-t')}\mathbf{z}_c\right\}f_{rs}\left(t'\right)dt'\right\}, \tag{J9}$$

where $\mathbf{a}_r^\dagger\mathbf{a}_s$ is the matrix given by

$$\left(\mathbf{a}_r^\dagger\mathbf{a}_s\right)_{kl}=\left\langle\Phi_k|a_r^\dagger a_s\Phi_l\right\rangle \tag{J10}$$

and its expectation value wrt $\mathbf{z}_c$ is the (s,r) matrix elements of the FORDM of the state $|\Phi\left(\mathbf{z}_c\right)\rangle$ i.e.

$$\rho_{sr}\left(\bar{\mathbf{z}}_c,\mathbf{z}_c\right)=\mathbf{z}_c^\dagger\left(\mathbf{a}_r^\dagger\mathbf{a}_s\right)\mathbf{z}_c. \tag{J11}$$

The linear response expression eq. (**??**) can succinctly written as

$$\delta^1 \mathcal{A}\left(\bar{\mathbf{z}}_c, \mathbf{z}_c, t\right)=\frac{-i}{\mathbf{z}_c^{\dagger} \mathbf{z}_c} \int_0^t \mathbf{z}_c^{\dagger}\left[\mathbb{A}_H\left(t-t^{\prime}\right), \mathbb{F}\left(t^{\prime}\right)\right] \mathbf{z}_c d t^{\prime}=-i \int_0^t \frac{\left\langle\Phi\left(\mathbf{z}_c\right)\right|\left[A_H\left(t-t^{\prime}\right), F\left(t^{\prime}\right)\right] \Phi\left(\mathbf{z}_c\right)\rangle}{\left\langle\Phi\left(\mathbf{z}_c\right) \mid \Phi\left(\mathbf{z}_c\right)\right\rangle} d t^{\prime}, \tag{J12}$$

where

$$\mathbb{F}\left(t^{\prime}\right)=\sum_{r, s} \mathbf{a}_r^{\dagger} \mathbf{a}_s f_{r s}\left(t^{\prime}\right) \tag{J13}$$

which should be compared to eq. (137). Recognizing that the derivatives of a classical observable $\mathcal{X}$ defined by eq. (**??**) are given by

$$\begin{aligned}
\frac{\partial \mathcal{X}(\bar{\mathbf{z}}, \mathbf{z})}{\partial \bar{\mathbf{z}}} & =\frac{1}{\mathbf{z}^{\dagger} \mathbf{z}}\{\mathbb{X}-\mathcal{X}(\bar{\mathbf{z}}, \mathbf{z}) \mathbb{I}\} \mathbf{z} \\
\frac{\partial \mathcal{X}(\bar{\mathbf{z}}, \mathbf{z})}{\partial \mathbf{z}} & =\frac{1}{\mathbf{z}^{\dagger} \mathbf{z}}\{\overline{\mathbb{X}}-\mathcal{X}(\bar{\mathbf{z}}, \mathbf{z}) \mathbb{I}\} \bar{\mathbf{z}}
\end{aligned} \tag{J14}$$

and that the matrix $\mathbb{X}$ is hermitian one can establish that the terms involving $\mathcal{X}(\bar{\mathbf{z}}, \mathbf{z}) \mathbb{I}$ cancel out and obtain that the linear response of the classical observable $\mathcal{A}$ about the point $\left(\bar{\mathbf{z}}_c, \mathbf{z}_c\right)$ is

$$\begin{aligned}
& \delta^1 \mathcal{A}\left(\bar{\mathbf{z}}_c, \mathbf{z}_c, t\right)= \\
& -i \sum_{r, s} \int_0^t\left\{\frac{\partial \mathcal{A}_H\left(\bar{\mathbf{z}}_c, \mathbf{z}_c, t-t^{\prime}\right)}{\partial \bar{\mathbf{z}}} \frac{\partial \rho_{s r}\left(\bar{\mathbf{z}}_c, \mathbf{z}_c\right)}{\partial \mathbf{z}}-\frac{\partial \rho_{s r}\left(\bar{\mathbf{z}}_c, \mathbf{z}_c\right)}{\partial \bar{\mathbf{z}}} \frac{\partial \mathcal{A}_H\left(\bar{\mathbf{z}}_c, \mathbf{z}_c, t-t^{\prime}\right)}{\partial \mathbf{z}}\right\} f_{r s}\left(t^{\prime}\right) d t^{\prime} .
\end{aligned} \tag{J15}$$

The expression in eq. (**??**) can be formulated in terms of the Poisson bracket induced by the Kähler metric for the $SU\left(\mathcal{H}^N\right)/U(1)$ CS-manifold, after noting that

$$\mathbf{z}^{\dagger}\{\mathbb{X}(t)-\mathcal{X}(t)\} \mathbf{z}=\mathbf{z}^{\dagger} \mathbb{X}(t) \mathbf{z}-\mathcal{X}(t) \mathbf{z}^{\dagger} \mathbf{z}=\mathcal{X}(t) \mathbf{z}^{\dagger} \mathbf{z}-\mathcal{X}(t) \mathbf{z}^{\dagger} \mathbf{z}=\mathbf{0}, \tag{J16}$$

which shows that $\mathbf{z}$ is always orthogonal to $\{\mathbb{X}(t)-\mathcal{X}(t)\} \mathbf{z}$ and hence

$$\mathbf{P}_{\mathbf{z}}^{\perp}\{\mathbb{X}(t)-\mathcal{X}(t)\} \mathbf{z}=\{\mathbb{X}(t)-\mathcal{X}(t)\} \mathbf{z}, \tag{J17}$$

as

$$\delta^1 \mathcal{A}(t)=-\sum_{r, s} \int_0^t\left\{\mathcal{A}_H\left(t-t^{\prime}\right), \rho_{s r}\right\} f_{r s}\left(t^{\prime}\right) d t^{\prime}=-\sum_{r, s} \int_{-\infty}^{\infty} \theta_{+}\left(t-t^{\prime}\right)\left\{\mathcal{A}_H\left(t-t^{\prime}\right), \rho_{s r}\right\} f_{r s}\left(t^{\prime}\right) d t^{\prime}, \tag{J18}$$

where

$$\mathcal{A}_H(\bar{\mathbf{z}}, \mathbf{z}, \tau)=\frac{\langle\Phi(\mathbf{z})| A_H(\tau) \Phi(\mathbf{z})\rangle}{\langle\Phi(\mathbf{z}) \mid \Phi(\mathbf{z})\rangle}=\frac{\left\langle\Phi\left(\mathrm{z}_0(\tau)\right)\right| A \Phi\left(\mathrm{z}_0(\tau)\right)\rangle}{\langle\Phi(\mathbf{z}) \mid \Phi(\mathbf{z})\rangle} . \tag{J19}$$

We have thus reformulated the definition of the response function from eq. (141) in terms of classical observables as

$$R_{Ars}(\tau) = \theta_+(\tau)\{\mathcal{A}_H(\tau), \rho_{sr}(0)\}. \tag{J20}$$

and obtain eq. (142) in the form

$$\delta^1 \mathcal{A}(t) = \sum_{k,l} \int_{-\infty}^{\infty} R_{Ars}(t-t') f_{rs}(t')\, dt'. \tag{J21}$$

In particular the response of any one particle expectation value can be expressed in terms of the response of the FORDM $\rho(0)$ given by

$$\delta^1 \rho_{ji}(t) = \sum_{k,l} \int_{-\infty}^{\infty} R_{ijrs}(t-t') f_{rs}(t')\, dt', \tag{J22}$$

where

$$\begin{aligned} R_{ijrs}(\tau) &= \theta_+(\tau)\{\rho_{jiH}(\tau), \rho_{sr}(0)\} \\ &= \theta_+(\tau) \begin{pmatrix} \frac{\partial\rho(\mathrm{z}_0(\tau),\bar{\mathrm{z}}_0(\tau))_{ji}}{\partial\bar{\mathbf{z}}} & \frac{\partial\rho(\mathrm{z}_0(\tau),\bar{\mathrm{z}}_0(\tau))_{ji}}{\partial\mathbf{z}} \end{pmatrix} \begin{pmatrix} \mathbf{0} & i\mathbf{P}^{\perp}_{\mathrm{z}_0(0)} \\ -i\mathbf{P}^{\perp}_{\bar{\mathrm{z}}_0(0)} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \frac{\partial\rho(\mathrm{z}_0(0),\bar{\mathrm{z}}_0(0))_{sr}}{\partial\bar{\mathbf{z}}} \\ \frac{\partial\rho(\mathrm{z}_0(0),\bar{\mathrm{z}}_0(0))_{sr}}{\partial\mathbf{z}} \end{pmatrix} \end{aligned} \tag{J23}$$

APPENDIX K: DIFFERENT TIME POISSON BRACKETS

Kramer and Saraceno [?] show that if the operators O_1 and O_2 belong to the representation, $\mathcal{R}_{\mathcal{H}^N}(\mathcal{G})$, of the Lie algebra, $\mathcal{G}$, of G on the Hilbert space $\mathcal{H}^N$, i.e.

$$O_k = \sum_{1\le j\le\nu} \alpha_{kj} X_j; \qquad k = 1, 2 \tag{K1}$$

and

$$\exp\left\{\sum_{1\le j\le\nu} \alpha_{kj} X_j\right\} \in \mathcal{R}_{\mathcal{H}^N}(G) \tag{K2}$$

then

$$\{\mathcal{O}_1, \mathcal{O}_2\}_{\mathbf{z}} = \frac{\langle\Phi(\mathbf{z})|\,[O_1, O_2]\,\Phi(\mathbf{z})\rangle}{\langle\Phi(\mathbf{z})|\,\Phi(\mathbf{z})\rangle}, \tag{K3}$$

where

$$\exp\left\{\sum_{1\le j\le\nu} \alpha_{kj} Q_j\right\} \in \mathcal{R}_{\mathcal{H}^N}(G/K) \tag{K4}$$

and

$$|\Phi(\mathbf{z})\rangle = \exp\left\{\sum_{1\leq j\leq\nu} z_j Q_j\right\} |\Phi_{ref}\rangle\,. \tag{K5}$$

This relationship can be made time dependent by

$$\{\mathcal{O}_1(t_1), \mathcal{O}_2(t_2)\}_{\mathbf{z}} = \frac{\langle\,\Phi(\mathbf{z})|\,[O_1(t_1), O_2(t_2)]\,\Phi(\mathbf{z})\rangle}{\langle\,\Phi(\mathbf{z})|\,\Phi(\mathbf{z})\rangle}, \tag{K6}$$

where $O_1(t_1)$ and $O_2(t_2)$ belong to $\mathcal{R}_{\mathcal{H}^N}(\mathcal{G})$. If we consider a linear time evolution given by

$$O_k(t_k) = e^{iht_k} O_k e^{-iht_k} \quad \in \mathcal{R}_{\mathcal{H}^N}(\mathcal{G})\,. \tag{K7}$$

expression eq. (**??**) can be developed to give

$$\begin{aligned}
\{\mathcal{O}_1(t_1), \mathcal{O}_2(t_2)\}_{\mathbf{z}} &= \frac{\left\langle\,\Phi(\mathbf{z})\right|\left\{e^{iht_1} O_1 e^{-ih(t_1-t_2)} O_2 e^{-iht_2} - e^{iht_2} O_2 e^{ih(t_1-t_2)} O_1 e^{-iht_1}\right\}\Phi(\mathbf{z})\right\rangle}{\langle\,\Phi(\mathbf{z})|\,\Phi(\mathbf{z})\rangle} \\
&= \frac{\left\langle e^{-iht_2}\Phi(\mathbf{z})\right|\left\{e^{ih(t_1-t_2)} O_1 e^{-ih(t_1-t_2)} O_2 e^{-iht_2} - e^{-iht_2} O_2 e^{ih(t_1-t_2)} O_1 e^{-ih(t_1-t_2)}\right\} e^{-iht_2}\Phi(\mathbf{z})\right\rangle}{\langle\Phi(\mathbf{z})|\,\Phi(\mathbf{z})\rangle} \\
&= \frac{\left\langle e^{-iht_2}\Phi(\mathbf{z})\right|[O_1(t_1-t_2), O_2]\,e^{-iht_2}\Phi(\mathbf{z})\right\rangle}{\langle\Phi(\mathbf{z})|\,\Phi(\mathbf{z})\rangle} = \frac{\langle\Phi(\chi(t_2))|\,[O_1(t_1-t_2), O_2]\,\Phi(\chi(t_2))\rangle}{\langle\Phi(\mathbf{z})|\,\Phi(\mathbf{z})\rangle} \\
&= \{\mathcal{O}_1(t_1-t_2), \mathcal{O}_2\}_{\mathbf{z}}\,,
\end{aligned} \tag{K8}$$

where

$$\exp\left\{\sum_{1\leq j\leq\nu} \chi_j(t)\,Q_j\right\} |\Phi_{ref}\rangle = e^{-iht} \exp\left\{\sum_{1\leq j\leq\nu} z_j Q_j\right\} |\Phi_{ref}\rangle \tag{K9}$$

and

$$\langle\,\Phi(\chi(t))|\,\Phi(\chi(t))\rangle = \langle\,\Phi(\mathbf{z})|\,\Phi(\mathbf{z})\rangle \quad \forall t. \tag{K10}$$

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